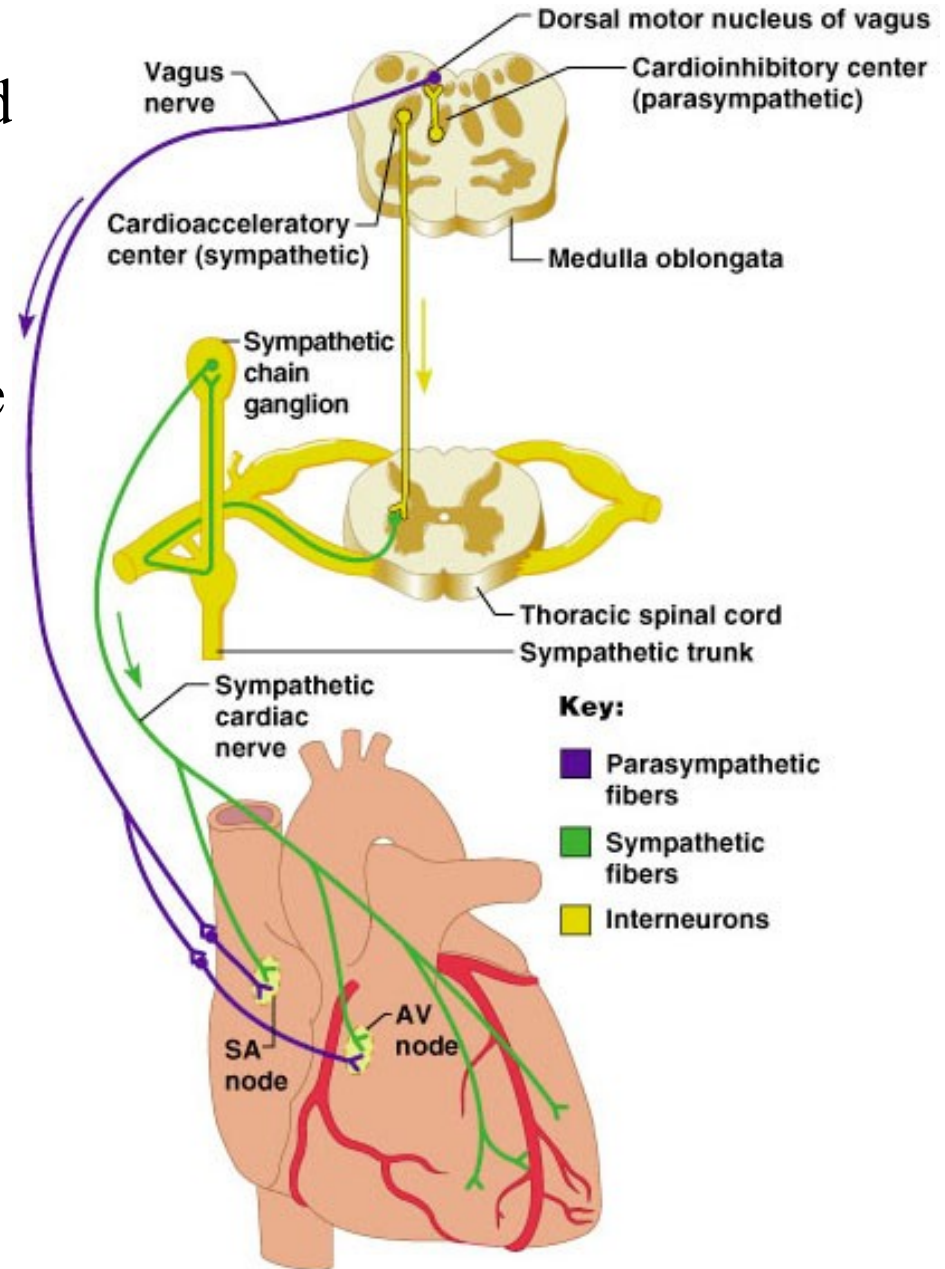


Innervation of the Heart

- Heart has dual autonomic innervation from both SNS and PNS with afferent and efferent components.
- The **sympathetic nerve supply** to the heart is controlled by the **medullary vasoconstrictor/ cardio accelerator center**
- Preganglionic sympathetic fibers arise from the lateral horn of the upper 5-thoracic spinal segments
- Postganglionic sympathetic fibers arise from the cervical and thoracic ganglia and proceed to supply atria, ventricles and nodal areas



Innervation of the heart cont'd...

- The parasympathetic nerve supply to the heart is controlled by **the vasodilator/ cardio inhibitor center**.
- Preganglionic parasympathetic fibers arise from cardio inhibitory center in the medulla and proceed as vagal fibers to relay in terminal ganglia in the wall of the atria
- Short postganglionic fibers arise from terminal ganglia and supply the atria, SA-node and the AV-node
- The right vagus has a strong influence on SA node, while the left vagus has dominant effect on AV node
- Ventricles are not supplied by vagus nerve

Afferent cardiac nerves include

- Pain receptors
- Stretch receptors
- Chemo receptors

which are visceral afferent fibers

transmitted↓through

Sympathetic/vagus

Electrical Activity of Cardiac Cells

- **Cardiac Excitability:** varies considerably depending on whether the action potentials are **fast responses** (from muscle of atrium and ventricle, and PNW) or **slow responses** (from SA Node, AV node).
- **Trans-membrane Potentials:** occurring across cell membranes include:
 - Resting Membrane Potential (RMP)
 - Action Potential (AP)

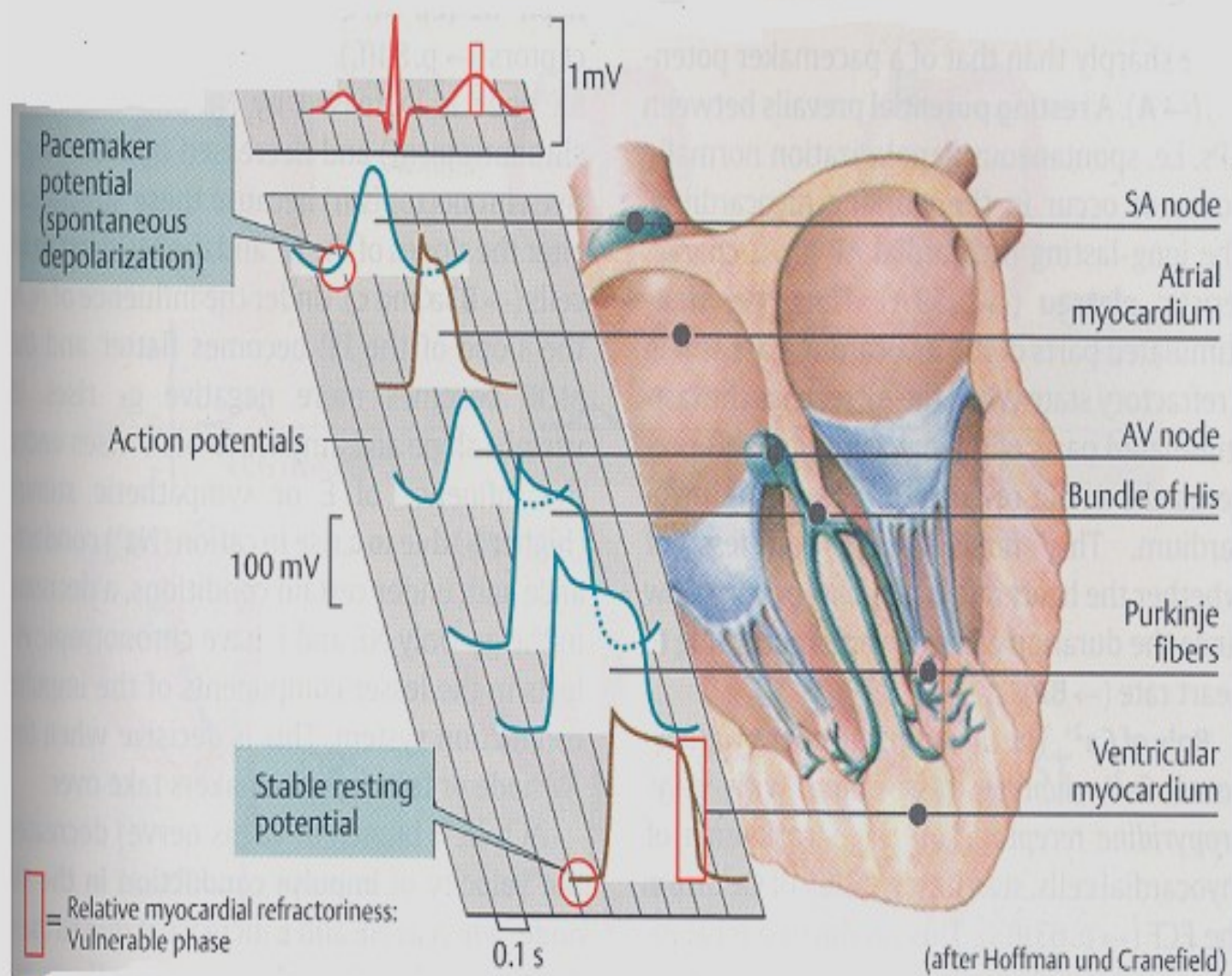


Fig..Cardiac excitation

Ionic Basis of Cardiac Action Potential

- Various phases of cardiac AP are associated with changes in the permeability of the cell membrane to, mainly, Na, K, and Ca ions.
- During AP, the influx of sodium into the cardiac cell occurs via two channels:
 1. **A fast channel** that accounts for the early influx of Na⁺ (atria, ventricles, and PNW)
 2. **A slow channel** that permits Ca⁺⁺ and some sodium to move down its concentration and electrical gradients into the cell (SA node and AV node).

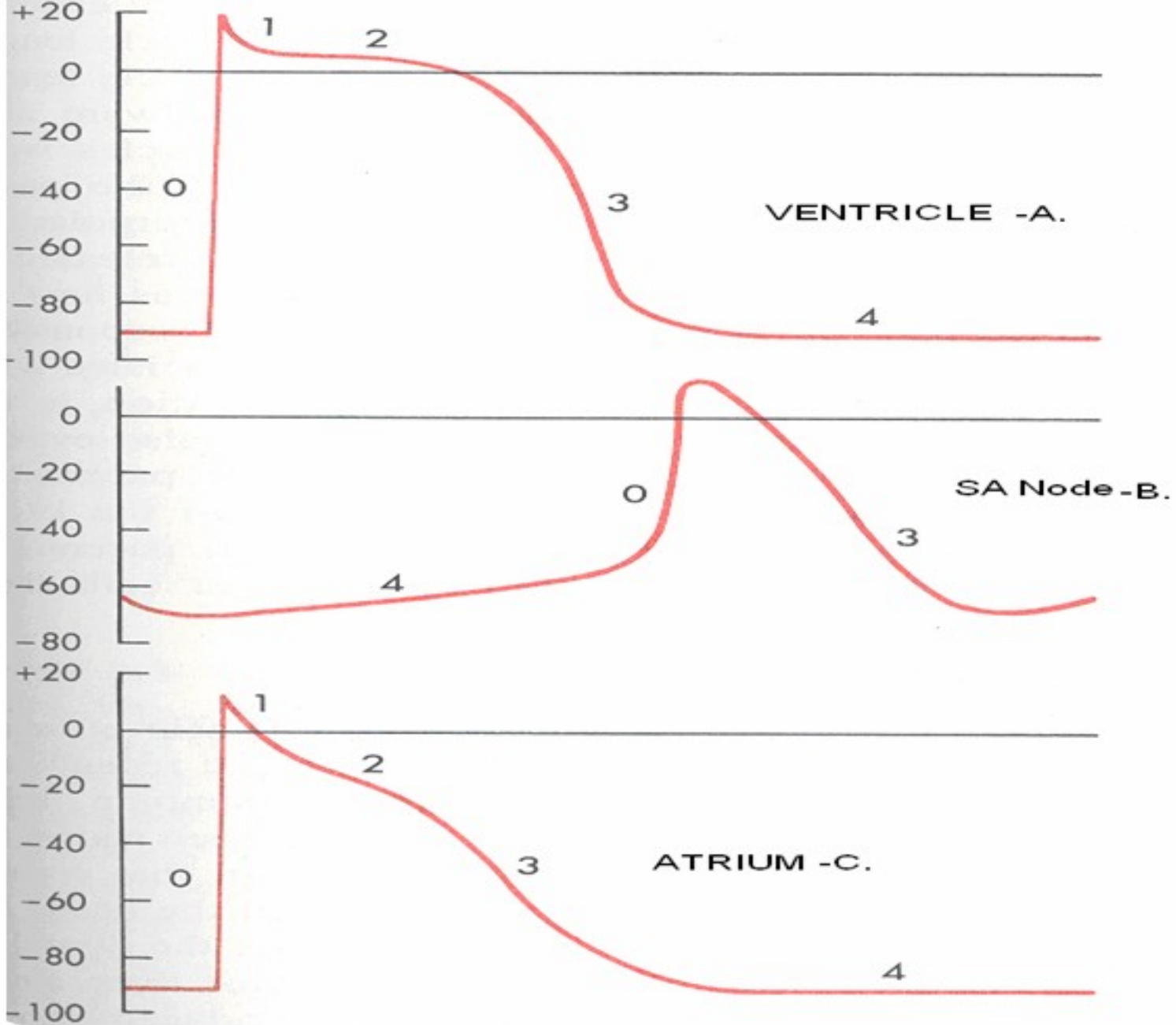


Fig..Action potentials (mv)

Electrophysiology of the heart

Phases and ionic basis of myocardial action potential

It has the following phases

Phase-0: Rapid depolarization

Caused by rapid Na-influx

Phase-1: Early brief repolarization

Caused by Cl^- influx

Phase-2: The plateau (prolonged depolarization)

Caused by Ca^{2+} influx via L-channels

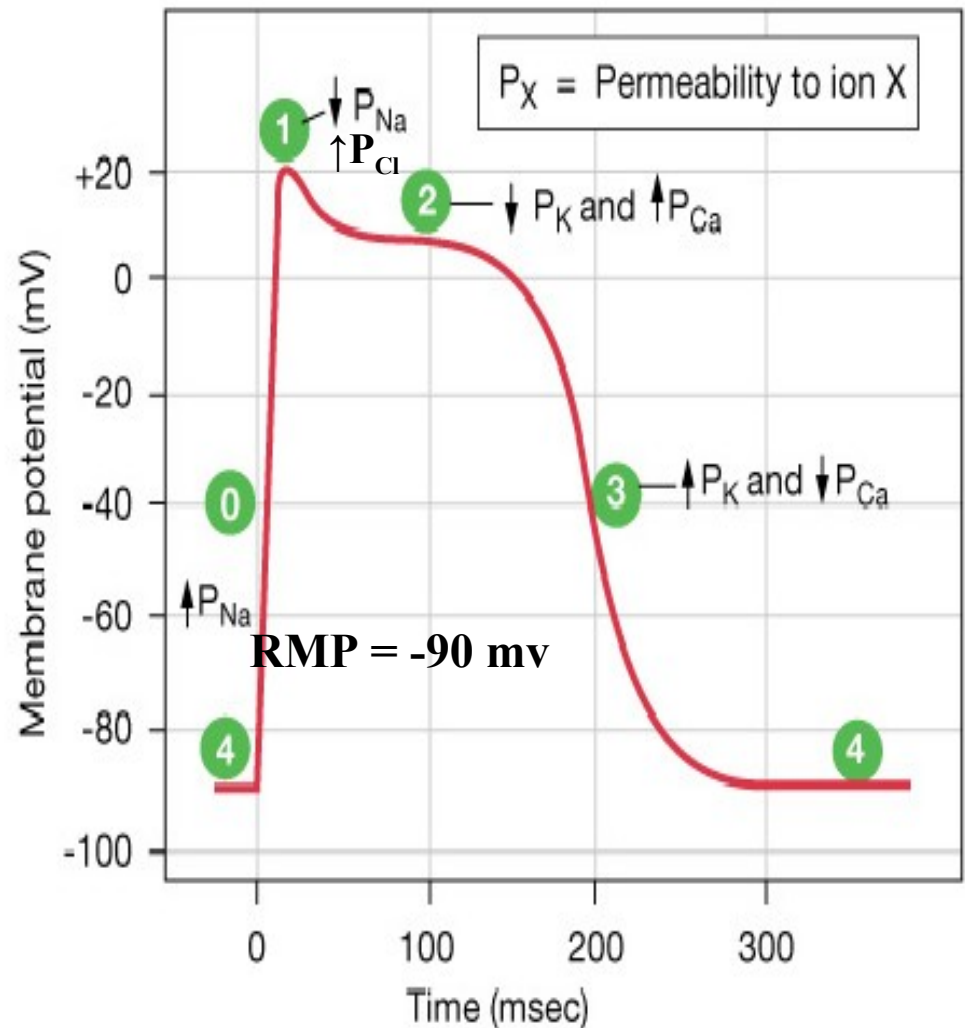
Phase-3: Repolarization

Caused by K^+ efflux

Phase-4: complete repolarization

RMP re-established

Caused by Na^+/K^+ -ATPase



Electrical Activity of the Pacemaker

- The pacemaker is composed of **small myocytes** with only scanty myofibrils and an electrically **unstable resting membrane potential (pre-potential, about -60 to -70mV)**.
- Following firing of an AP, membrane potential decreases gradually from a basal value of -60 or -70 mV (maximum diastolic potential) to a critical firing level of -40 to -45 mV.
- A wave of depolarization spreads across the two atria passing from cell to cell at a rate of about 1m/s and initiating atrial systole.

Electrical Activity ...cont'd

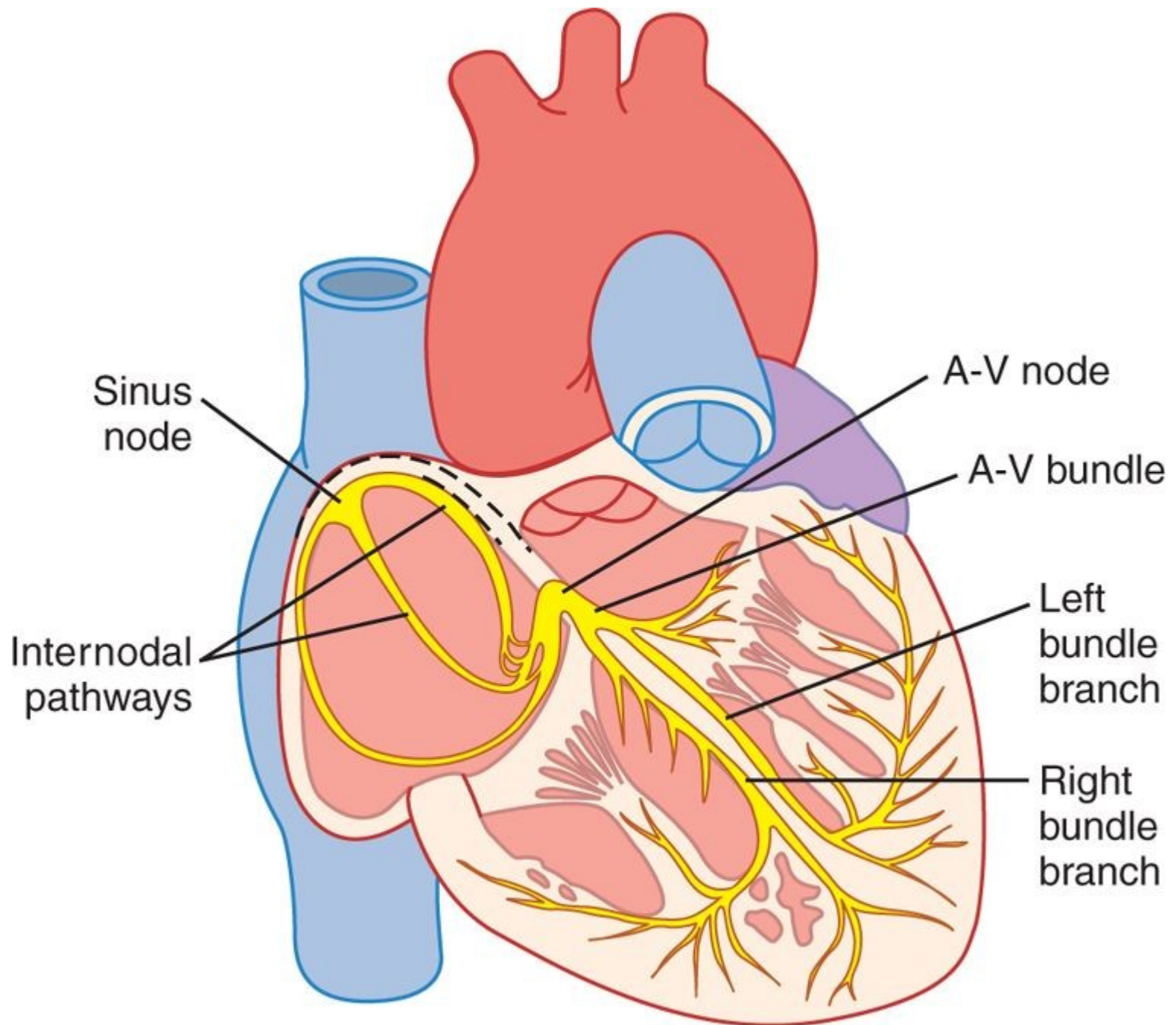
- The **pre-potential** is primarily due to a slow decrease in K^+ permeability and reduced K^+ efflux.
- As K^+ efflux decreases, membrane begins to depolarize forming the first part of the pre-potential.
- Pre-potential is altered by sympathetic and parasympathetic stimulation or other factors like drugs

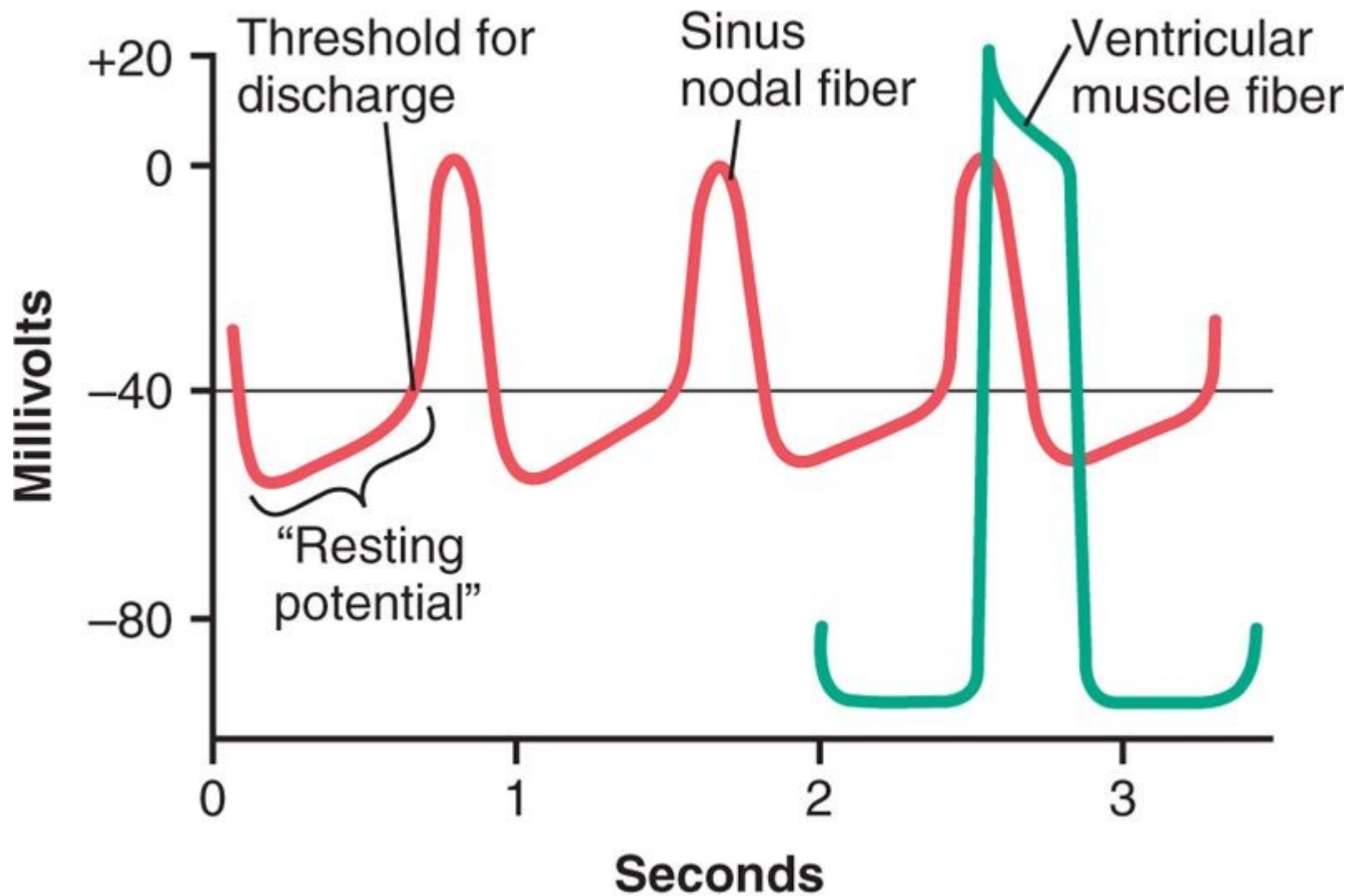
Electromechanical Properties of The Heart

- Cardiac function is **not fully dependent on intact nervous pathways**.
- It continues to beat after full denervation because of the intrinsic properties it has.
 - A. Automaticity (Myogeny):** Ability to start pace making is most highly developed in nodal tissues but can be observed in any piece of cardiac tissue.
 - Inherent rates:
 - SA node = 70-80 bpm, AV node = 60-65 bpm
 - Atria = 60-65 bpm, Ventricles = 30-40 bpm

...cont'd

- Evidence of these rates is obtained from ligature experiments in animals.
- With intact sinus rhythm, HR of a resting
 - adult human is 50 to 100 beats/min;
 - in small mammals like the **shrew**, it is **faster, 600 bpm**.
 - In large mammals like the **elephant**, it is **slower, 25 bpm**.





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...cont'd

- SA node is understood to be primary pacemaker by the following findings:
 - i. First region to display electrical activity is SA node.
 - ii. Crushing and localized cooling of SA node leads to bradycardia
 - iii. Topical application of drugs and humoral agents to SA node leads to alteration of heart rate

B. Conductivity:

- Most highly developed in PNW which is feebly contractile.
- Conduction velocity is some 6 times that of the rest of myocardium.

C. Contractility: Atrial and ventricular muscles are:

1. Highly specialized for contraction.
2. Follow the ALL-or-NONE response pattern.

D. Refractoriness: Cardiac cell is unresponsive to further stimuli. During the phase of contraction, the heart is unexcitable (refractory) to stimulation. There are two degrees of refractoriness:

- **Relative refractoriness:** Only APs of smaller amplitudes and rates of rise can be generated.
- **Absolute refractoriness:** No other AP can be triggered, even by extremely strong stimuli. Because of this refractoriness, the heart muscle cannot be tetanized.

Effect of ANS on the Electromechanical activities of the Heart

- NS controls various aspects of the characteristics of the heart including the **frequency and vigor** (chronotropic and inotropic effects) at which it beats.
- The frequency of pacemaker (SA node) firing is influenced by underlying changes in ionic conductance and ionic flow through the plasma membrane.
- It is controlled by the activity of both divisions of the ANS
 - Stimulation of symp. NS $\rightarrow \uparrow$ HR
 - Stimulation of parasymp. NS $\rightarrow \downarrow$ HR

Excitation-contraction coupling in cardiac muscle

- It is a mechanism by which an action potential causes contraction of myocardium.
- Calcium ions provide the link b/n **electrical excitation** and **muscle contraction** as shown in the figure below.
- The influx of Ca^{++} from the ECF during excitation (plateau) triggers the release of Ca^{++} from the SR.
- The free cytosolic Ca^{++} activates contraction of the myofilament (systole).
- Relaxation (diastole) occurs as a result of uptake of Ca^{++} by the SR and extrusion of intracellular Ca^{++} by $\text{Na}^+ - \text{Ca}^{++}$ exchange

- It has the following sequences
Auto-rhythmicity → Membrane depolarization → T-tubule depolarization → Release of Ca^{2+} from SR + from T-tubule (ECF) → Ca^{2+} activates contractile molecules → Sliding of filaments → Contraction
- The action potential of cardiac muscle has a **prolonged refractory period**.
- Advantage: prevents tetanic contraction of the heart

Important terms

- **Ventricular volumes:** The volume of blood in the ventricles
- **Ventricular end diastolic volume (VEDV):** The volume of blood in the ventricle at the end of ventricular diastole (relaxation phase)

$$EDV = 120-140$$

- **Ventricular end systolic volume (VESV):** The volume of blood that remains in the ventricle at the end of ventricular systole (contraction phase).

$$VESV = 50-60 \text{ ml}$$

Important terms...cont'd

- **Stroke volume (SV):** the volume of blood ejected from the ventricle during ventricular systole.

$$SV = VEDV - VESV, \quad 70 - 80 \text{ ml}$$

- **Cardiac output:** the volume of blood ejected from the heart per minute.

$$CO = SV \times HR = 6 \text{ L/min}$$

- **Ejection fraction:** the blood proportion that enters the ventricles during diastole to the amount ejected.

$$EF = SV/VEDV, \quad 60\% - 70\%$$

Heart Rate

- HR is the number of cardiac cycles per minute
- Normal HR: 60 to 100 beats/minute
 - < 60 beats/minute, bradycardia but normal in athletes
 - > 100 beats/minute, tachycardia

HR varies with the following factors

1. Age: higher in new born infants (120 b/min)
2. Sex: higher in females (85 b/min)
3. Time of the day: $\downarrow$ morning, $\uparrow$ evening
4. Resting and sleep: decreased
5. Physical training: low in athletes (45-60 b/min)
6. Body position: $\uparrow$ standing, $\downarrow$ supine positions

How to count HR?

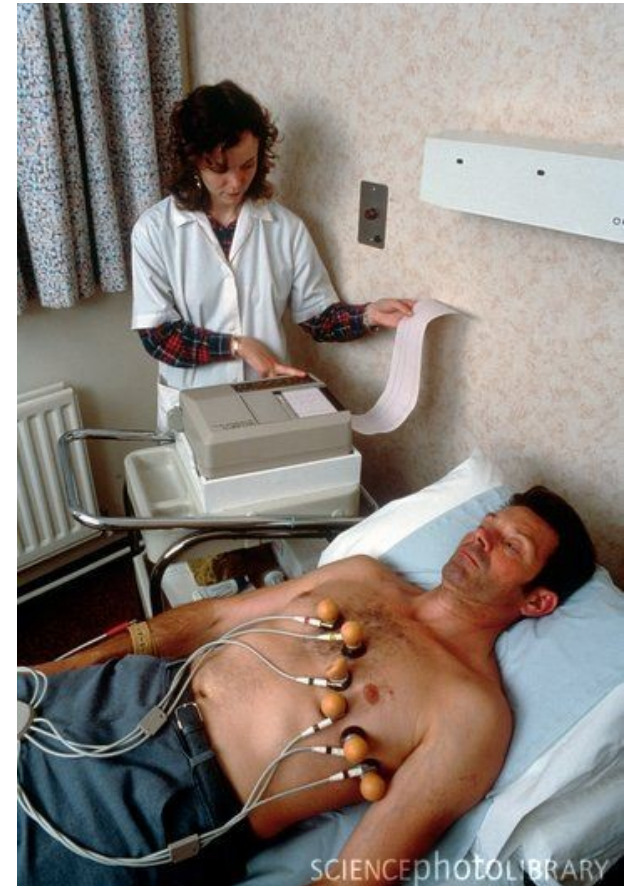
- Counting arterial pulsation, heart sound and ECG cycles

Fundamentals of Electrocardiography

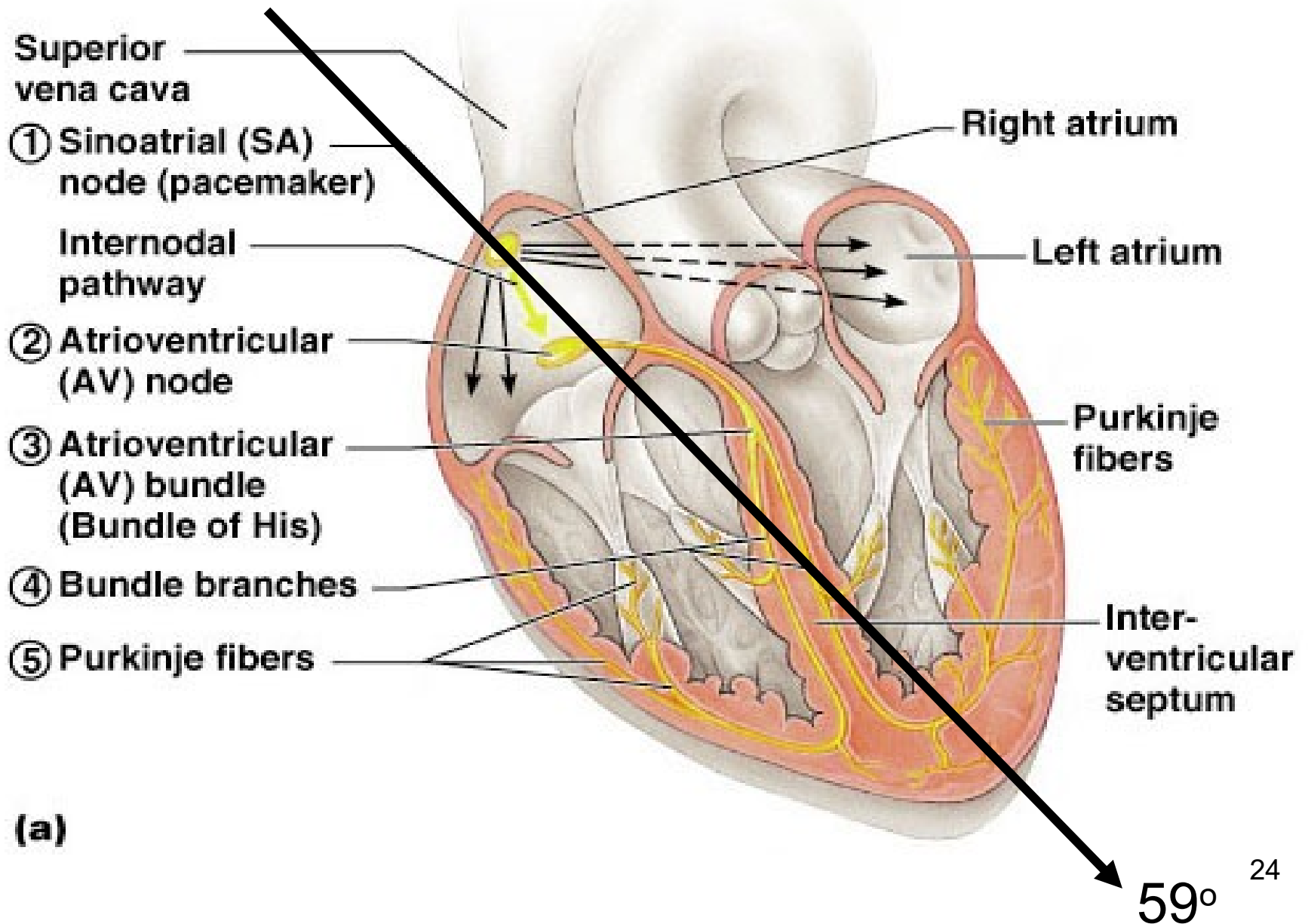
- ECG is the electrical activity of the heart that is conducted through the body fluid and recorded from the surface of the body using highly sensitive electrodes.
- Recorded from specific sites of the body in graphic form (Fig. below) **relating voltage (vertical axis) with time (horizontal axis).**
- It is the sum of all the depolarization occurring in the cardiac muscle fibers during each cardiac cycle
- Developed by W. Einthoven in Leiden and A. Waller in London in 1909.

ECG...cont'd

- ❖ Differences of electric potentials are recorded b/n pairs of points (**electrodes**) on the external surface of the body or b/n skin loci and indifferent or reference electrode.
- ❖ It provides events about arrhythmia, hypertrophy, myocardial damage, ischemia and necrosis
- ❖ Electrical axis of the heart is 59°



Heart Physiology: Sequence of Excitation



ECG...cont'd

ECG Conventions

1. 1mV input→10mm deflection
 2. Paper speed → 25mm/sec.
 3. Recording points → wrist, ankle, skin on chest
 4. Right leg → ground(earth)
- ECG: Fluctuations of potential that represent the **algebraic sum of the action potentials of myocardial fibres recorded extracellularly during the cardiac cycle.**

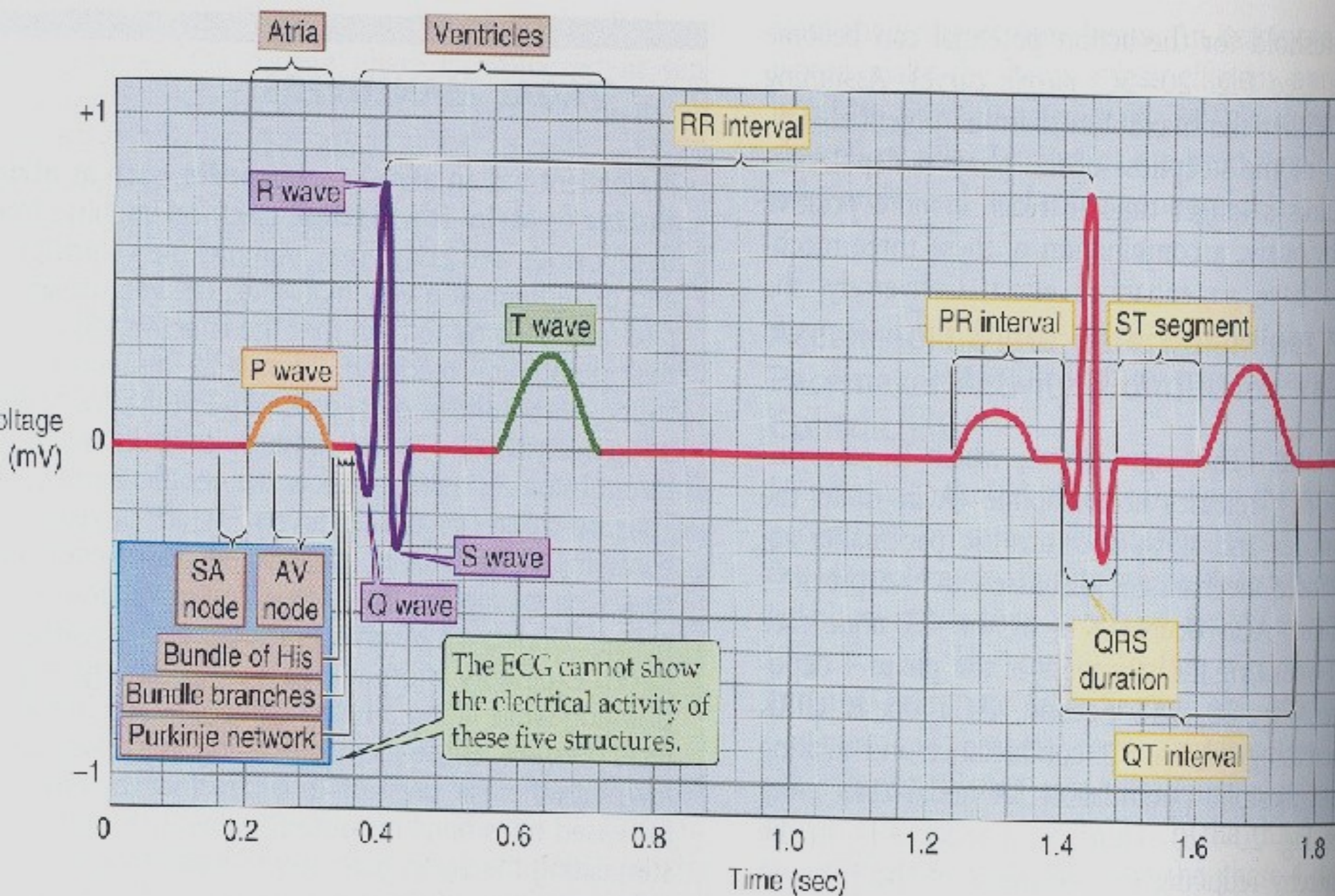
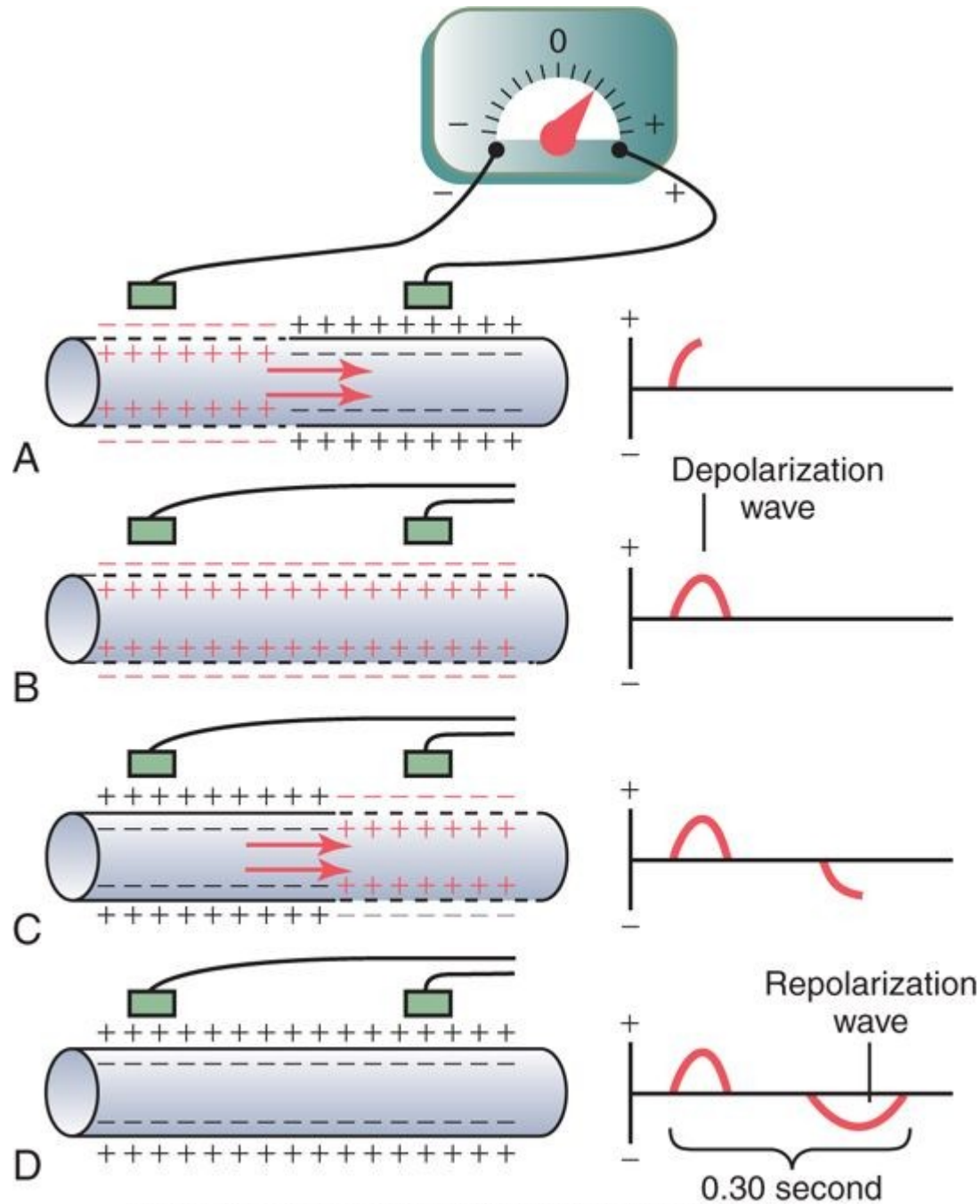
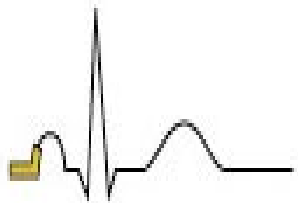
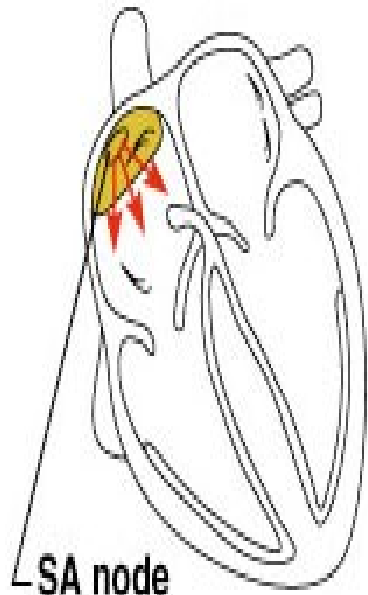


Fig. ECG

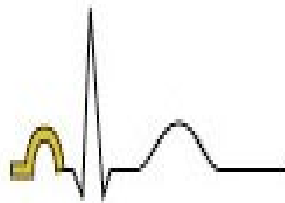
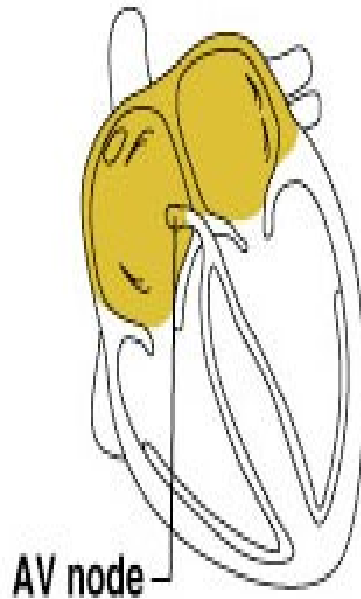


Heart Excitation Related to ECG

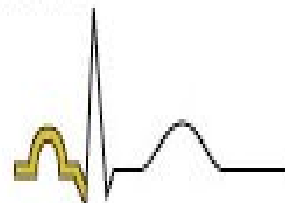
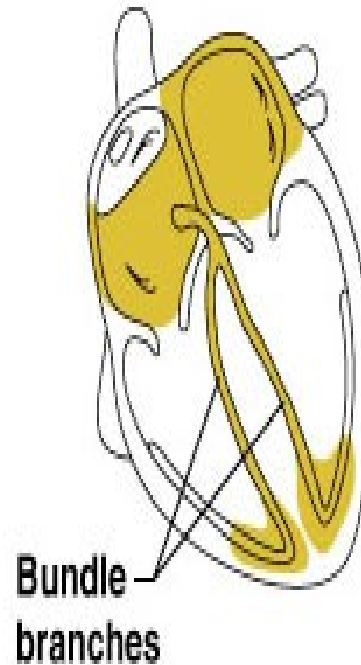
SA node generates impulse; atrial excitation begins



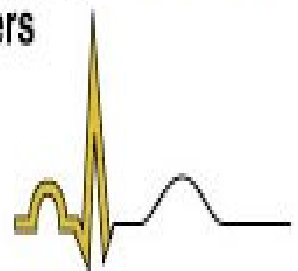
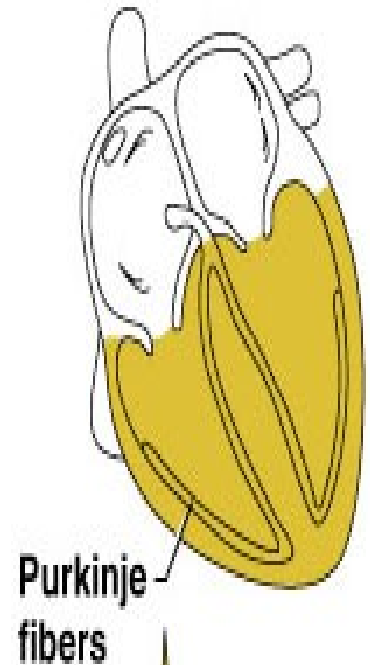
Impulse delayed at AV node

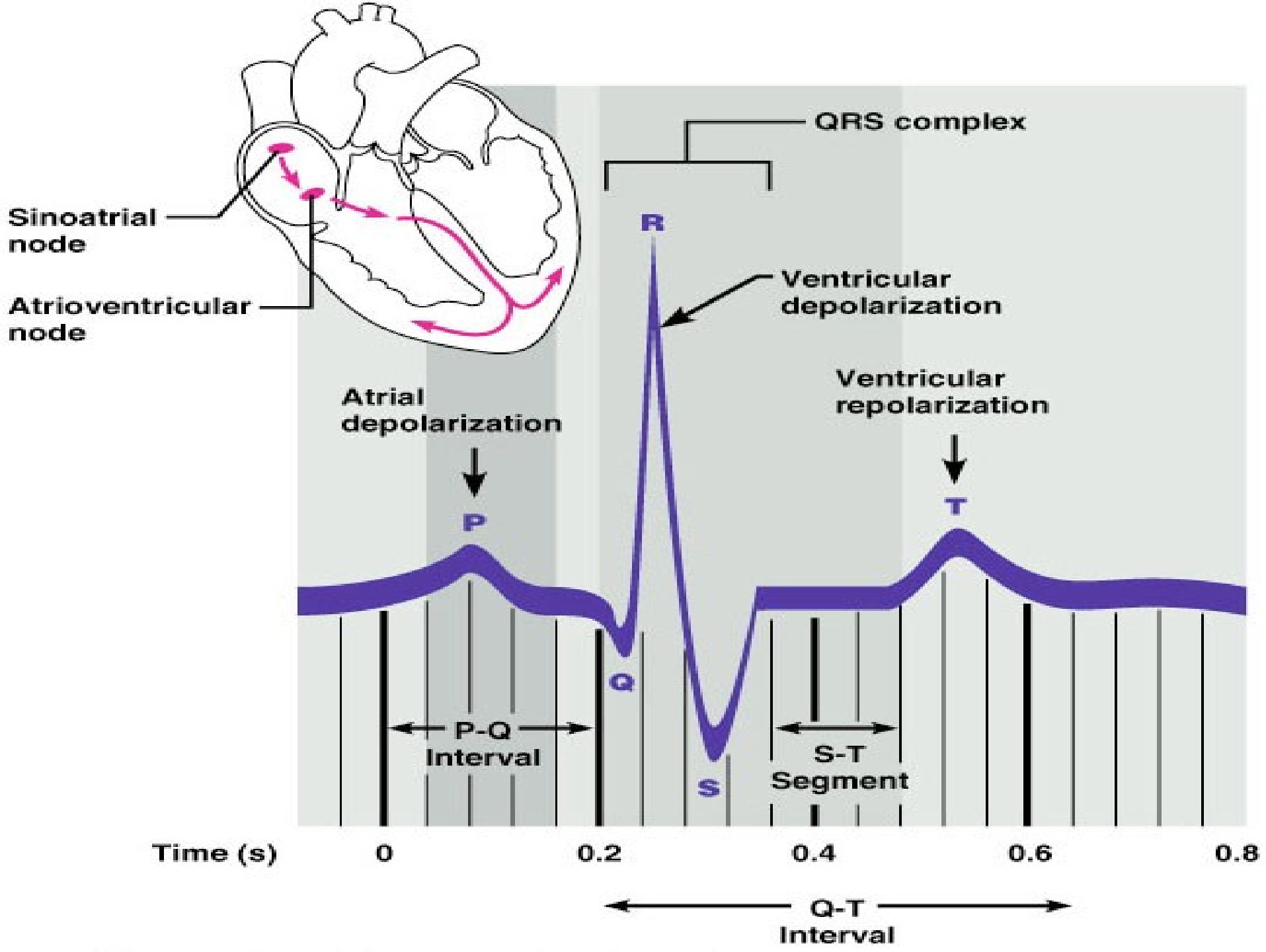


Impulse passes to heart apex; ventricular excitation begins



Ventricular excitation complete



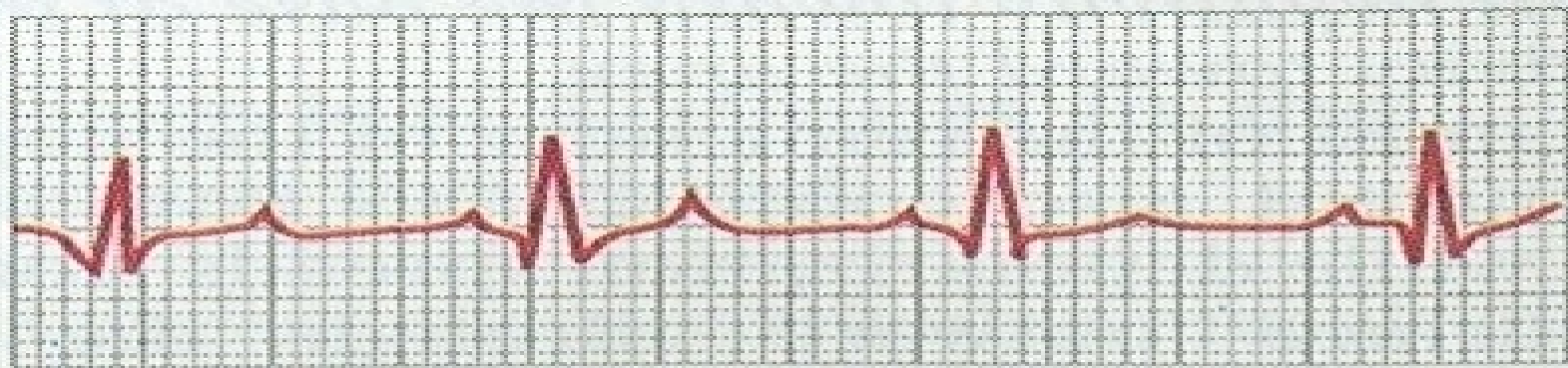
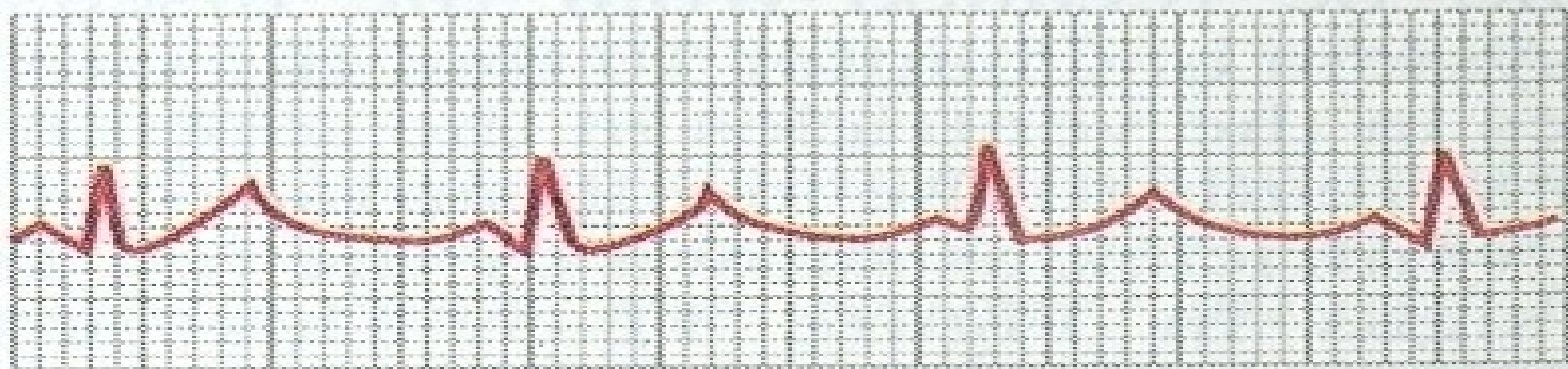


ECG Recordings

- There are 3 types of recording:
 1. Bipolar Limb Leads
 2. Augmented (Unipolar) Limb Leads
 3. Precordial (Chest) Leads

A. Bipolar Limb Leads

- Record voltage b/n two electrodes (leads) placed on the wrists and legs.
- These leads include:
 1. **Lead I= LA-RA**: electric potential d/c b/n Lft arm & Rt arm
 2. **Lead II=LL-RA**: electric potential d/c b/n Lft leg & Rt arm
 3. **Lead III= LL-LA**: electric potential d/c b/n Lft arm & Lft leg.
- Each lead shows waves of depolarization and repolarization: P-wave, QRS complex, T-wave and occasionally U-wave.



Bipolar Limb L...cont'd

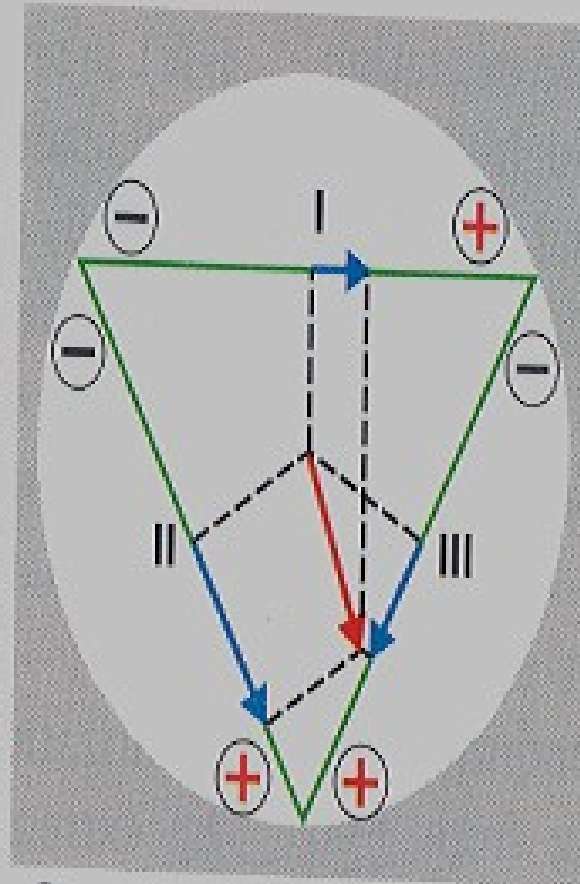
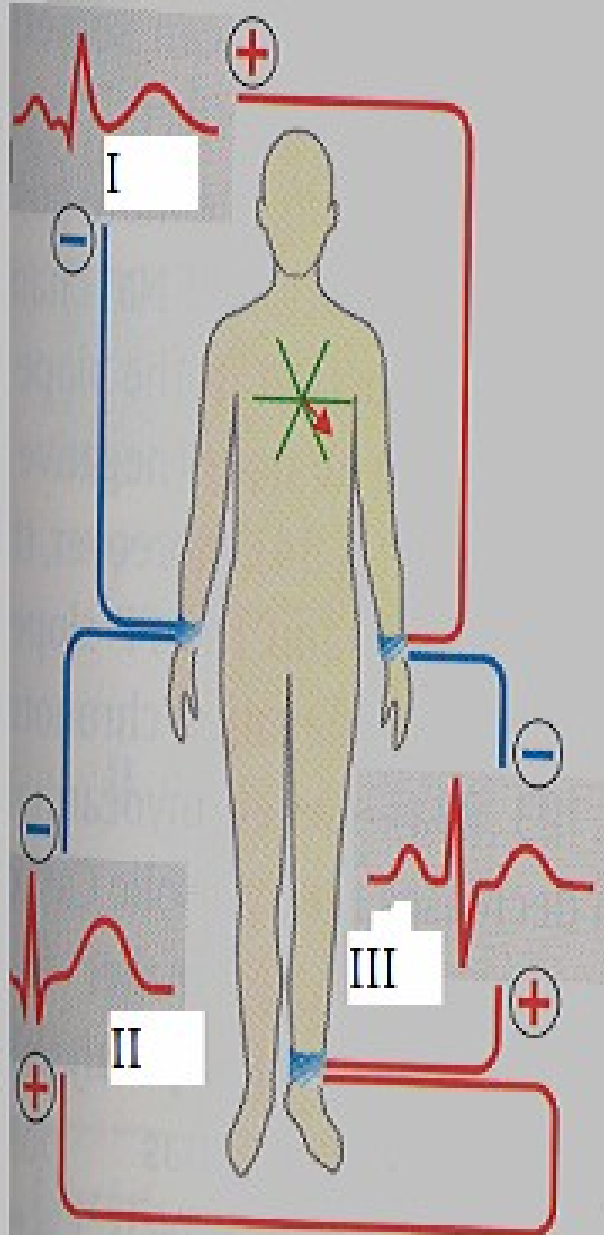
- **Einthoven triangle:**

- Drawn around the heart.
- The triangle shows that the **two arms and the left leg form apices of a triangle surrounding the heart.**
- The two arms with electrodes connect electrically with the fluid around the heart and the left leg with another electrode also connects with the fluid. (Fig. below)

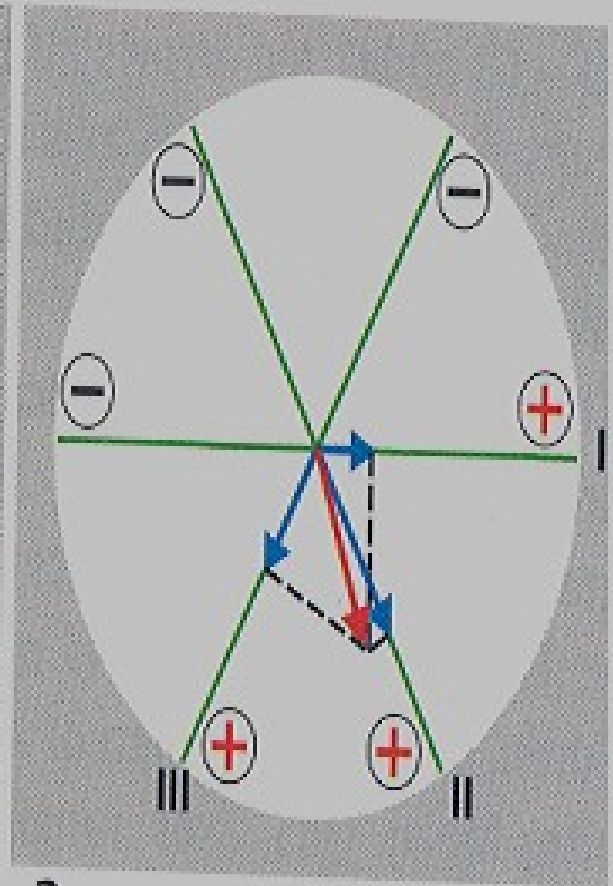
- **Einthoven's law:**

- If electrical potentials of any two of the three leads are given, the 3rd one can be determined.
- The relation, $\text{Lead II} = \text{Lead I} + \text{Lead III}$, can be used to estimate electric potential of each of the leads.

Einthoven leads I, II and III (bipolar)

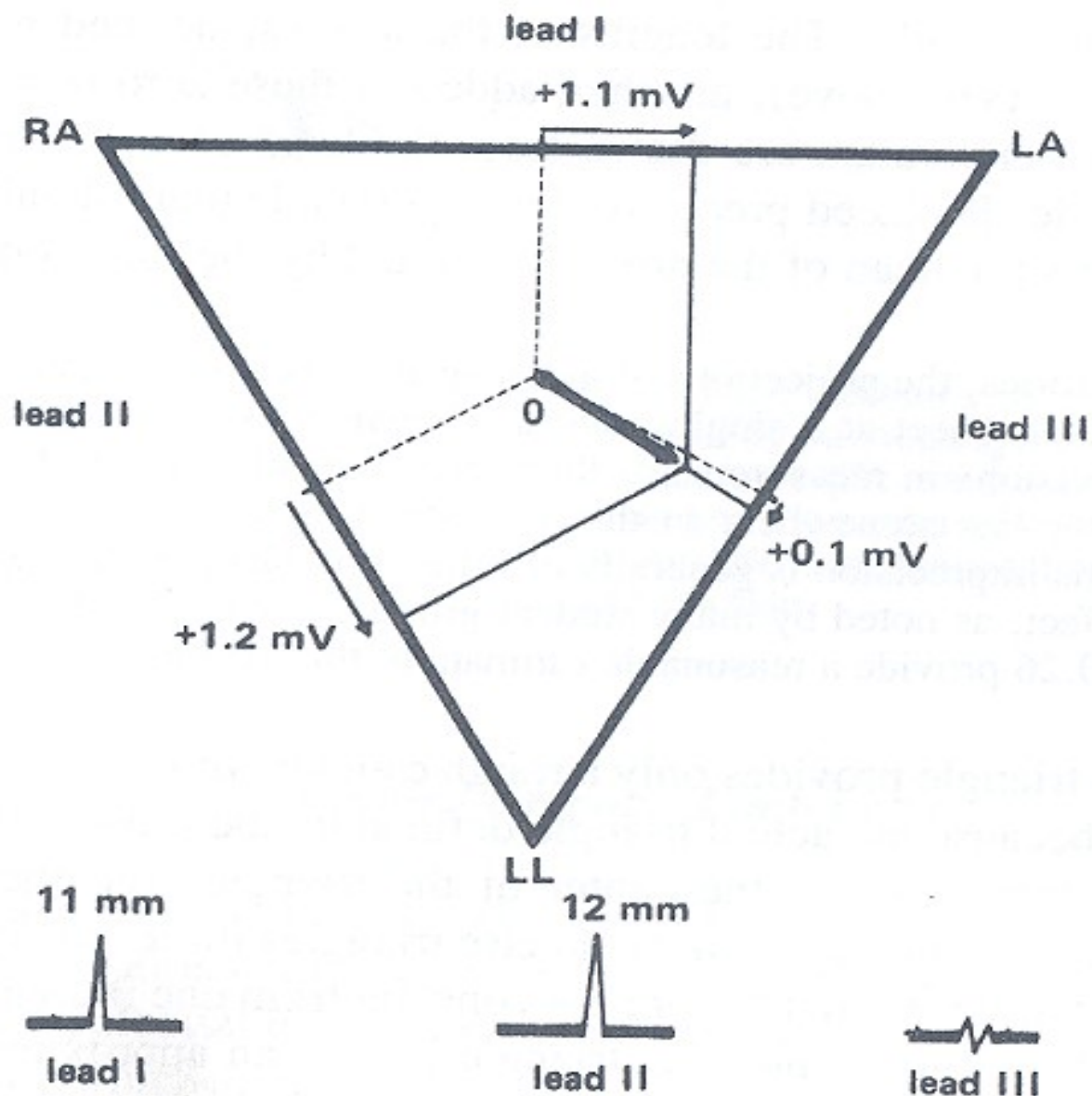


2



3

Triaxial system



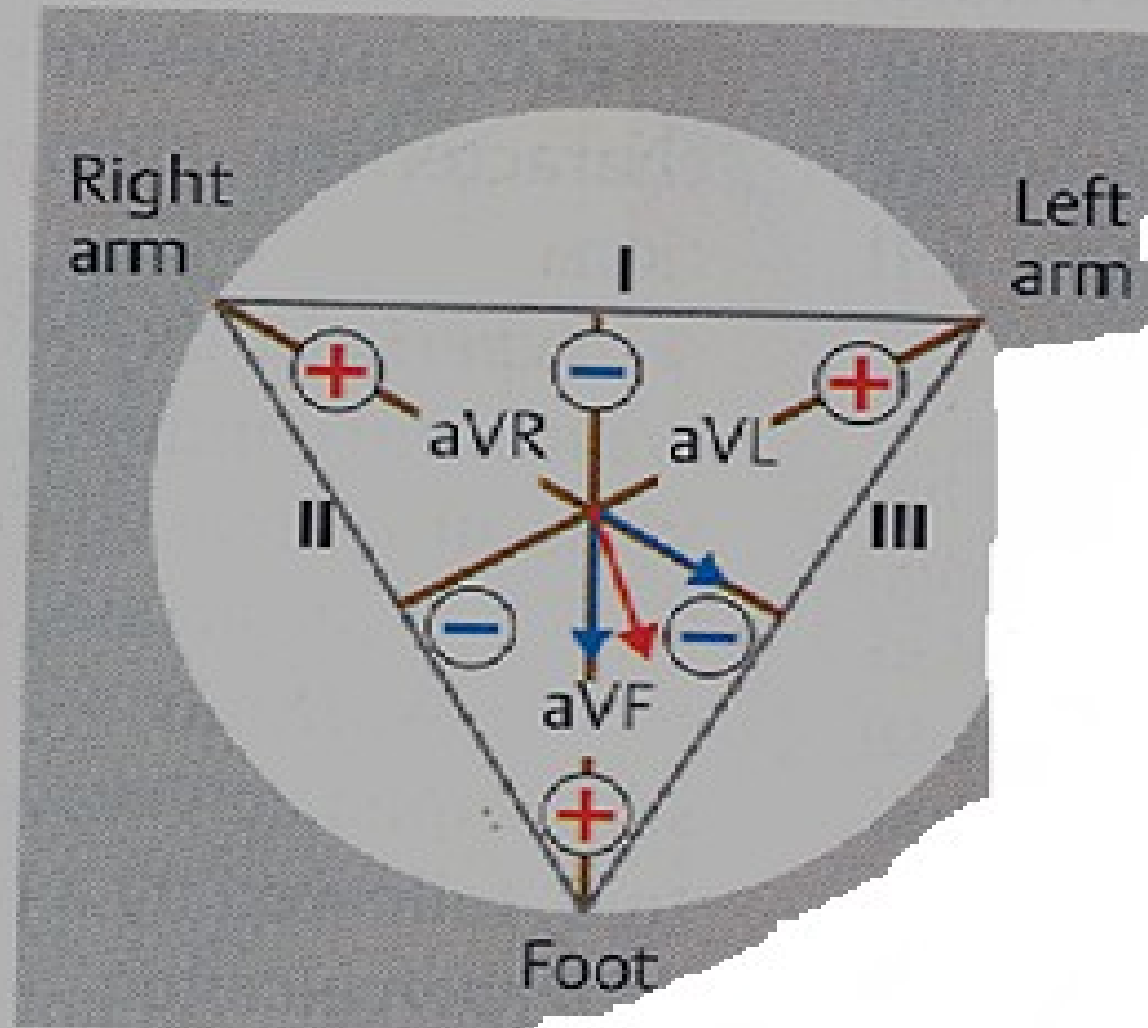
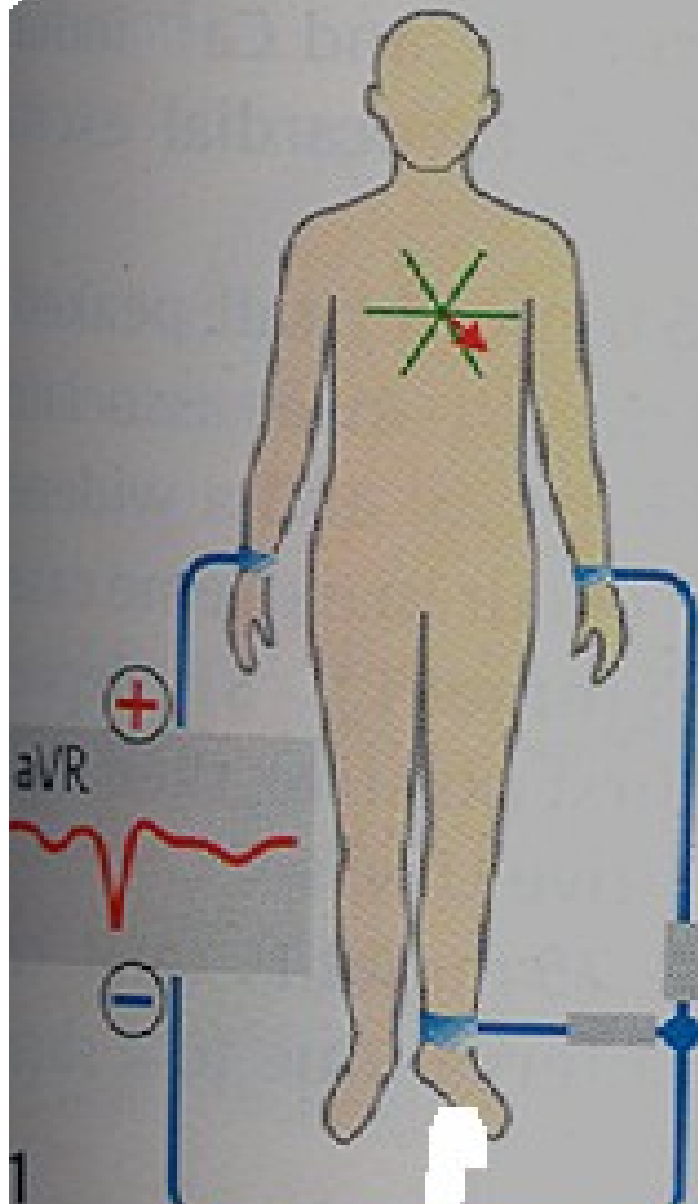
A normal mean QRS vector projected on the Einthoven triangle, each side of which represents one of the three bipolar limb leads (I, II, III).

ECG Recordings...cont'd

B. Unipolar (Augmented) Limb Leads

- Voltage is recorded b/n a single “exploratory electrode” placed on the body and an electrode that is built into the electrocardiograph and maintained at zero potential (ground).
- In this system, two of the limbs are connected through electrical resistances to the **negative (indifferent) terminal** of the electrocardiograph and the 3rd limb is connected to the **positive terminal**.
- When positive terminal is placed on right arm, the lead is designated as **aVR**; when on the left arm **aVL**, and when on left leg, **aVF** (Fig. below).

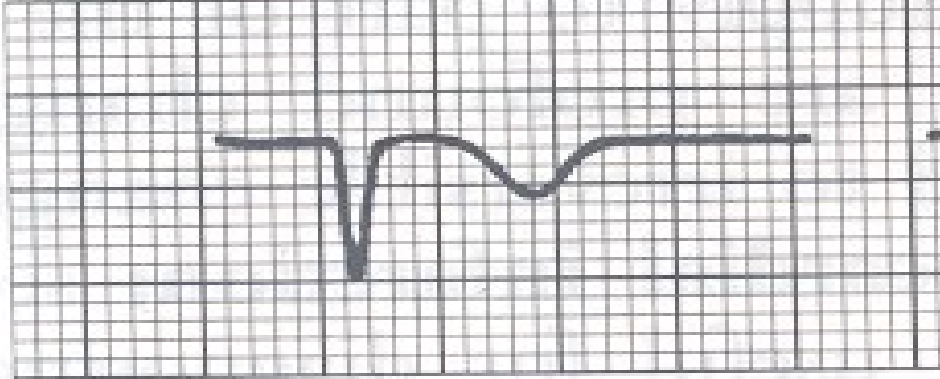
Goldberger limb leads (unipolar)



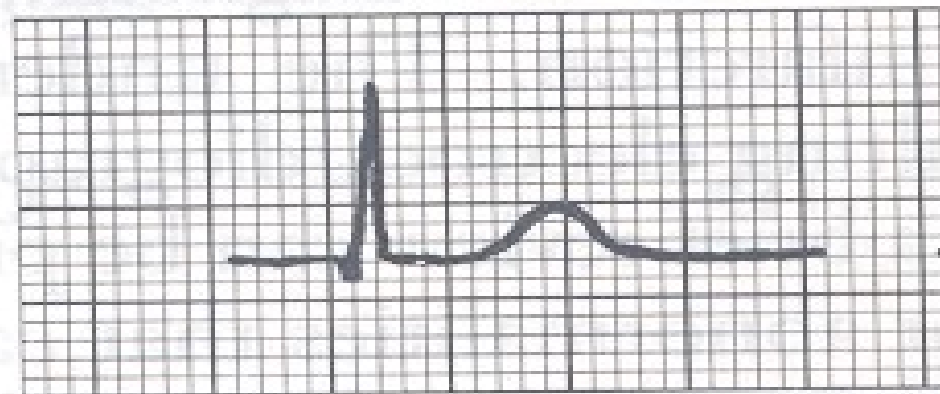
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Hexaxial system

aVR



aVL



aVF

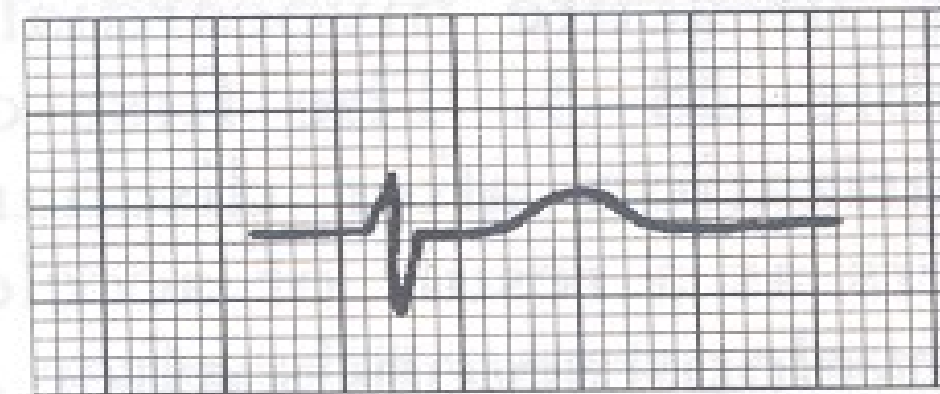
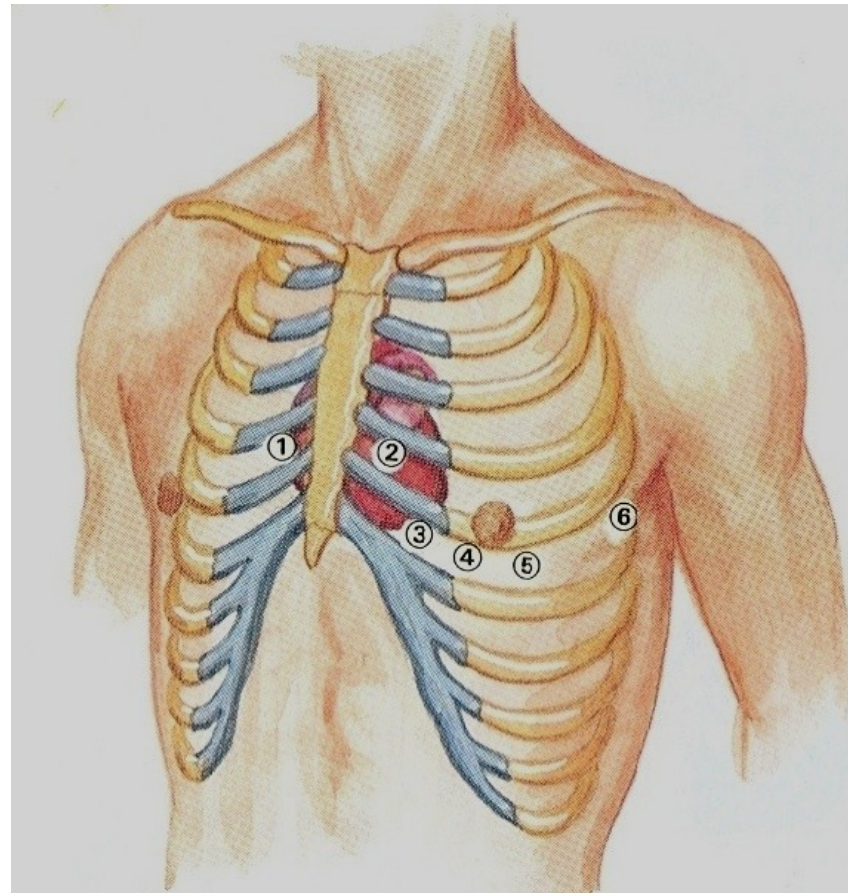


Fig..Unipolar (Goldberg) limb leads (augmented)

ECG Recordings...cont'd

C. Precordial (Chest) Limb Leads

- Unipolar leads labeled V1 through V6.
- One electrode is connected to the **positive terminal** of the Electrocardiograph and the **negative electrode (indifferent electrode)** is connected through equal electrical resistances to the right arm, left arm, and left leg.
- Electrodes are placed on the chest.



...cont'd

- Heart surfaces are close to the chest wall and each chest lead records mainly the electrical potential of the cardiac muscle immediately beneath the electrode,
 - i.e., relatively minute abnormalities in the ventricles can cause marked changes in the ECG.
- QRS in
 - V1, V2, are negative because the chest electrodes are nearer the base of the heart (direction of electronegativity).
 - V3 is in b/n.
 - QRS of leads v4-v6 are positive because they are nearer the apex (direction of electro positivity).

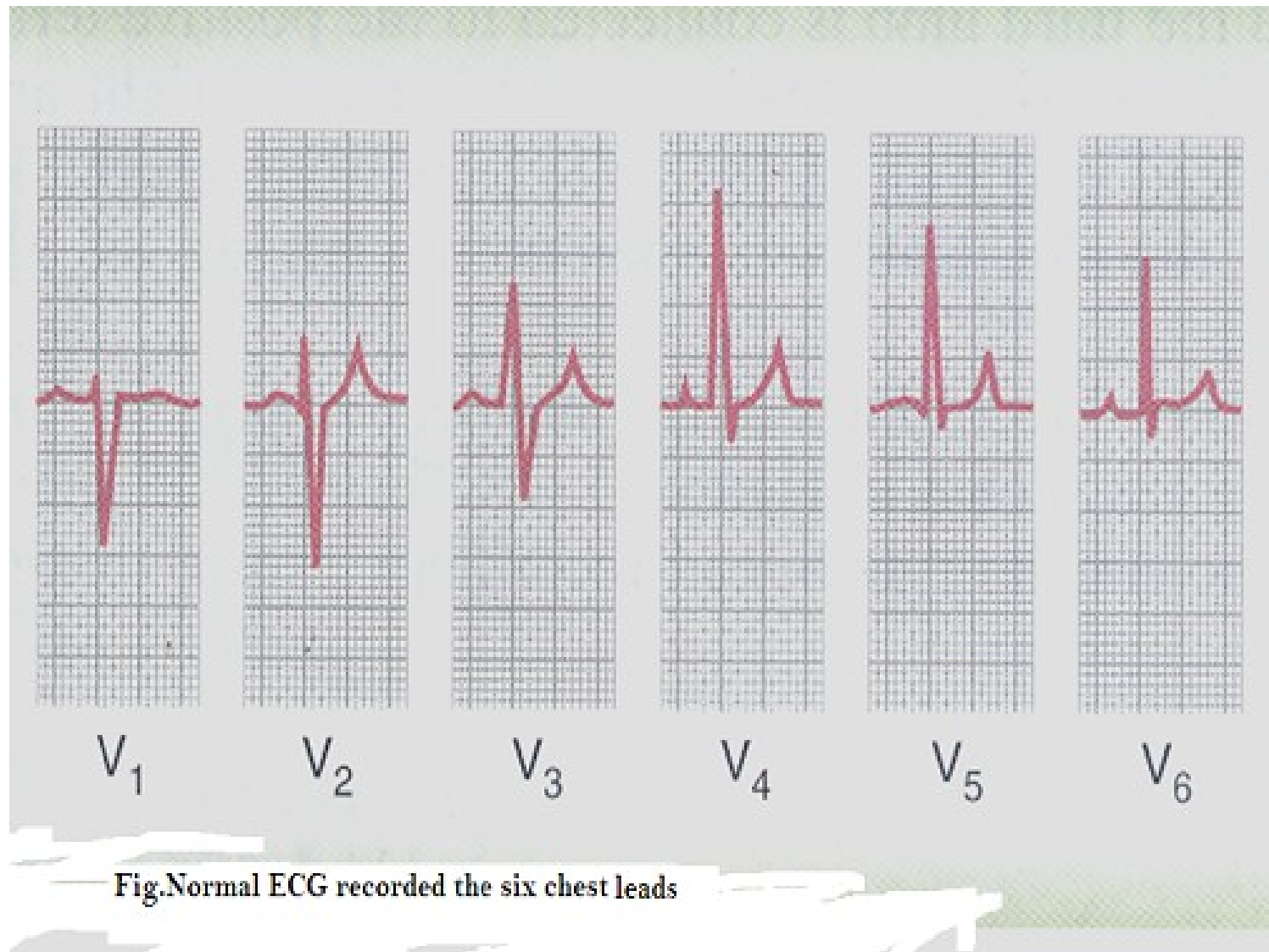


Fig. Normal ECG recorded the six chest leads

Standard chest leads

Lead	Position of exploring electrode
V ₁	Fourth intercostal space just to the right of the sternum
V ₂	Fourth intercostal space just to the left of the sternum
V ₃	Midway between V ₂ and V ₄
V ₄	Fifth intercostal space at the left midclavicular line
V ₅	Left anterior axillary line horizontally to the left of V ₄
V ₆	Midaxillary line horizontally to the left of V ₄ and V ₅

Vectorial Analysis of Potentials Recorded in different Leads

- Current flows mainly downward from the **base of the ventricles toward the apex** than in the upward direction.
- This is **shown by the summated vector** (long black arrow below) of the potential generated.

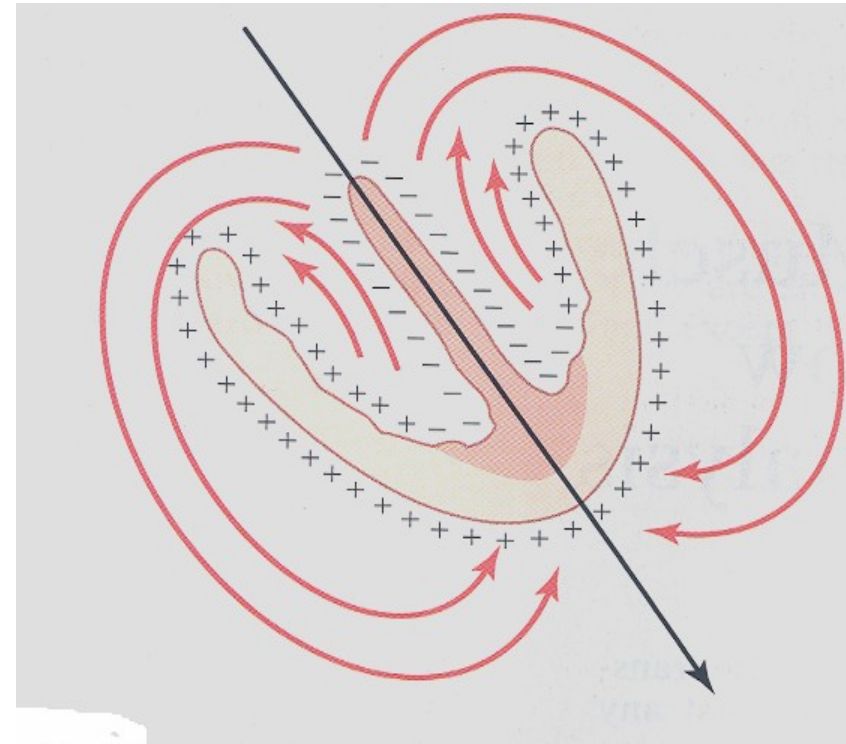


Fig..Mean vector through the partially depolarized ventricles

Axis of the three bipolar and three unipolar limb leads

- The limb leads are placed on a **Cabrera Circle** with their relative axis angles and charges.

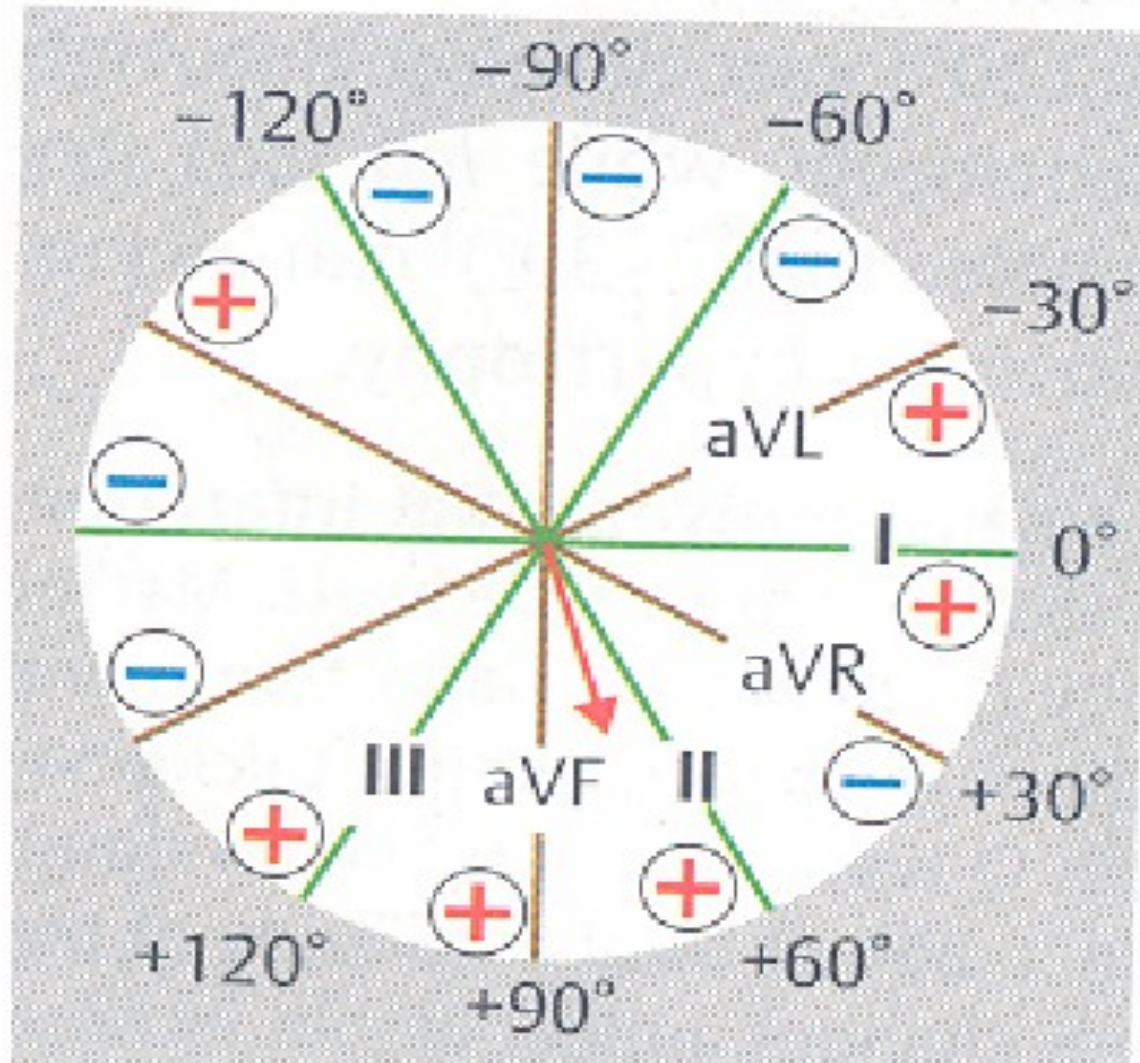


Fig..Cabrera circle

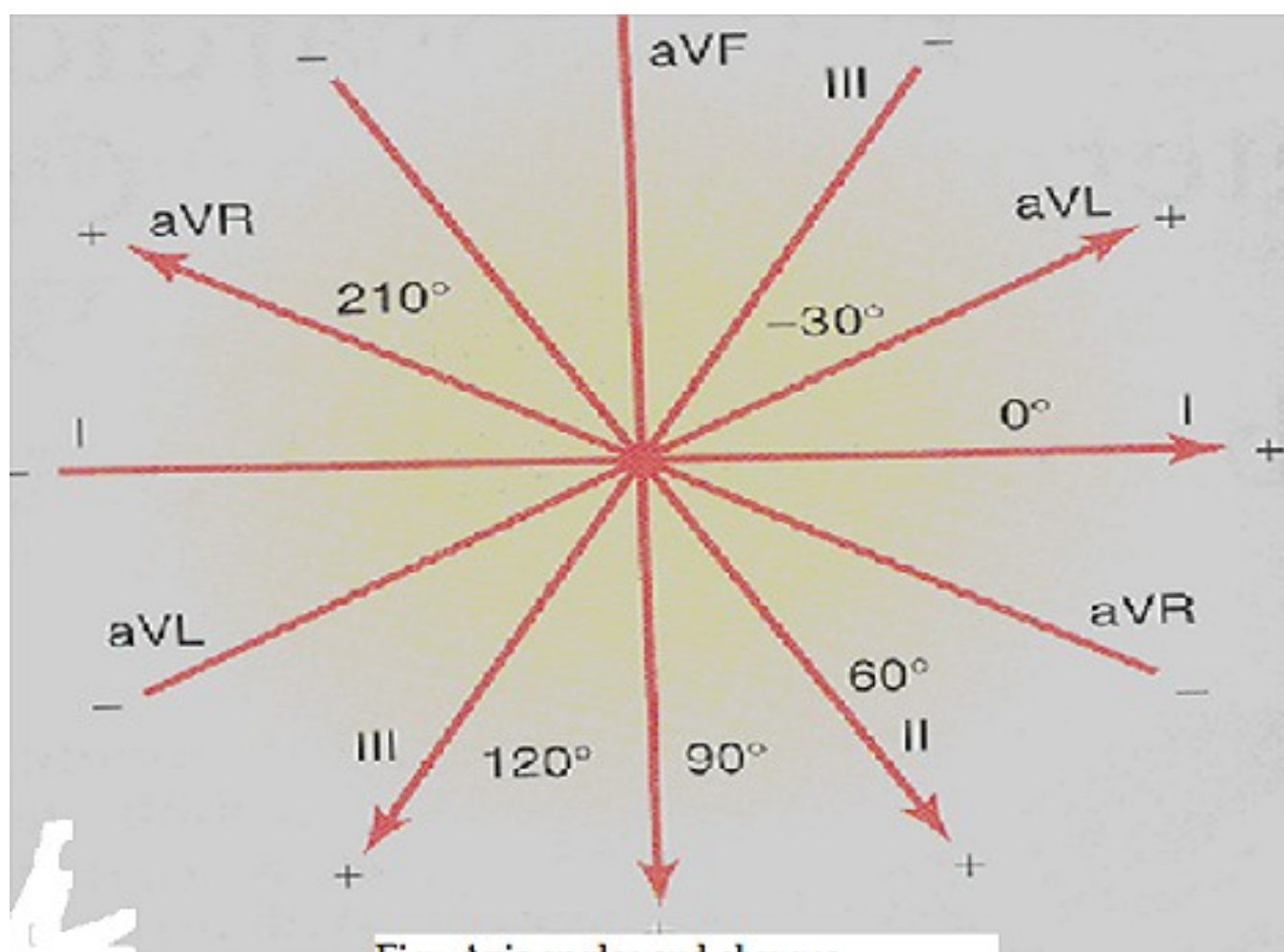


Fig.. Axis angles and charges

Axis of the...cont'd

- The direction from **positive** to **negative** electrode is called **axis of the lead**.
- For example, as shown above:
 - In **lead I**, the electrodes lie in the horizontal direction with the positive electrode to the left; the axis of lead I is, therefore, 0 degree.
 - In **lead II**, electrode placed on RA connected to the torso in upper right hand corner and LL connected in the lower left-hand corner $\rightarrow +60^\circ$
 - By a similar analysis, **Lead III** $\rightarrow +120^\circ$
 - **aVR** $\rightarrow -210^\circ$,
 - **aVL** $\rightarrow -30^\circ$
 - **aVF** $\rightarrow +90^\circ$

Mean Electrical Axis of the Heart

- Mean electrical axis of the ventricles = vector showing current flows from base of ventricles (-ve) toward apex (+ve).
- This amounts to 59-60°.
- Electrical axis is determined from standard limb leads (fig..below).
- A positive vector in a lead (shown in axis) causes recording in the ECG above the zero line, while a negative vector causes ECG recording below the zero line.

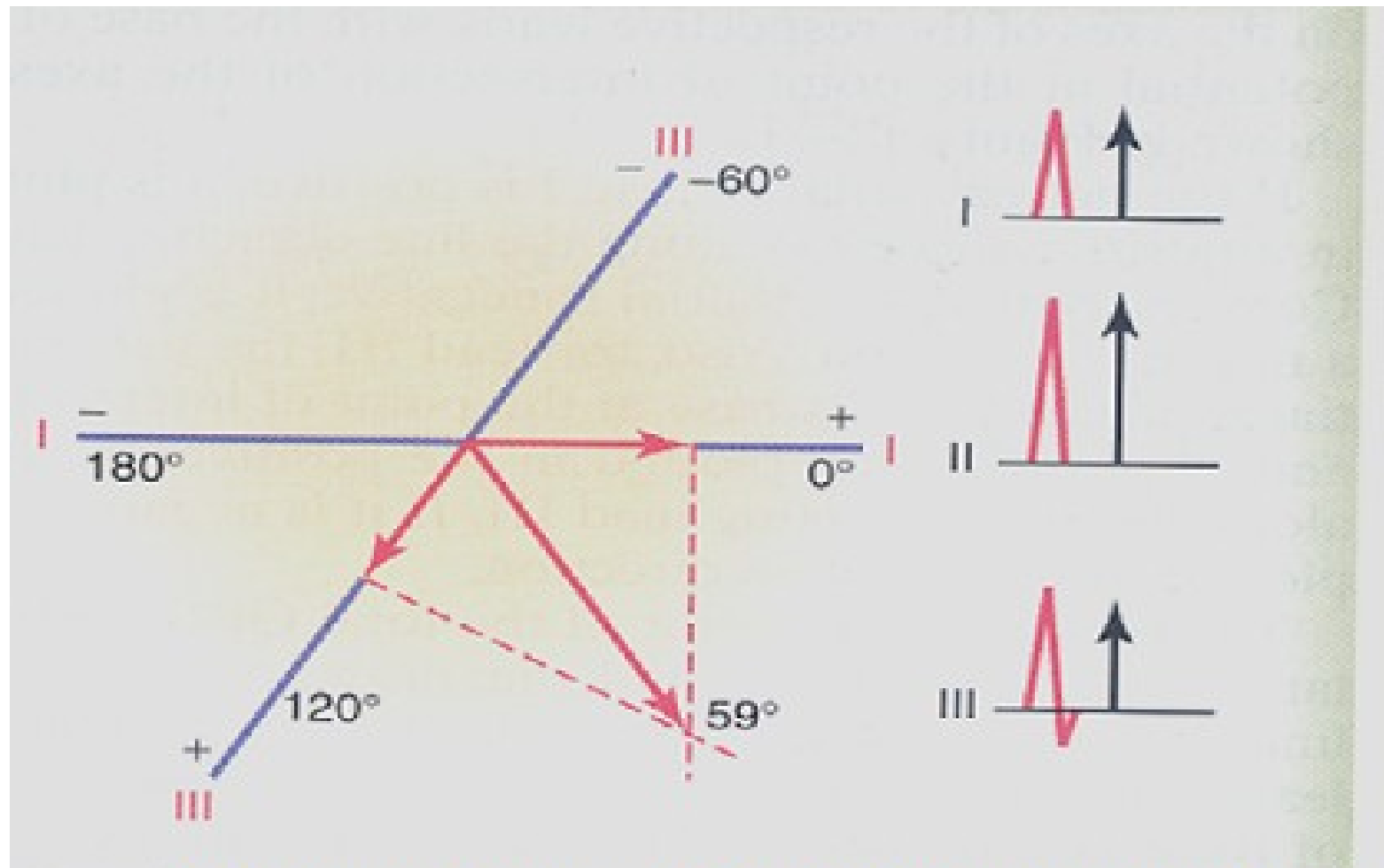


Fig..Plotting the electrical axis of the ventricles from two ECG leads

...cont'd

- Negative potential subtracted from positive potential.
- Vectors of L I and L III(+ or -) plotted along lines depicting the leads and with the base at the point of intersection.
- Actual vector determined by drawing perpendicular lines for the apices of two vectors of L I and L III.
- Point of intersection of two perpendicular lines, by vectorial analysis = Apex of the actual mean QRS vector in the ventricles (Fig...above).

...cont'd

- The mean QRS vector in the frontal plane is normally b/n -30° and $+110^\circ$.
- Mean QRS vectors:
 - greater than $+110^\circ$ are said to represent **right axis deviation**,
 - those less than -30° are designated as **left axis deviation**.
- Mean vectors for ECG deflections other than the QRS, for e.g. mean T-wave vectors, can be calculated.

Application of ECG

- By analyzing electric potential fluctuations, the physician can get some insight into:
 - anatomical orientation of the heart,
 - heart rate determination
 - relative size of heart chambers,
 - a variety of disturbances of rhythm and conduction
arrhythmia and conduction block
 - extent, location and progress of ischaemic damage
(myocardial infarction)
 - hypertrophy
 - pericarditis and myocarditis

Some Examples of ECG Abnormalities

A. Axis Deviation

- Occurs when the mean electrical axis is deviated either **to the left** ($<-30^\circ$) or **to the right** (>90 to 110°).

Causes:

- Change in the **position** of the heart
 - **Left axis deviation:** expiration, supine position, obesity
 - **Right axis deviation:** inspiration, upright position, tall and lanky stature, overload and hypertrophy of right ventricle.
- Pathologic conditions:** congenital heart disease, right ventricular hypertrophy, acquired disease of myocardium: ischaemic heart disease, degeneration of the conducting system, e.g. BBB.

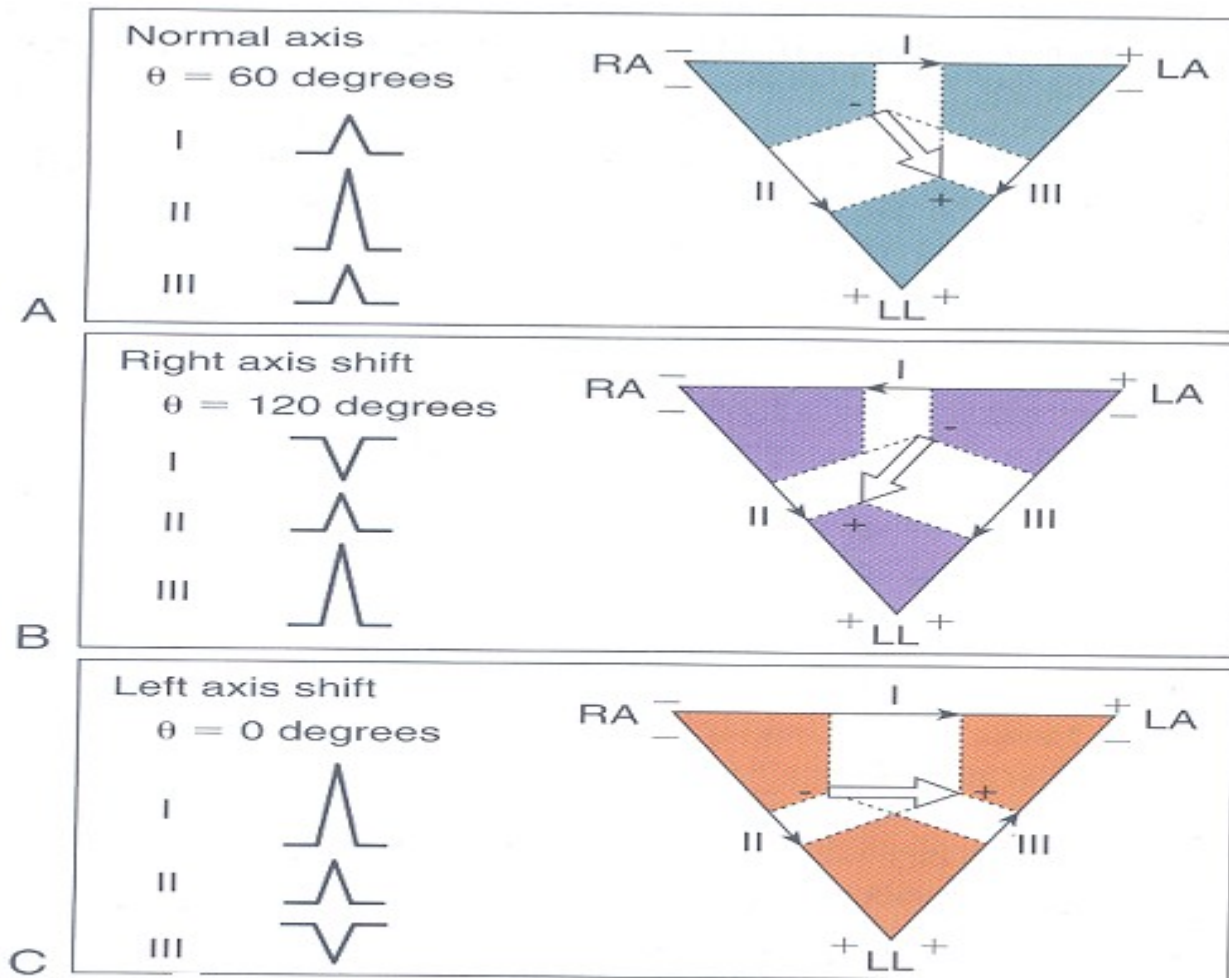


Fig.. Magnitude and direction of the QRS complexes in limb leads I, II, and III when the mean electrical axis (Q) is 60 degrees (A), 120 degrees (B), and 0 degrees (C).

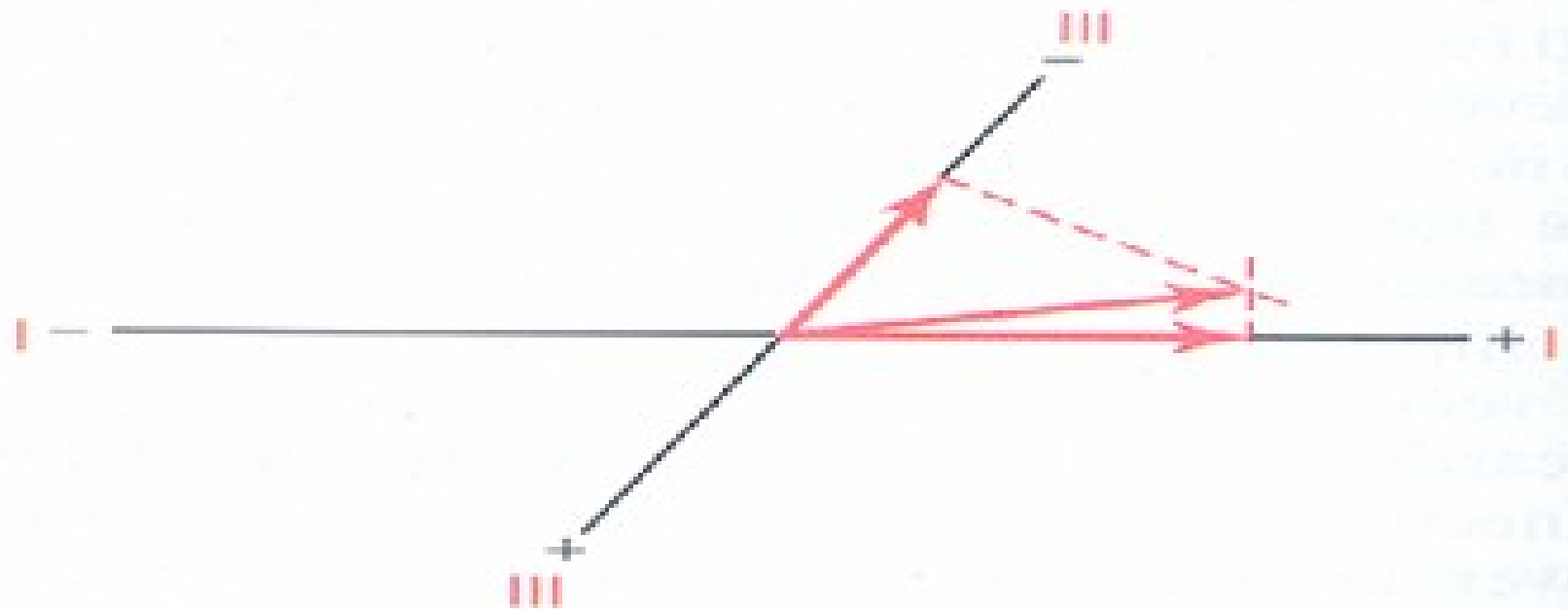
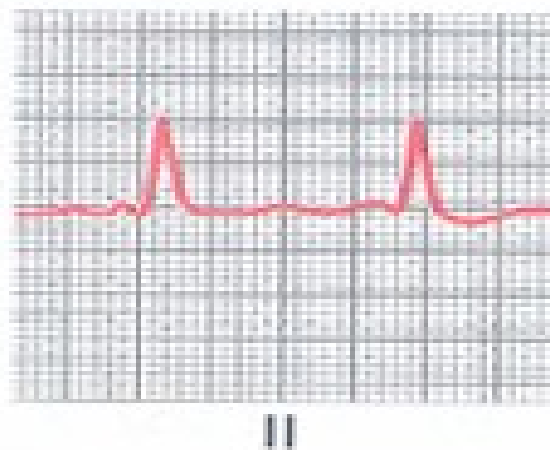
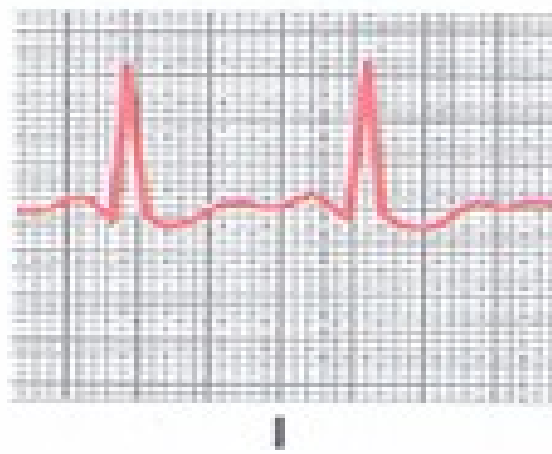


Fig..Left axis deviation in a hypertensive heart (hypertrophic left ventricle with slightly prolonged QRS complex)

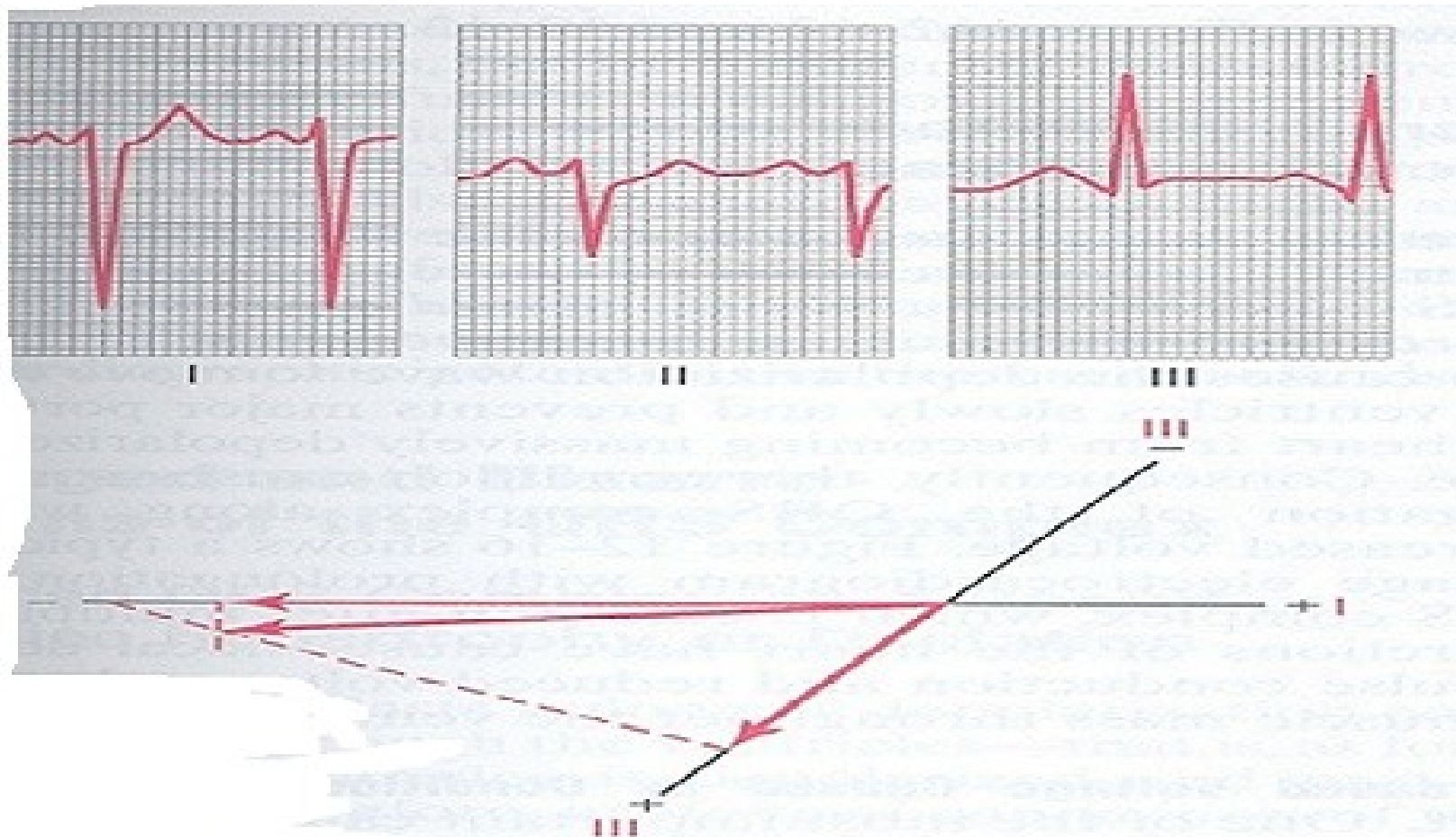


Fig..High voltage ECG in congenital pulmonary valve stenosis with right ventricular hypertrophy. Intense Rt axis dev. and slightly prolonged QRS complex.

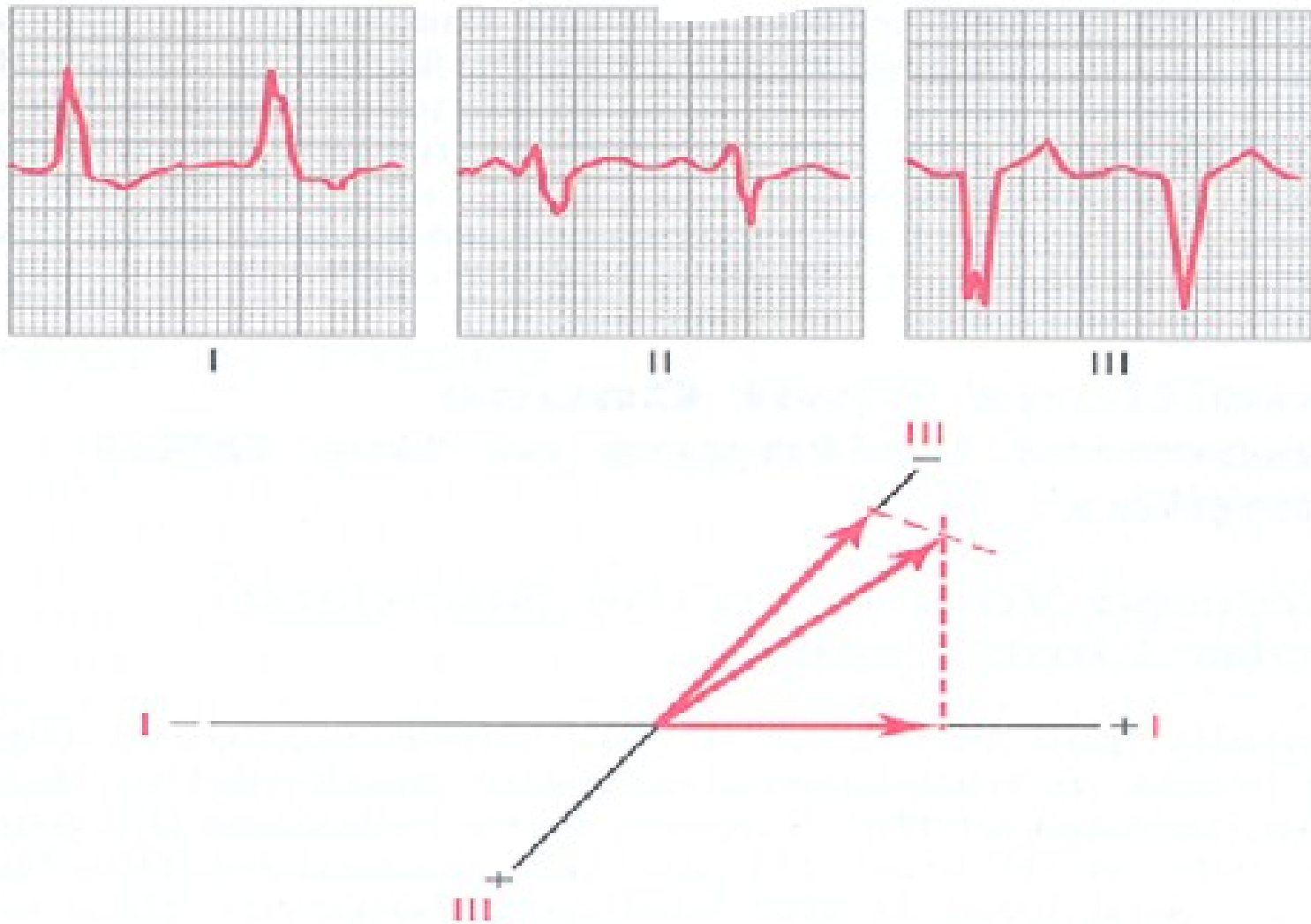


Fig..Left axis dev.caused by left BBB and markedly prolonged QRS complex

Some Examples of ECG ...cont'd

B. Conduction Block

- This may occur anywhere between the S.A node and the ventricles.
- A block between the SA node and the AV node occurs very rarely. One of the more common sites of block is A-V node.
- Three degrees of heart block are usually distinguished.

a. First degree heart block

- PR interval prolonged, >0.21 sec, usually due to A-V nodal delay.
- May be caused by increased vagal tone and systemic diseases
- Has no effect on the pumping ability of the ventricles

Some Examples of ECG ...cont'd

b. Second Degree Heart Block

- Associated with organic heart disease
- Not all P waves are followed by QRS complexes.

1. Partial progressive AV block, normal waves.

- PR interval increases progressively leading to dropped beat (Wenckebach phenomenon, Mobitz type I, 2:1 block)

2. Constant block : PR interval relatively fixed; the atrio-ventricular rate is a constant small number ratio, e.g. 2:1, 3:1,

3. Periodic block, Mobitz type II, PR waves not fixed. Ratio of transmission: 6: 5, 8:7, etc.

Some Examples of ECG ...cont'd

c. **Third Degree (complete) Heart Block**

- Impulses completely interrupted between atria and ventricles

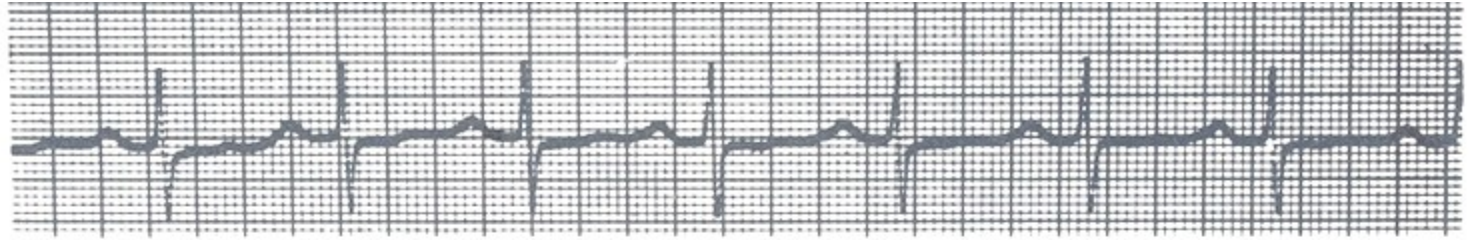
1. QRS complexes dissociated from P waves
2. Ventricular rate much slowed (35-40 beats/min)
3. Fainting often occurs due to cerebral ischaemia

Cause: organic heart disease is often the cause.

Significance:

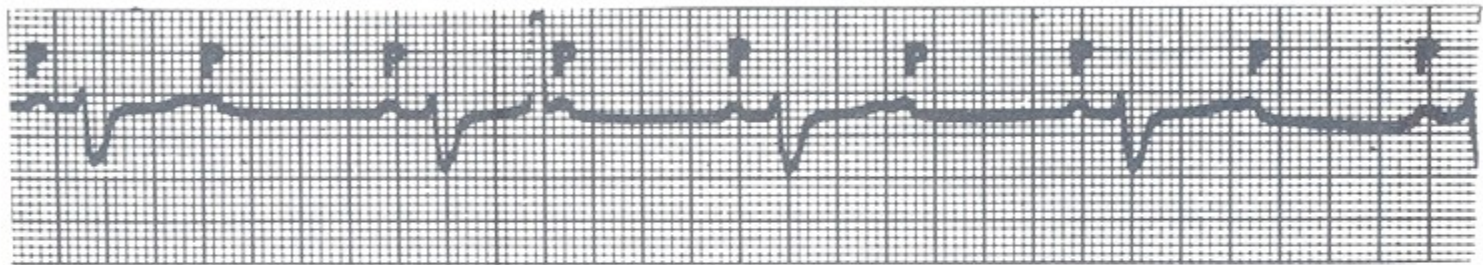
- ✓ May be associated with prolonged ventricular stand still.
- ✓ Cardiac arrest, if prolonged, can lead to cerebral ischaemia, syncope, or death.
- ✓ This condition is termed **Stokes-Adams Syndrome**.

A



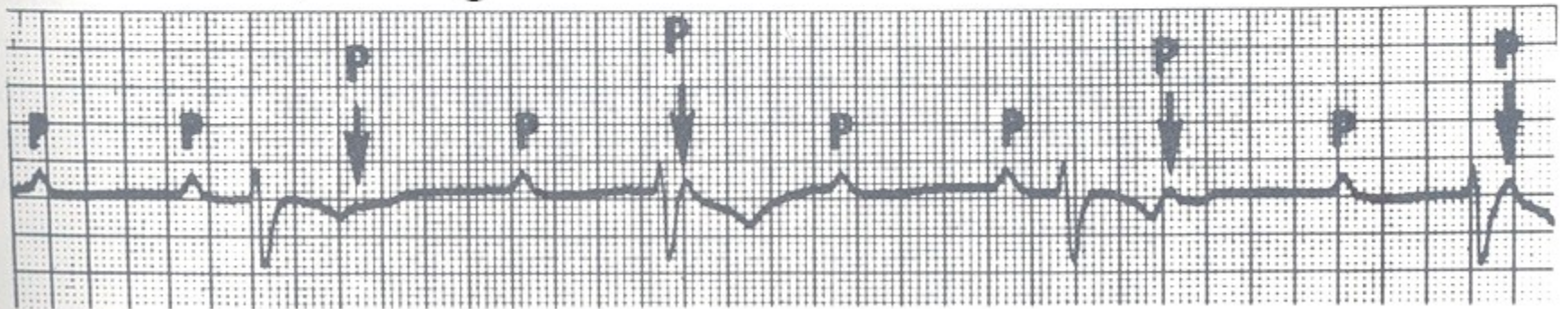
1st degree heart block

B



2nd degree heart block, 2:1

C



3rd degree heart block

Some Examples of ECG ...cont'd

- d. Wolf-Parkinson-White syndrome** (Ventricular pre-excitation or accelerated conduction)
 - results from aberrant conduction pathway b/n atria and the ventricles.
 - It eliminates the normal delay of impulse transmission.
- e. Bundle branch block:**
 - blockage of the right or left common bundles or the left anterior or posterior fascicles.

Some Examples of ECG ...cont'd

C. Arrhythmias

- Impulse initiation or impulse propagation are disturbed.
- Disturbances of impulse initiation include those that arise from the SA node and those that originate from various ectopic foci.
- Conduction blocks and reentrant rhythms are the principal disturbances of impulse propagation.

Some Examples of ECG ...cont'd

C. Arrhythmias

Type	Arrhythmias of the Heart
1. Extra systole (premature beat)	• Abnormal contraction produced during diastole • Caused by an ectopic foci • It could be nodal, atrial, or ventricular extra systole
2. Paroxysmal atrial tachycardia (PAT)	• Atrial beat >160 beats/min • A complication of MI; broad and bizarre QRS complexes • Caused by ectopic focus
3. Paroxysmal ventricular tachycardia (PVT)	• Ventricular beat =200 beats/min, • Caused by ectopic focus

4. Atrial flutter	• Fast and regular atrial beats (200-400 b/min). • Different parts contracting • Caused by re-entry, a wave of excitation propagating continuously within a closed circuit- a ring-like pathway around ends of vena cavae; 2:3, 3:1 block
5. Atrial fibrillation	• Fast and irregular uncoordinated atrial beats (400-600 b/min), dissociated from QRS • Caused by multiple re-entry. • Small and numerous circuits, “F” waves in ECG. • No p- wave. • May be caused by: mitral valve dx, coronary atherosclerosis, thyrotoxicosis, etc.
6. Ventricular fibrillation	• Fast and irregular uncoordinated QRS and ventricular beats (400-600b/min) • Caused by multiple re-entry • May lead to cardiac stand still • Circulation stops • May be caused by electrocution • May lead to death if cardiac massage is not performed

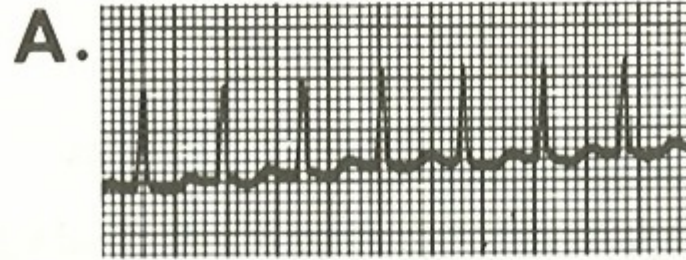
Some Examples of ECG ...cont'd

D. Voltage Abnormalities

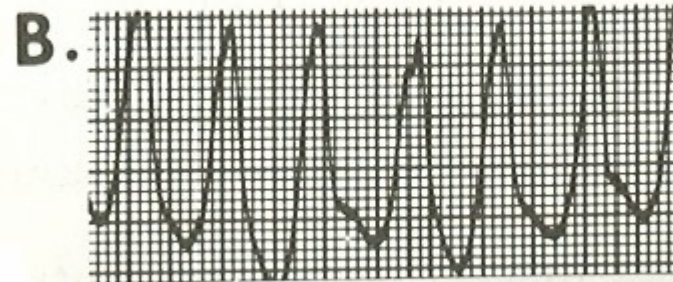
- May be determined by making use of the convention for voltage calibration
 - i. Increased voltage
 - If QRS of 3 standard leads is greater than 4mV, then the heart is said to be hypertrophic.
 - ii. Prolonged duration of QRS is caused by:
 - Delayed conduction of impulses
 - Block of Purkinje system
 - iii. Spiked T-wave caused by hyperkalaemia
 - iv. Decreased voltages are caused by:
 - Muscle abnormalities-like myocardial infarction
 - Conduction problem like 1st to 3rd degree heart block
 - Ant. rotation of cardiac apex.

Normal		
Digoxin therapy		• ST "scooped" depression • Mild PR prolongation
Hyperkalemia		• Tall "peaked" T-wave
Severe hyperkalemia		• Flattened P • Widened QRS
Hypokalemia		• ST depression, flattened T • Prominent U wave
Hypercalcemia		• Shortened QT interval
Hypocalcemia		• Prolonged QT interval

ST segment and T wave abnormalities caused by different conditions



Paroxysmal atrial tachycardia



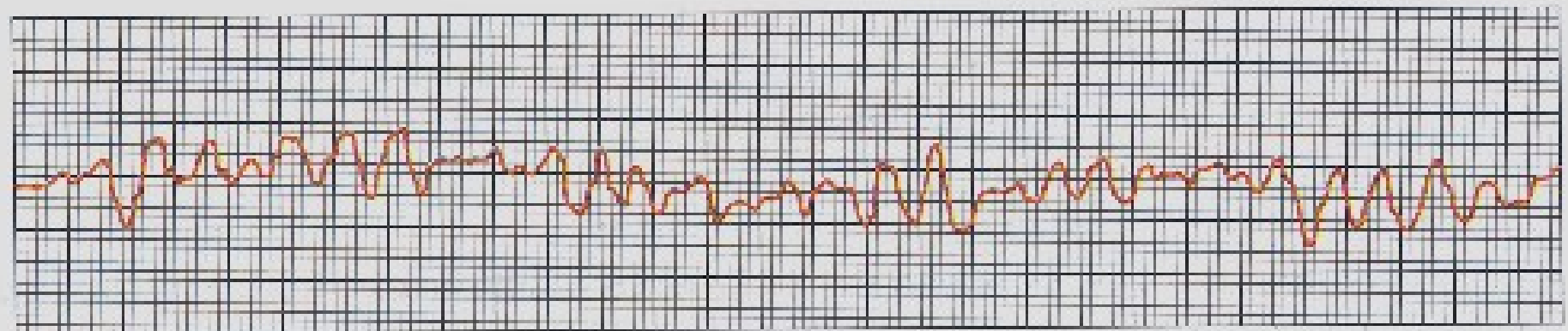
Ventricular tachycardia



Atrial Flutter



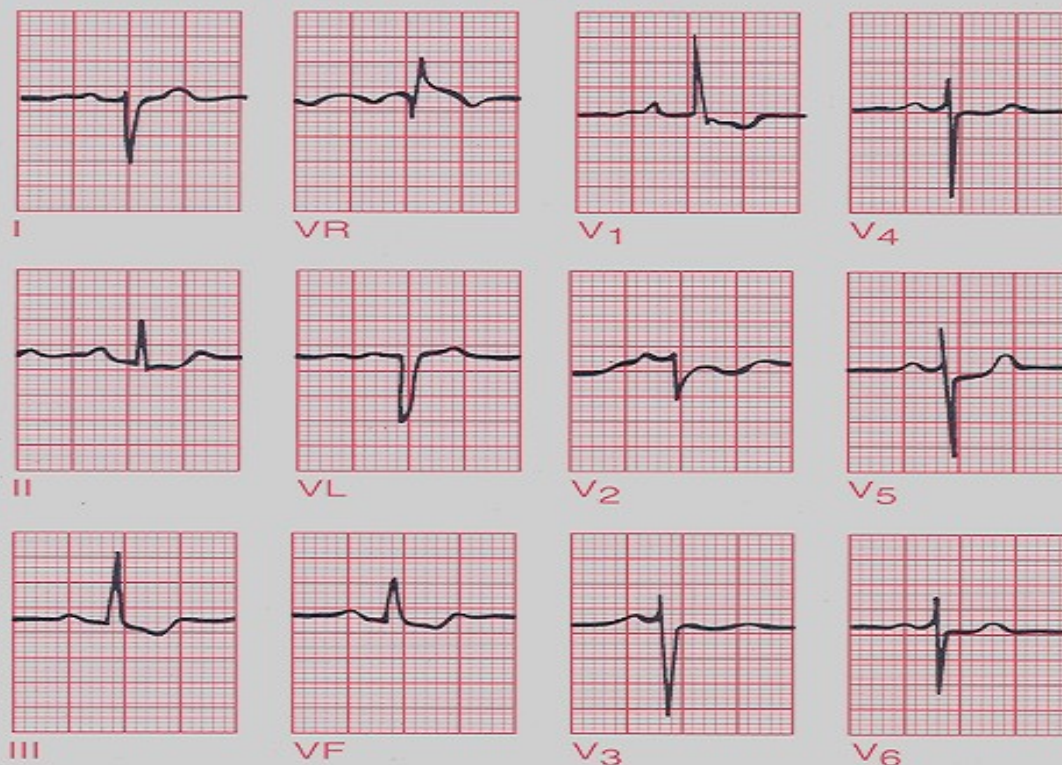
Atrial fibrillation



Ventricular fibrillation

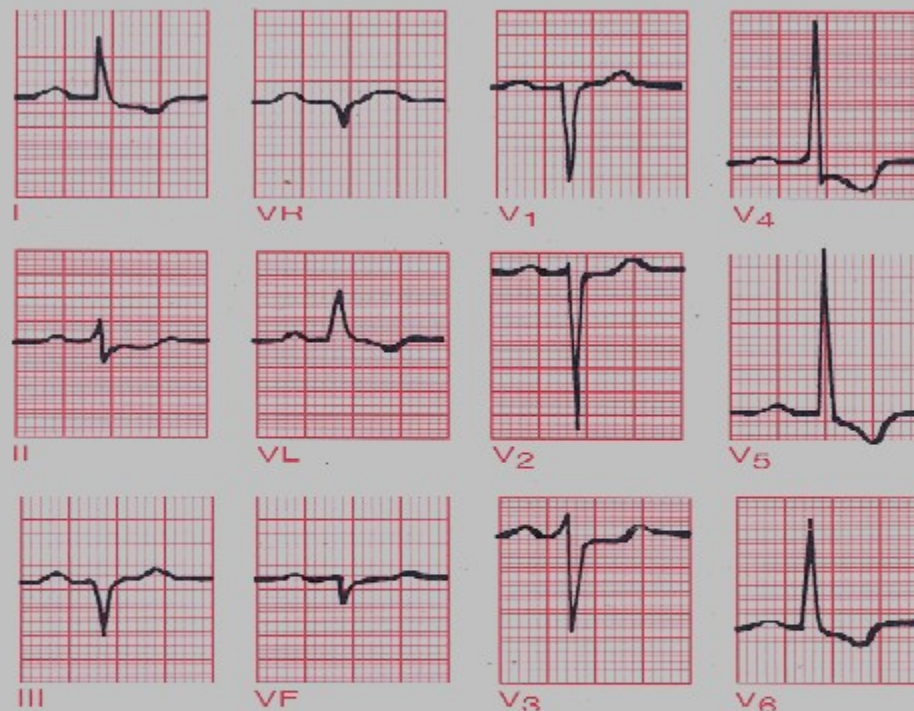
Fig..Fibrillation ECG

RIGHT VENTRICULAR HYPERTROPHY



Note: Sinus rhythm
 Peaked P wave in V₁
 Right axis deviation (deep S wave in lead I)
 Tall R wave in V₁
 Deep S wave in V₆
 Clockwise rotation (no septal complex seen)
 Inverted T waves in V₁ and V₂.

LEFT VENTRICULAR HYPERTROPHY



Note: Sinus rhythm

Normal axis (the R and S waves are equal in lead II, so the axis is at the limit of normality)

Height of R wave in V_5 is 35 mm

Deep S waves in V_1 and V_2

Inverted T waves in leads I, VL, V_1 – V_6 .

Some Examples of ECG ...cont'd

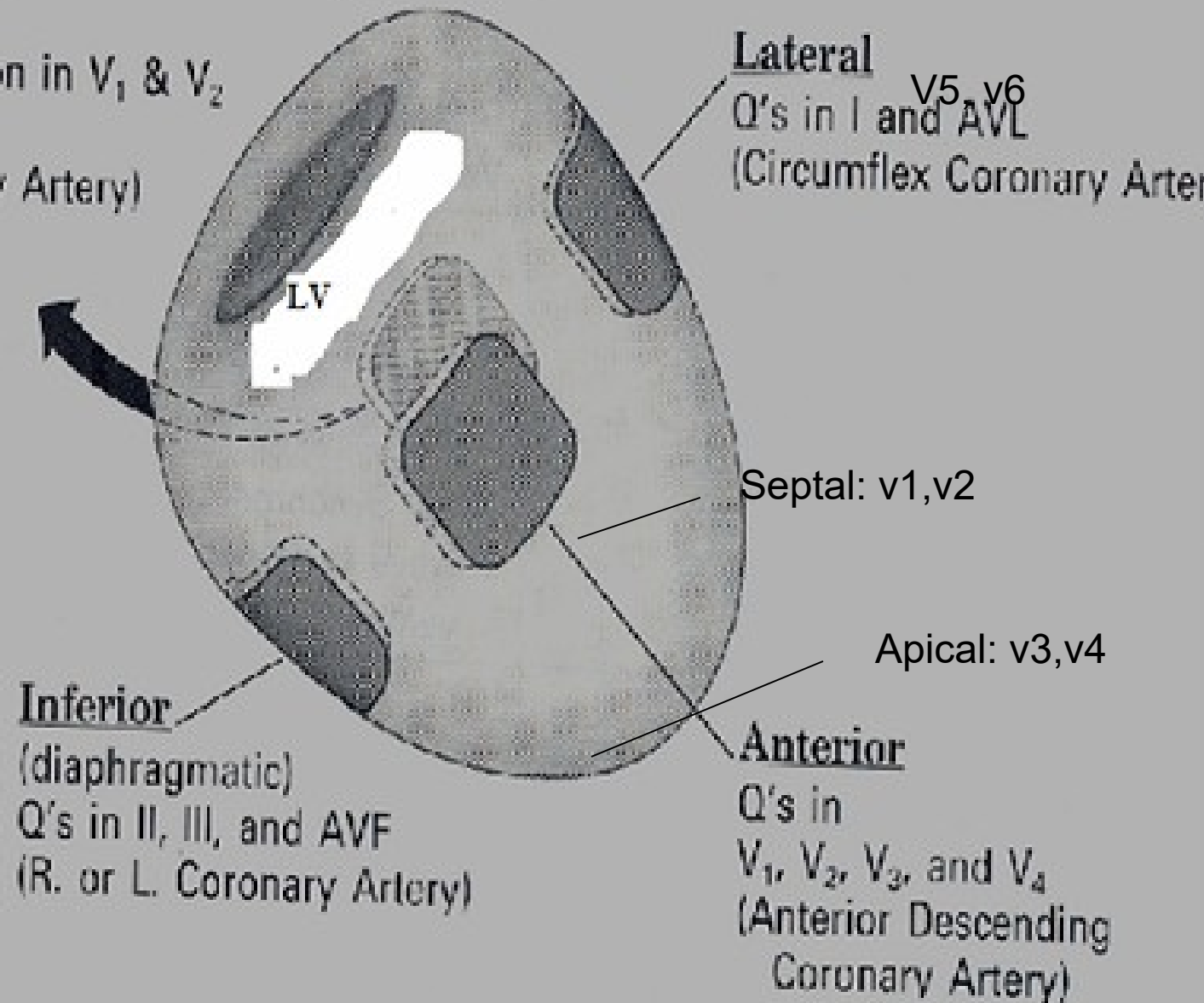
D. Myocardial Infarction

- A sudden, irreversible, ischaemic injury due to coronary arterial narrowing or occlusion with sustained damage to a portion of the myocardium.
- Associated with electrocardiographic, metabolic, and haemodynamic changes

Location of infarction

Posterior

- large R with ST Depression in V_1 & V_2
- mirror test (Right Coronary Artery)



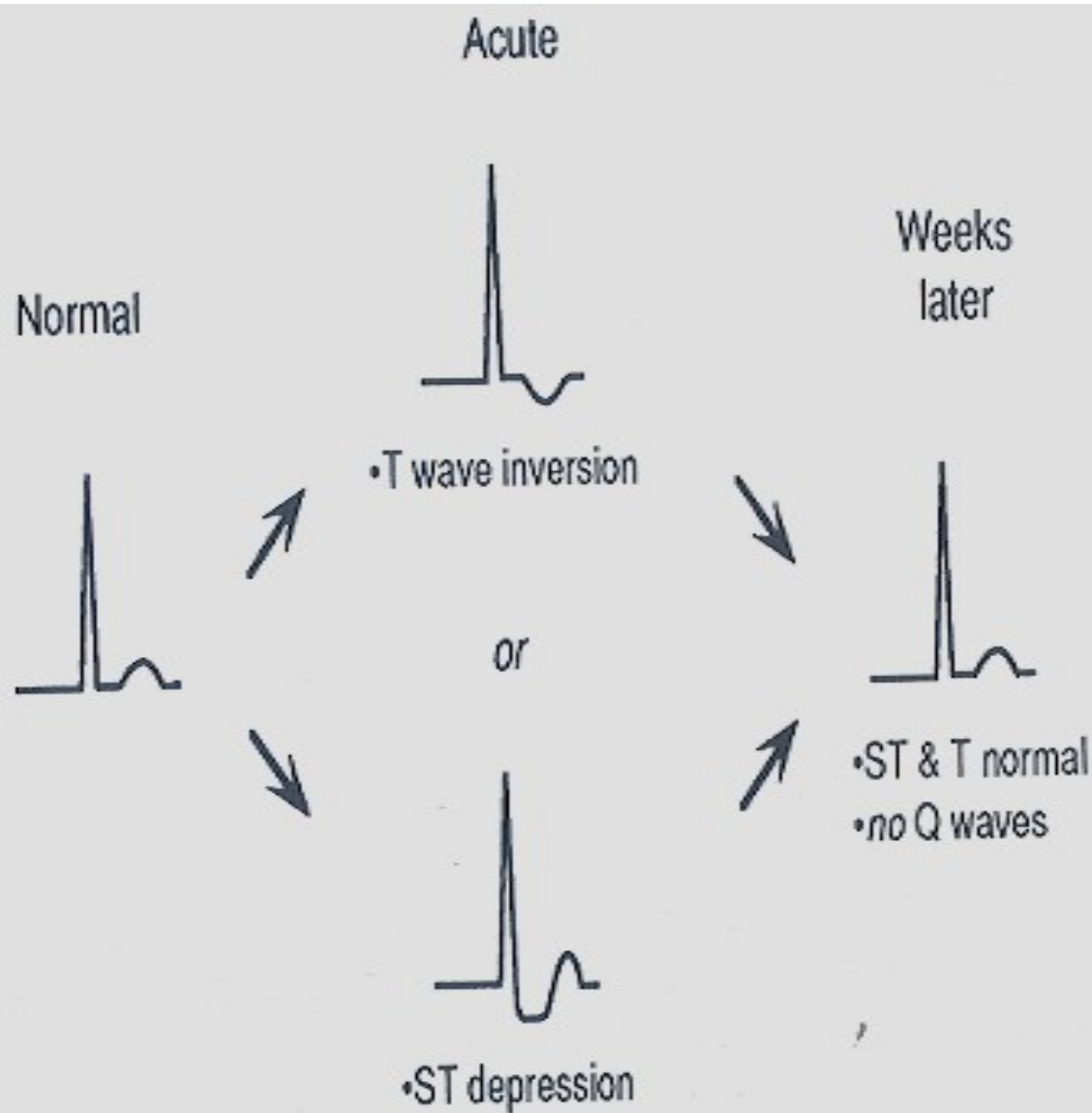


Fig.. ECG abnormalities in unstable angina and non-ST-elevation myocardial infarction.

ST-Elevation Myocardial Infarction

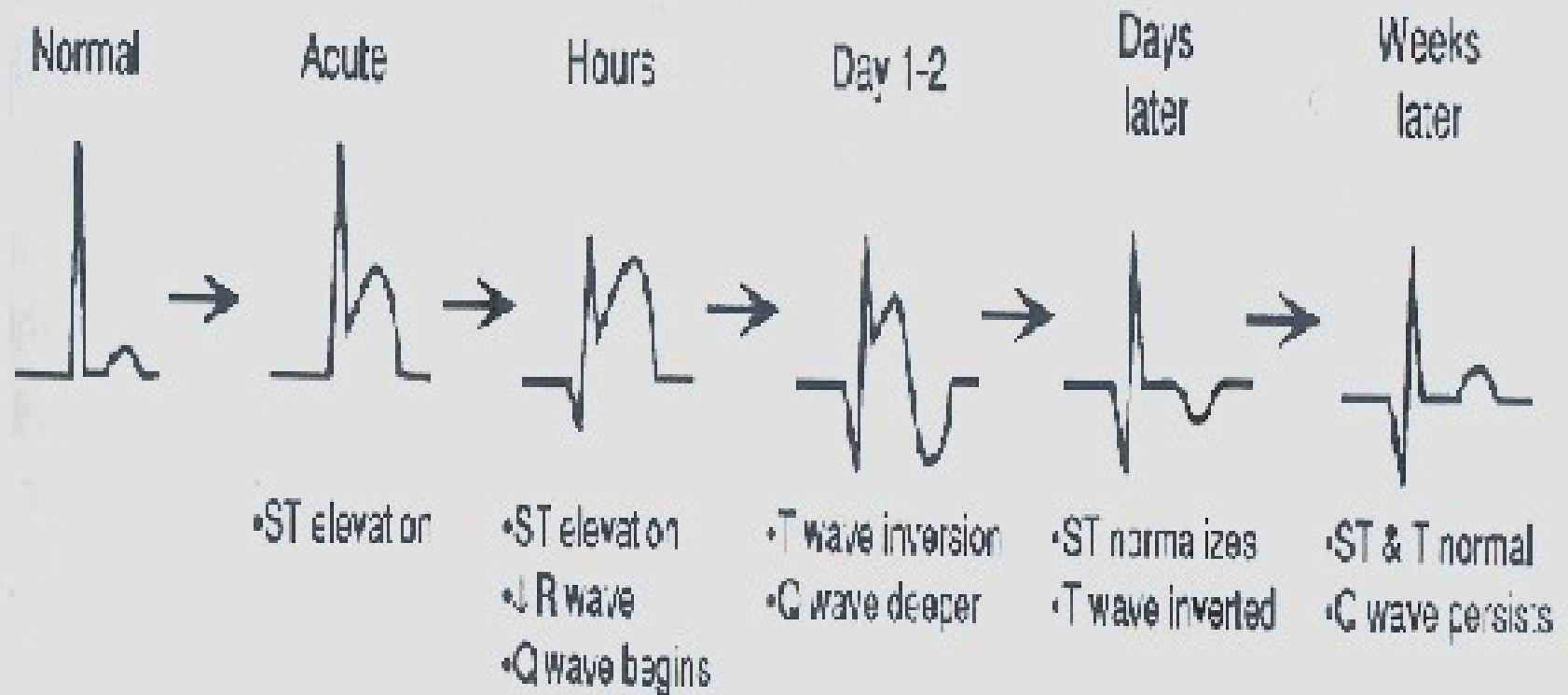


Fig... ECG evolution during ST-elevation myocardial infarction.

Some Examples of ECG ...cont'd

D. Congestive Heart Failure

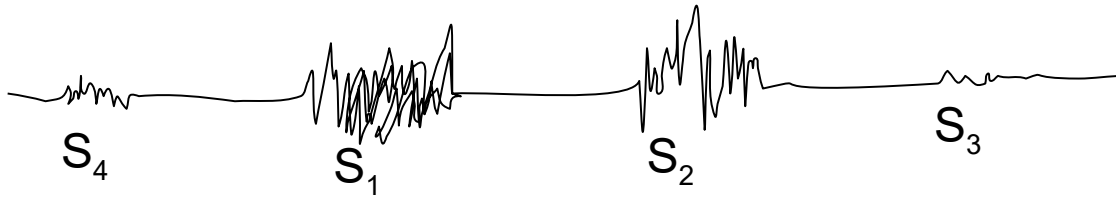
- A Heart is unable to pump blood at a rate commensurate with systemic metabolic requirements.
- Causes may be **cardiac or extra cardiac**.
 - i. **Cardiac** causes may be:
 - Myocardial contractile failure due to **ischaemic heart dx, myocarditis or cardiomyopathy**
 - A disorder which prevents proper filling or emptying of heart such as **valvular stenosis or regurgitation or pericardial dx**.
 - ii. **Extra cardiac** causes
 - High pressure overloads such as **hypertension or high output overloads such as renal failure, anaemia, AV fistula or thyrotoxicosis**

Cardiac valves

- All valves consist of connective tissue (not cardiac muscle tissue) and, therefore, open & close passively.
- Valves open & close in response to changes in pressure:
- AV valves - open when pressure in the atria is greater than pressure in the ventricles (i.e., during ventricular diastole) & closed when pressure in the ventricles is greater than pressure in the atria (i.e., during ventricular systole)
- Semi lunar valves - open when pressure in the ventricles is greater than pressure in the arteries (i.e., during ventricular systole) and closed when pressure in the pulmonary trunk & aorta is greater than pressure in the ventricles (i.e., during ventricular diastole)

Heart Sounds

- Heart sounds (lubb-dubb) are associated with **closing** of heart valves
- 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)
- Means of identification
 1. Auscultation:
 - direct/immediate auscultation
 - Stethoscope mediated auscultation
 2. Phonocardiographic based recording



S_1 : is always audible, has a LUBB-sound

S₂: is always audible, has DUBB-sound

Continuous heart sound: Lubb-Dubb, Lubb-Dubb

Duration of Lubb is shorter than that of Dubb

S₃: is audible in children and in adults during exercise 77

S₄: is audible very rarely

Heart Sounds (cont'd)

S_1 : First heart sound

Caused by sudden closure of AV-valves.

Timing: occurs at the beginning of ventricular systole

Best site of auscultation

- 5th LIS, at mid-clavicular line and 4th LIS, near the sternum

S_2 : Second heart sound

Cause: Sudden closure of semilunar valves

Timing: occurs at the beginning of ventricular diastole

Best site of auscultation = 2nd RIS, near the sternum and 2nd LIS,
near the sternum

Heart Sounds (cont'd)

S₃: Third heart sound

Cause: rapid filling of the ventricles with blood during ventricular diastole

- It is audible in children and in adults during exercise

Best site of auscultation: at the apex of the heart beneath the xiphoid process

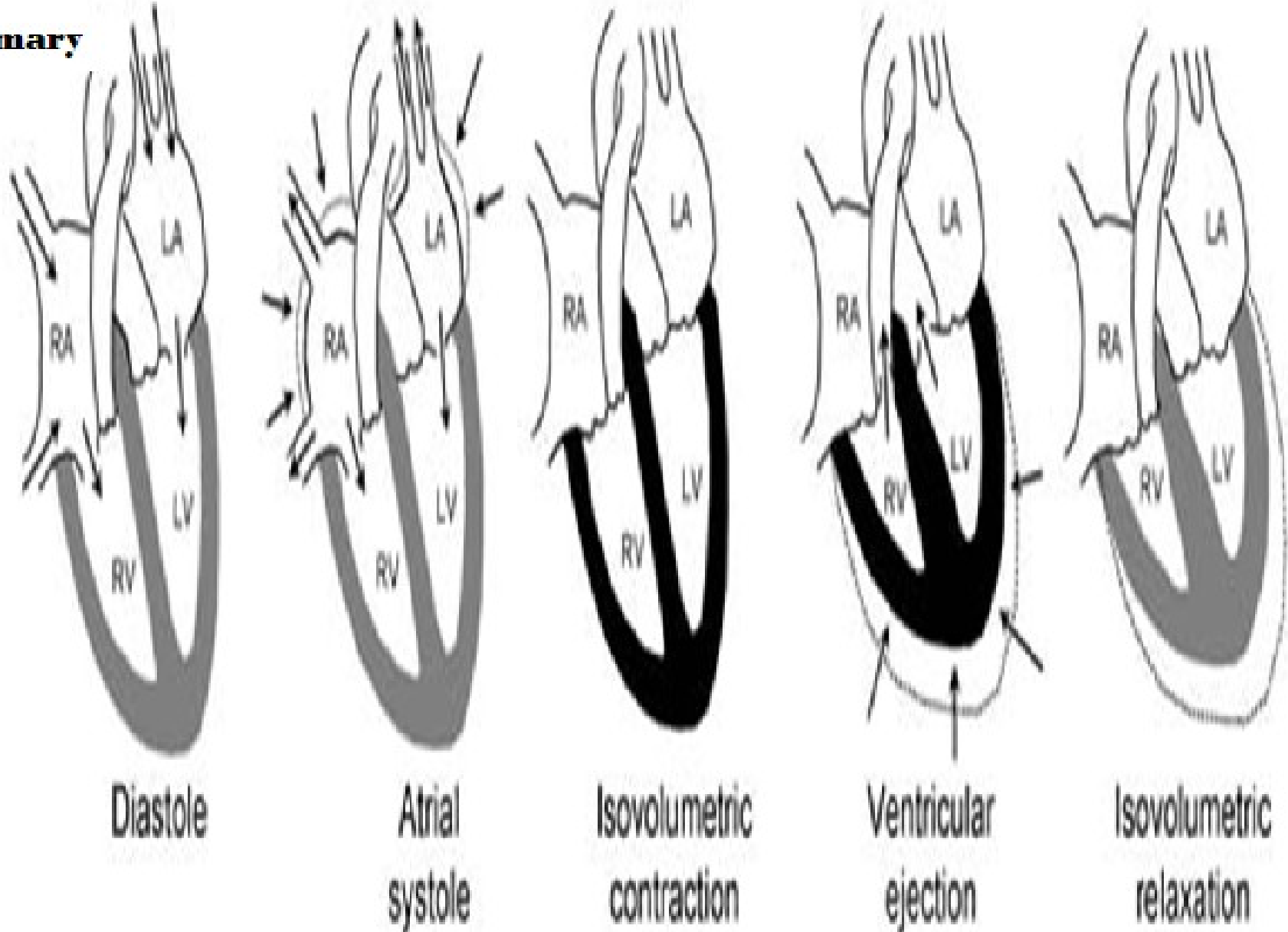
S₄: Fourth heart sound

Cause: rapid ventricular filling during atrial systole

Timing: during atrial systole

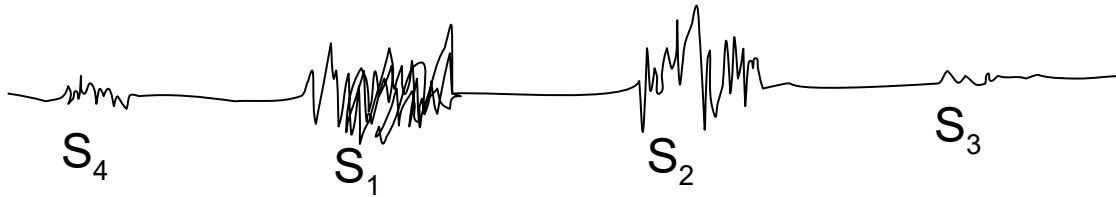
Best site of auscultation: at the apex of the heart

Summary



Heart Sounds

- Heart sounds (lubb-dubb) are associated with **closing** of heart valves
- 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)
- Means of identification
 1. Auscultation:
 - direct/immediate auscultation
 - Stethoscope mediated auscultation
 2. Phonocardiographic based recording



S_1 : is always audible, has a LUBB-sound

S₂: is always audible, has DUBB-sound

Continuous heart sound: Lubb-Dubb, Lubb-Dubb

Duration of Lubb is shorter than that of Dubb

S₃: is audible in children and in adults during exercise

S₄: is audible very rarely

Heart Sounds (cont'd)

S₁: First heart sound

- Slightly prolonged “lub” sound
 - **Caused by** sudden closure of AV-valves.
 - **Timing:** occurs at the beginning of **ventricular systole**
- 10-13 vibrations of different frequencies
- Audible above 40 cps frequency
- **Associated with isometric contraction of ventricles**
- Intensity directly proportional to rate of IVP rise
- Best heard over **cardiac apex**
- Asynchronous closure of tricuspid and mitral valves in health → splitting of S₁

Heart Sounds (cont'd)

S₂: Second heart sound

- Shorter, high pitched “dub” sound
- Caused by sudden closure of semi-lunar valves and vibration of the valves and large arteries
- Occurs at the beginning of ventricular diastole,
- 5-10 vibrations of different frequencies audible at 100-200 cps frequency
- Less than 0.05 sec duration
- Associated with isometric relaxation of ventricles...minor contribution to sound
- Intensity directly proportional to rate of IVP fall
- Asynchronous closure of aortic and pulmonic valves during inspiration....
- Increased filling of RV→delay in closure of pulmonary valve →splitting of 2nd heart sound.
- BBB→Split 2nd heart sound

Heart Sounds (cont'd)

S₃: Third heart sound

- 30 cps frequency; hence usually inaudible
- Detected by phonocardiograph
- 0.04 sec duration
- Develops 0.10 to 0.20 sec after 2nd heart sound
- Caused by rapid filling of the ventricles with blood during ventricular diastole
- Audible in children and in adults during exercise

Heart Sounds (cont'd)

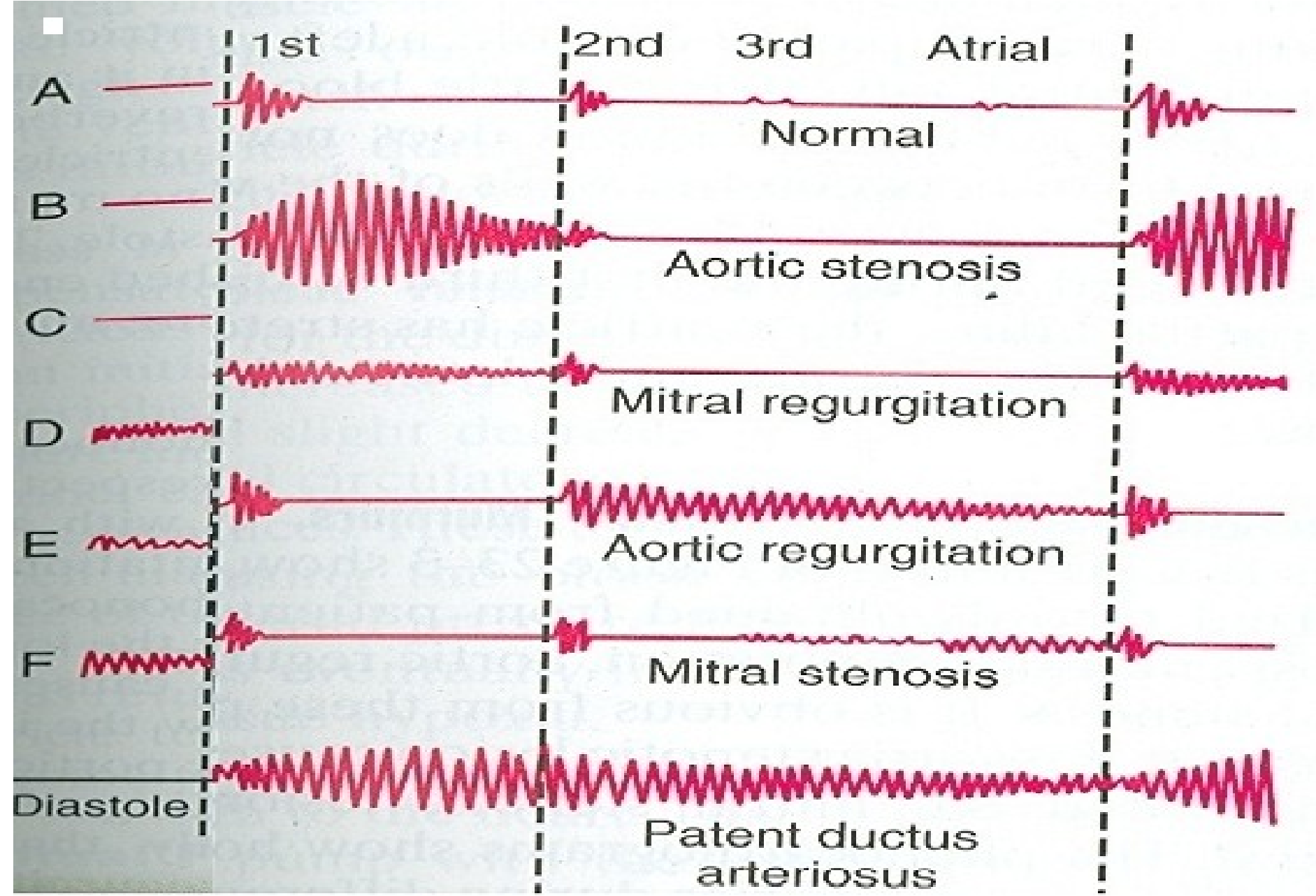
S₄: Fourth heart sound

- Very rarely audible
- 20 cps frequency
- Detected by phonocardiograph
- 0.04 sec duration atrial systole → blood flow on ventricular walls → vibrations
- Caused by rapid ventricular filling during atrial systole
- Occurs during atrial systole

Abnormal Heart sounds

- **Heart murmurs:** Abnormal sounds heard in various parts of the vascular system.
 - a. Systolic murmurs:
 - AV-valve insufficiency
 - Semilunar valve stenosis
 - High CO (physiological)
 - b. Diastolic murmurs
 - Semilunar valve insufficiency
 - Av-valve stenosis
- Valvular insufficiency causes regurgitation

Graphic presentation of phonocardiograms of murmurs under various conditions is as shown below.



Cardiac cycle

- Activities in the heart in a single beat
- Contraction (systole) and relaxation (diastole) of cardiac chambers
- A single cardiac cycle comprised of
 - Atrial systole and atrial diastole
 - Ventricular diastole + ventricular systole
- 75 cycles completed per minute
- Duration of each cycle = 0.8 second
 - Ventricular diastole = 0.5 second
 - Ventricular systole = 0.3 second

Means of exploring activities accomplished in a cardiac cycle

1. Auscultation of heart sounds, phonograph
2. ECG tracing
3. Measuring aortic pressure
4. Measuring atrial pressure
5. Measuring ventricular pressure and volumes

Phases of the Cardiac Cycle

Cardiac cycle has 3 major phases

- I. Atrial systole
- II. Ventricular systole
- III. Diastole of the whole heart

I. Atrial systole

- Duration: 0.15 second
- Valvular events: SL-valves closed, AV-valves opened
- Phonocardiogram: S4
- ECG: P-wave, PR-segment and Q-wave are recorded
- Ventricular volume : increased
- Ventricular pressure: increased

Phases of the Cardiac Cycle (cont'd)

II. Ventricular systolic phase (contraction)

- ☐ Duration: 0.3 sec., has 3 phases

A. Isovolumic phase

- ☐ Ventricles contract with no change in volume (EDV=130 ml)
- ☐ Ventricular pressure rises from 4-80 mm Hg
- ☐ All valves are closed
- ☐ S_1 is produced, QRS- complex is recorded

B. Rapid ejection phase

- ☐ Maximum contraction of the ventricle
- ☐ Ventricular pressure rises from 80-120 mm Hg
- ☐ SL-valves opened
- ☐ Blood will be ejected (70% of SV), ↓ Ventricular Volume
- ☐ Aortic pressure increases
- ☐ Atrial pressure decreases

C. Slow ejection phase

- ☐ The remaining 30% of the SV is ejected
- ☐ Ventricles start to be relaxed

Phases of the cardiac cycle (cont'd)

Ventricular diastole: duration 0.5 sec, 4-phase

A. Protodiastolic phase

- ❖ A transitional phase
- ❖ Ventricular pressure drops rapidly
- ❖ SL-valves are closed and S2 is produced
- ❖ Aortic pressure declines

B. Isovolumic relaxation phase

- ❖ Ventricular pressure drops from 80 to 4 mm Hg
- ❖ 4-valves are closed, no change in volume
- ❖ T-wave completes, ventricles relax

Phases of the cardiac cycle (cont'd)

C Rapid ventricular filling phase

- ❖ Atrial pressure is higher than ventricular pressure
- ❖ AV-valves opened
- ❖ Ventricular pressure falls near 0 mm Hg
- ❖ Ventricular volume rises sharply
- ❖ 80% of the filling blood run to the relaxed ventricle

D. Slow filling phase 20% of EDV fills the ventricle, pumped by atrial contraction & ventricular volume still increasing, S_4 produced

START

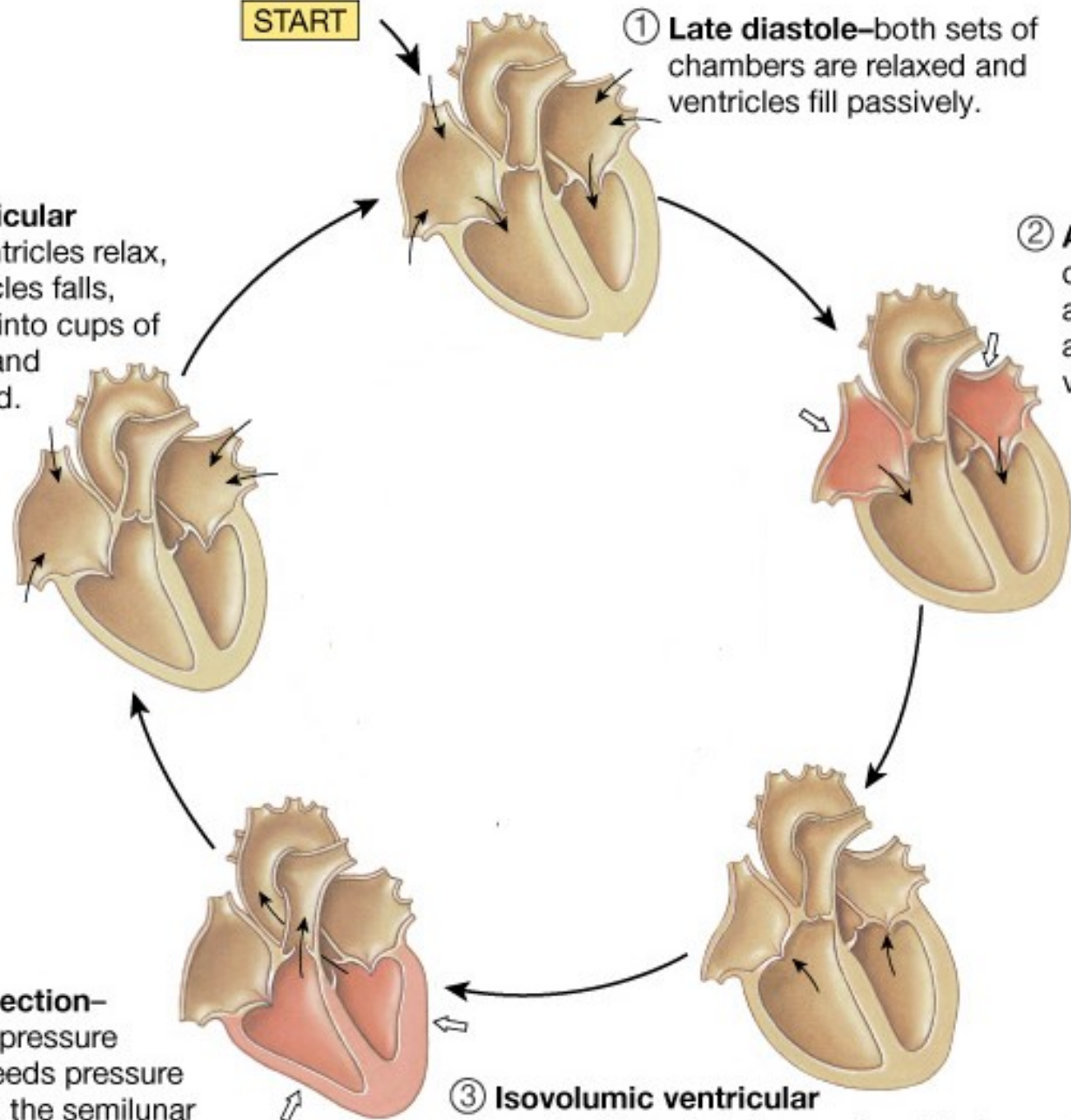
① **Late diastole**—both sets of chambers are relaxed and ventricles fill passively.

② **Atrial systole**—atrial contraction forces a small amount of additional blood into ventricles.

③ **Isovolumic ventricular contraction**—first phase of ventricular contraction pushes AV valves closed but does not create enough pressure to open semilunar valves.

④ **Ventricular ejection**—as ventricular pressure rises and exceeds pressure in the arteries, the semilunar valves open and blood is ejected.

⑤ **Isovolumic ventricular relaxation**—as ventricles relax, pressure in ventricles falls, blood flows back into cups of semilunar valves and snaps them closed.



Cardiac Output (CO)

Cardiac Output:

- It is the amount of blood pumped by each ventricle in one minute
- It is the product of heart rate (HR) and stroke volume (SV)
 - HR is the number of heart beats per minute
 - SV is the amount of blood pumped out by ventricle with each beat
 - $SV = \text{ejection fraction} \times EDV$
 - **Ejection fraction** = 2/3 of blood contained and ejected by the heart

...cont'd

Cardiac reserve:

- It is the difference between maximal exercise and resting CO
 - CO = 5-6 liters per minute at rest
 - CO = 20-40 liters per minute at maximum exercise

E.g. 35L-5 L =30L

Cardiac Index:

- It is the amount of blood pumped of the left ventricle each minute

CI = Cardiac output ÷ Body surface area

E.g. 5.5 L/min ÷ 1.7 = 3.23 L/m²

Effect of Venous Return (VR) on C.O

Venous return is:

1. volume of blood flowing from veins into RA/min
 2. 5.5L/min at rest; about 35L/min in well trained athletes during strenuous exercise
 3. influenced by RAP like C.O as shown on cardiac function curves (Fig..)
- **C.O = VR** at the “operating point” of the cardiac function curves (Fig..).
 - Any factor influencing C.O or VR shifts the “operating point” of C.O/VR intersection to a new position.

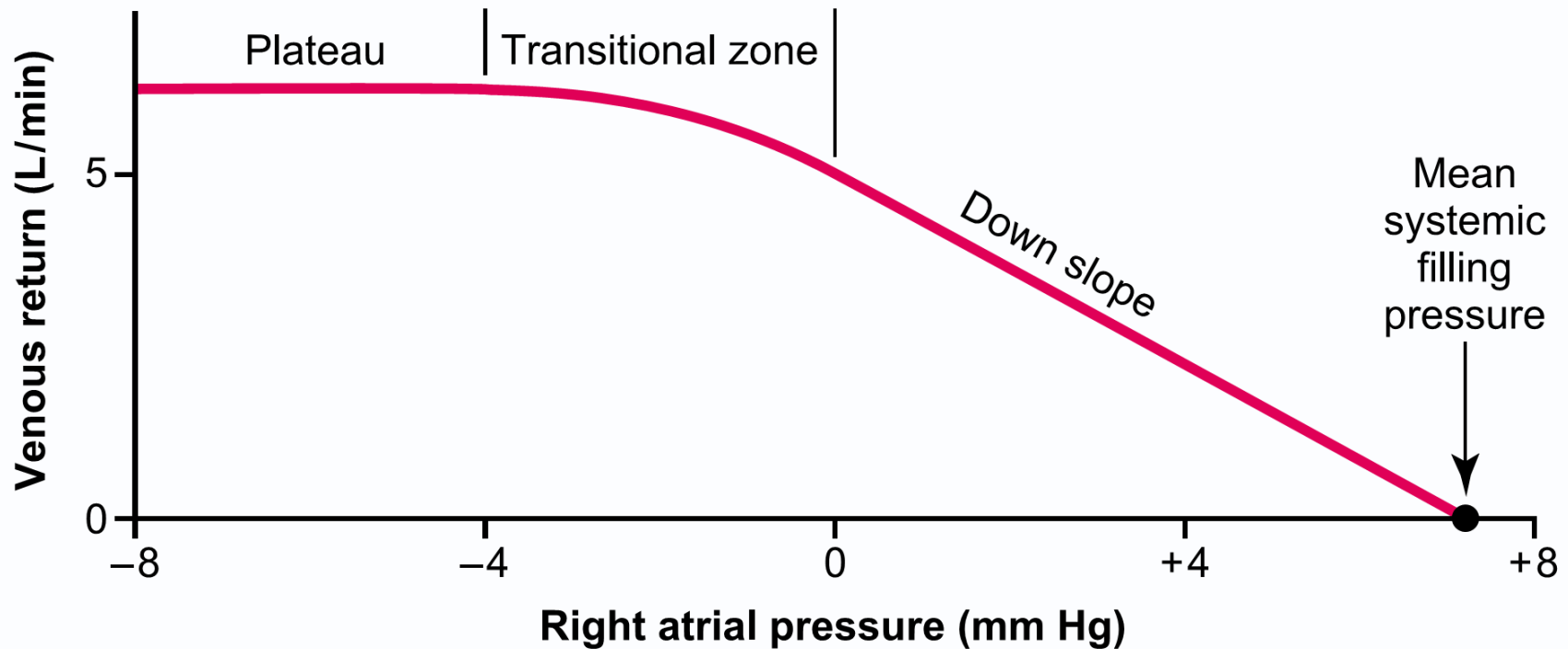
Venous return curve

- *venous return is the quantity of blood flowing from the veins into an atrium each minute*
- three principal factors affect venous return to the heart from the systemic circulation:
 1. *right atrial pressure*, which exerts a backward force on the veins to impede flow of blood from the veins into the right atrium
 2. *degree of filling of the systemic circulation* (measured by the *mean systemic filling pressure*), which forces the systemic blood toward the heart
 3. *resistance to blood flow between the peripheral vessels and the right atrium*

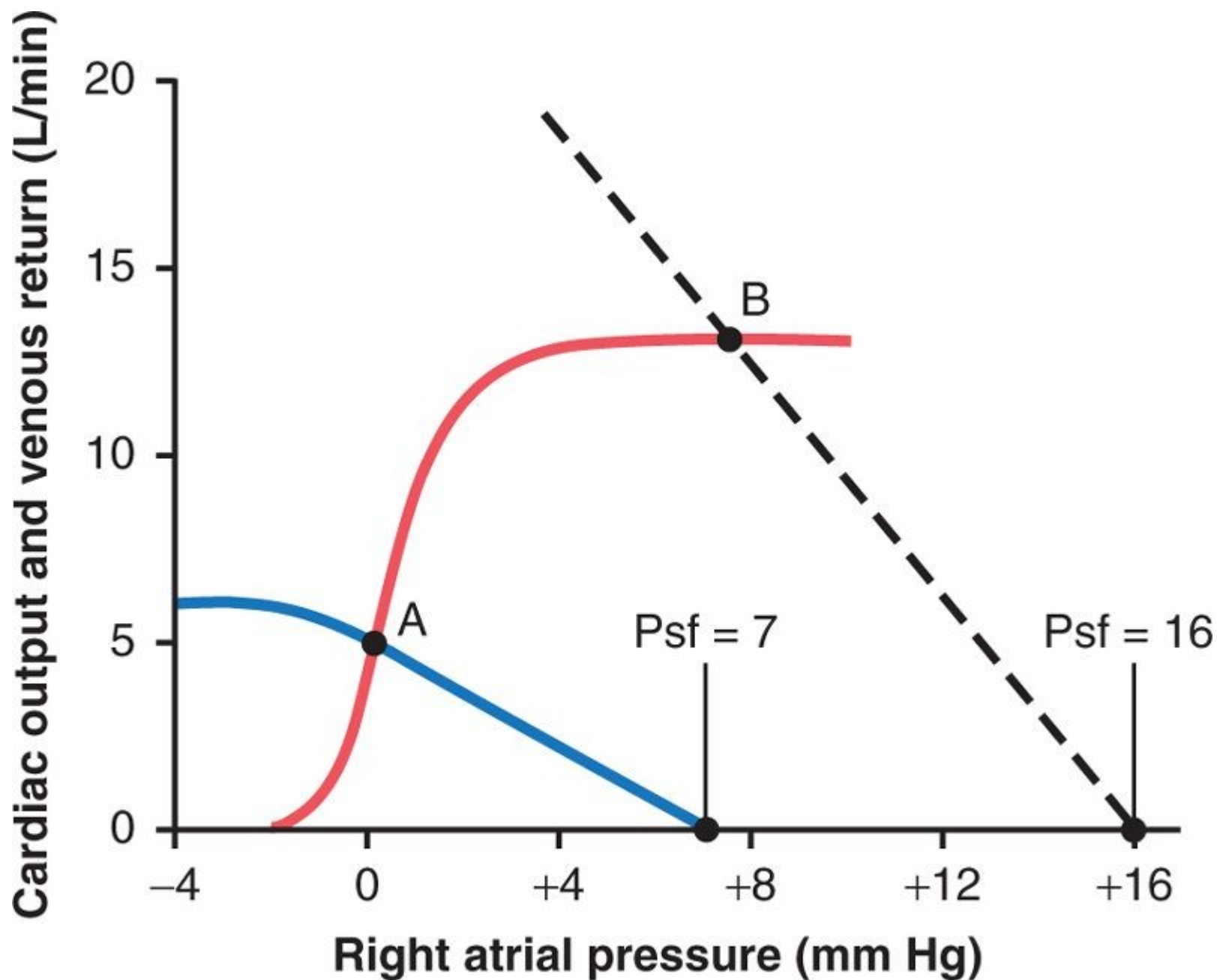
Venous return curve cont'd

- *venous return curve* relates venous return also to right atrial pressure
- *venous return curve* shows that when heart pumping capability becomes diminished and causes *the right atrial pressure* to rise, the backward force of the rising atrial pressure on the veins of the systemic circulation *decreases venous return of blood to the heart*
- *mean systemic filling pressure (P_{sf})* is the pressure measured everywhere in the systemic circulation after blood flow has been stopped by clamping the large blood vessels at the heart
- the mean systemic filling pressure is almost always nearly equal to the *mean circulatory filling pressure*
- the normal mean systemic filling pressure is *7 mm Hg*

Venous return curve cont'd



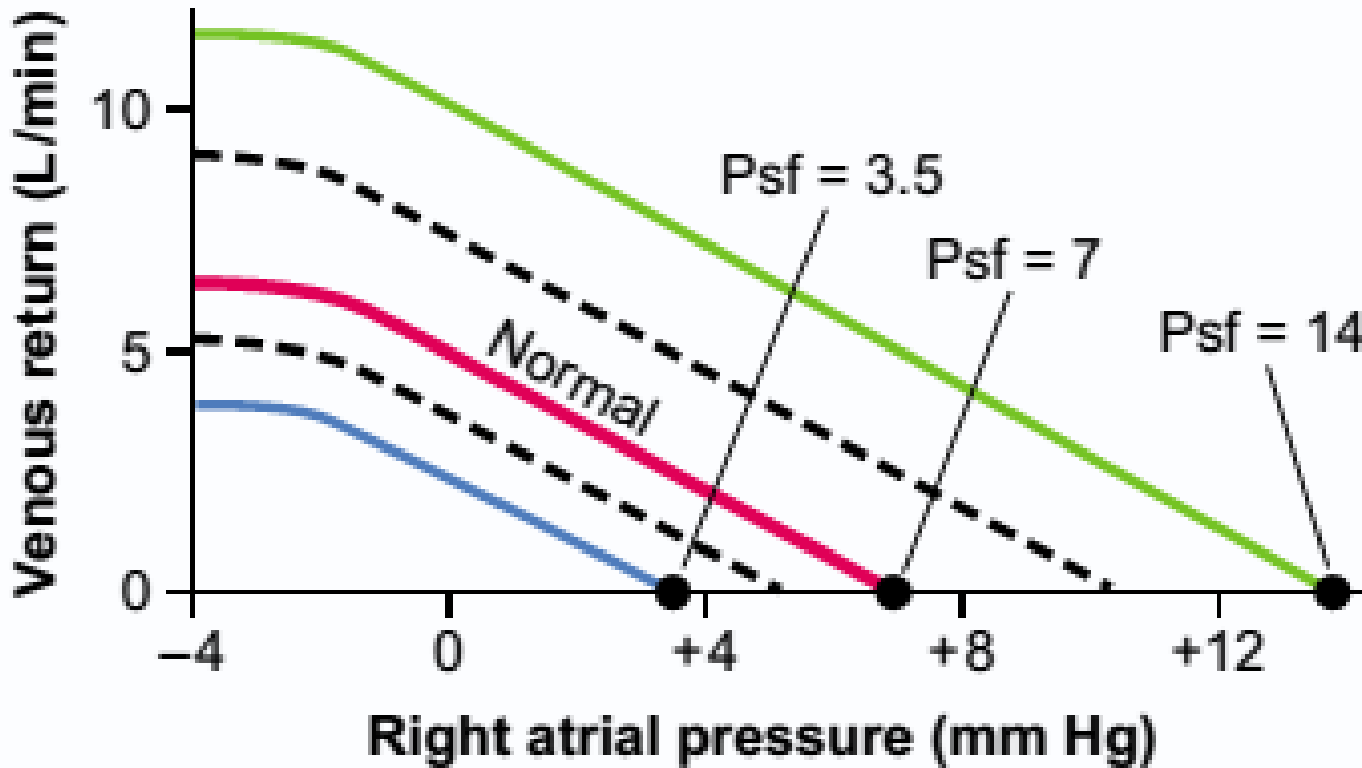
Normal venous return curve



Venous return curve cont'd

- the greater the mean systemic filling pressure the more the venous return curve shifts upward and to the right
- the lower the mean systemic filling pressure, the more the curve shifts downward and to the left
 - ✓ the greater the system is filled, the easier it is for blood to flow into the heart
 - ✓ the less the filling, the more difficult it is for blood to flow into the heart
- the greater the difference between the *mean systemic filling pressure* and the *right atrial pressure*, the greater becomes the venous return
 - ✓ therefore, the difference between these two pressures is called the *pressure gradient for venous return*

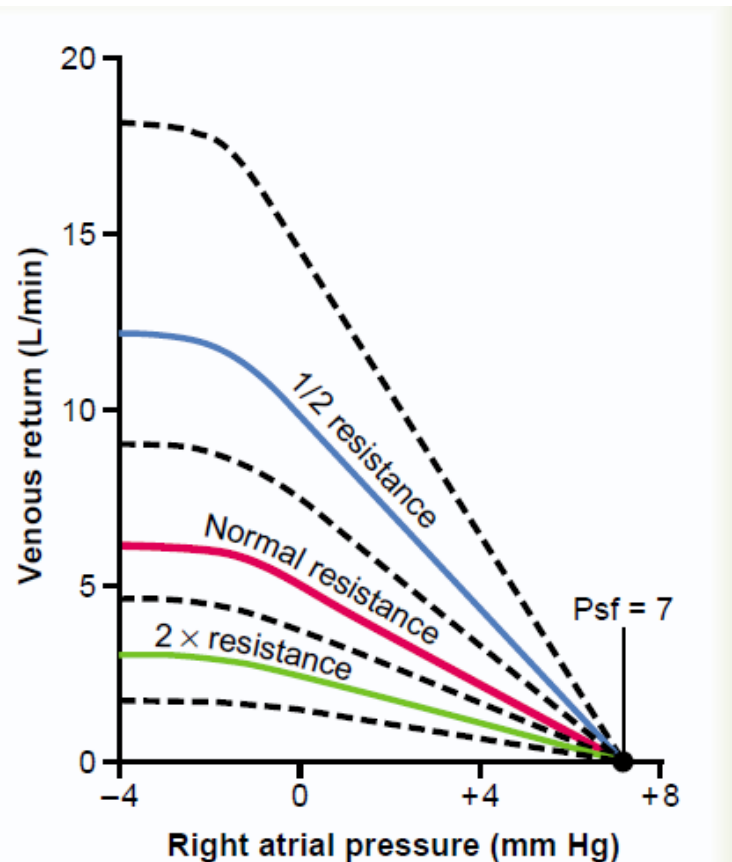
Venous return curve cont'd



Effect of altering the P_{sf} on Venous return curve

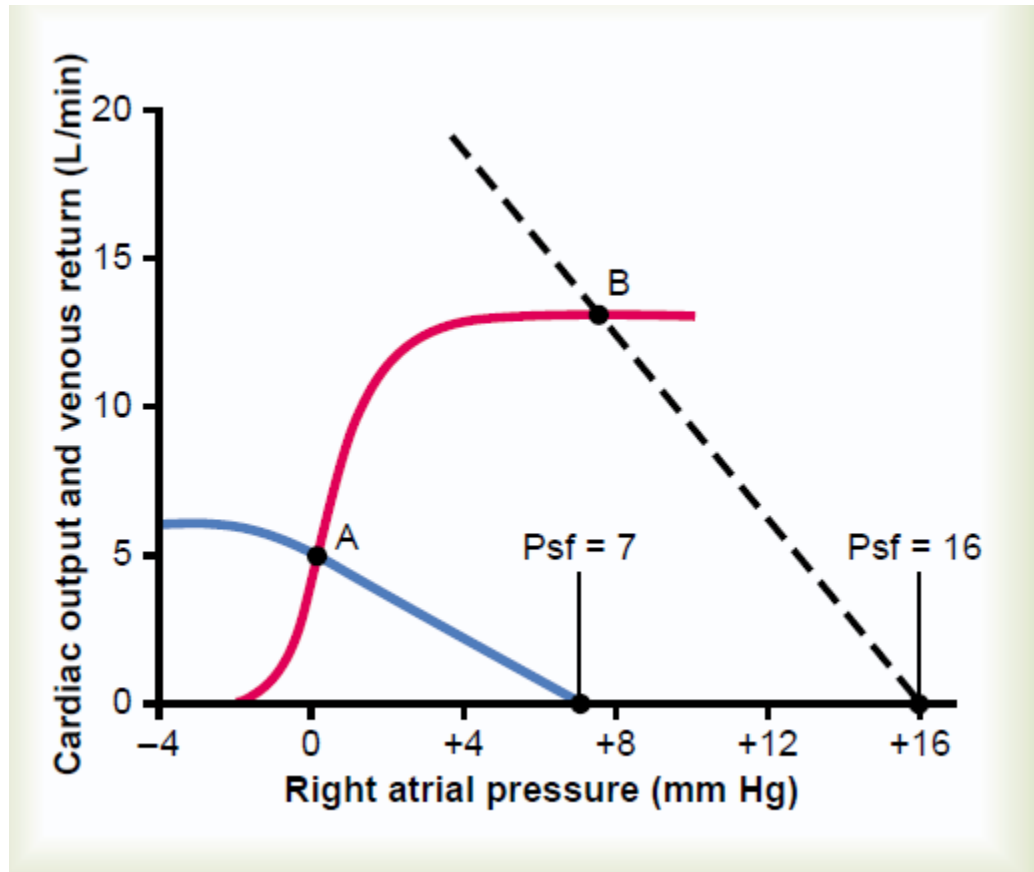
Venous return curve cont'd

- in the same way that mean systemic filling pressure represents a pressure pushing venous blood from the periphery toward the heart, there is also resistance to this venous flow of blood
- most of the resistance to venous return occurs in the veins, although some occurs in the arterioles and small arteries as well

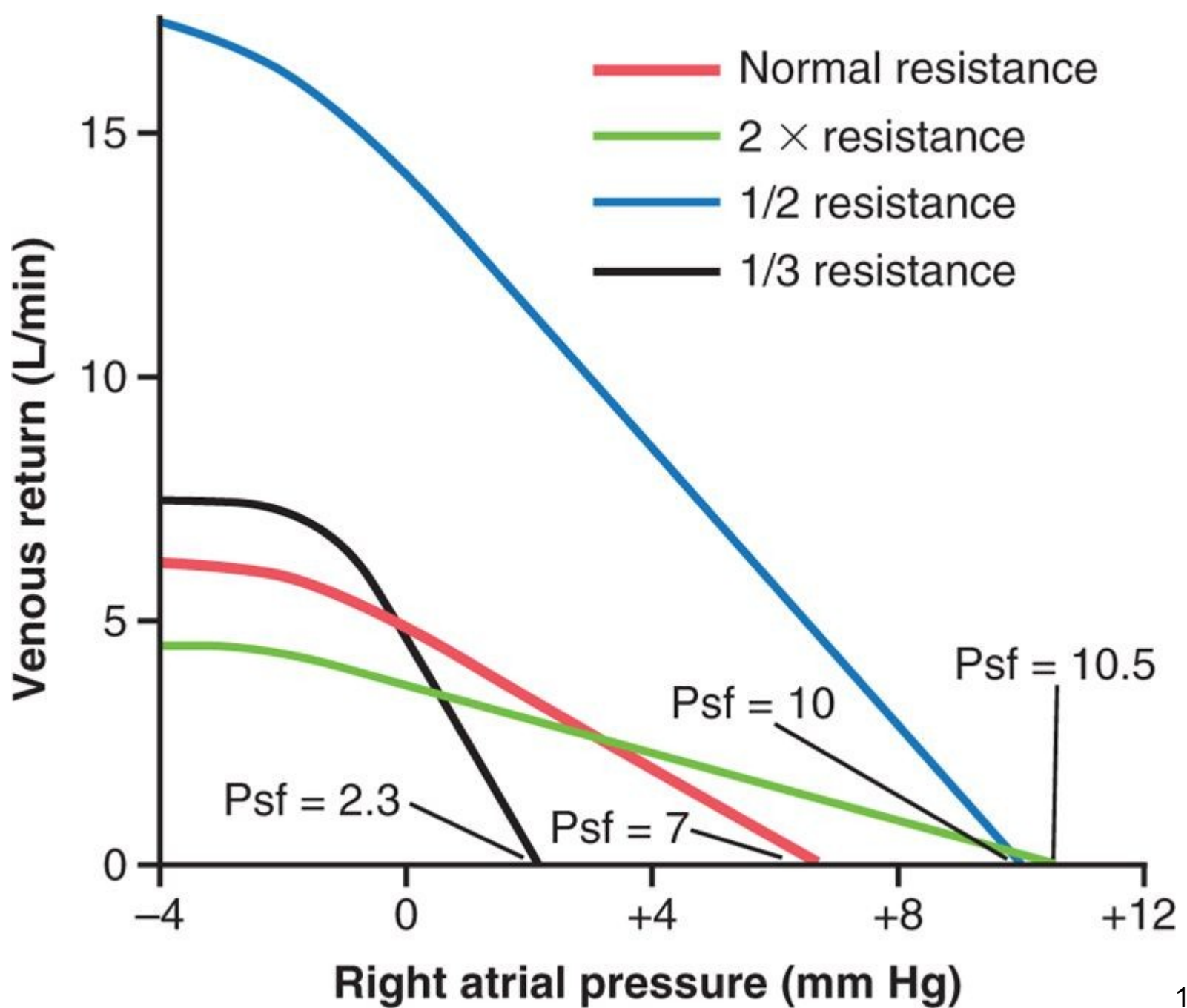


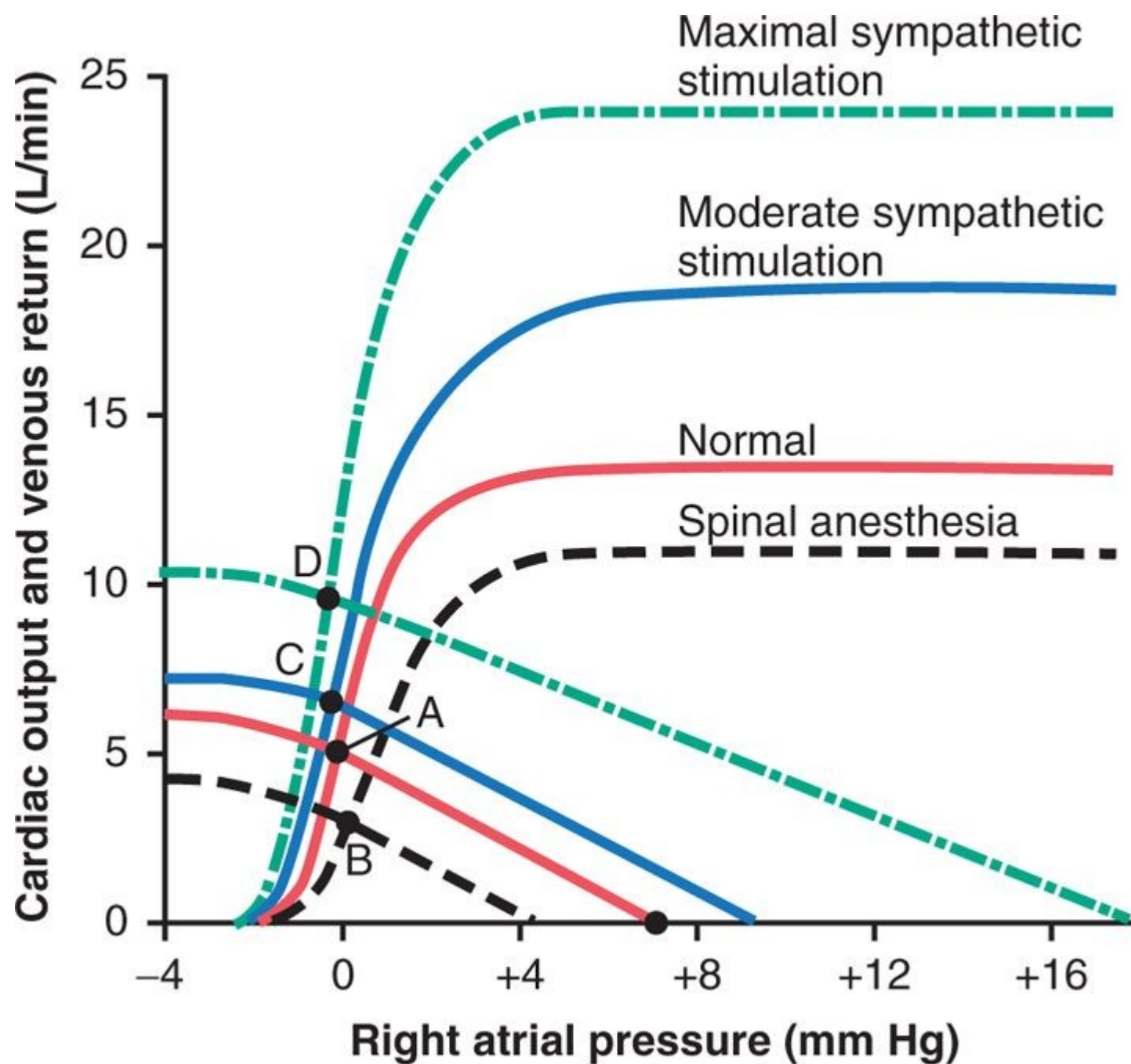
Venous return curves depicting the effect of altering the resistance to venous return

Venous return curve cont'd



- ✓ in the complete circulation, the heart and the systemic circulation must operate together
- ✓ the venous return from the systemic circulation must equal the cardiac





Factors affecting Venous Return

1. Mean Systemic Filling Pressure (MSFP)

- Pressure measured every where in the systemic circulation 1 min after blood flow has been stopped by clamping large blood vessels near the heart.
- It is 7 mmHg and indicates the degree of filling of the systemic circulation
- It is the driving force for VR.

2. Right Atrial Pressure(RAP)

- Mean pressure in the right atrium (Central venous pressure) = 2 mmHg
- Pressure gradient = $MSFP - RAP$
 $\uparrow RAP \rightarrow \downarrow VR$

Factors affecting ...cont'd

3. Resistance to venous return (RVR)

- Resistance blood meets (1mmHg/L/min) during its flow from arterial side to RA
- Occurs mainly at arterioles
- $VR = MSFP - RAP / RVR = MAP - RAP / TPR$

4. Sympathetic stimulation →

↑VR by inducing vasoconstriction, improved cardiac pumping power and arteriolar dilatation

5. Blood volume

↑Blood volume → ↑VR by ↑MSFP and vice versa.

6. Respiratory movements

- VR increases with inspiration and decreases with expiration

Factors affecting ...cont'd

7. Arteriolar dilatation $\rightarrow \downarrow RVR$ and $\uparrow VR$

8. Capillary dilatation $\rightarrow \uparrow$ vascular capacity, $\downarrow MSFP$ and $\downarrow VR$

9. Skeletal muscle contraction

- Squeezes veins b/n muscles $\rightarrow \uparrow MSFP \rightarrow \uparrow VR$

10. Gravity

- Standing motionless for some time $\rightarrow$ pooling of blood in lower extremities $\rightarrow \downarrow MSFP \rightarrow \downarrow VR \rightarrow \downarrow C.O \rightarrow$ hypotension $\rightarrow$ brain ischaemia $\rightarrow$ **syncope**.

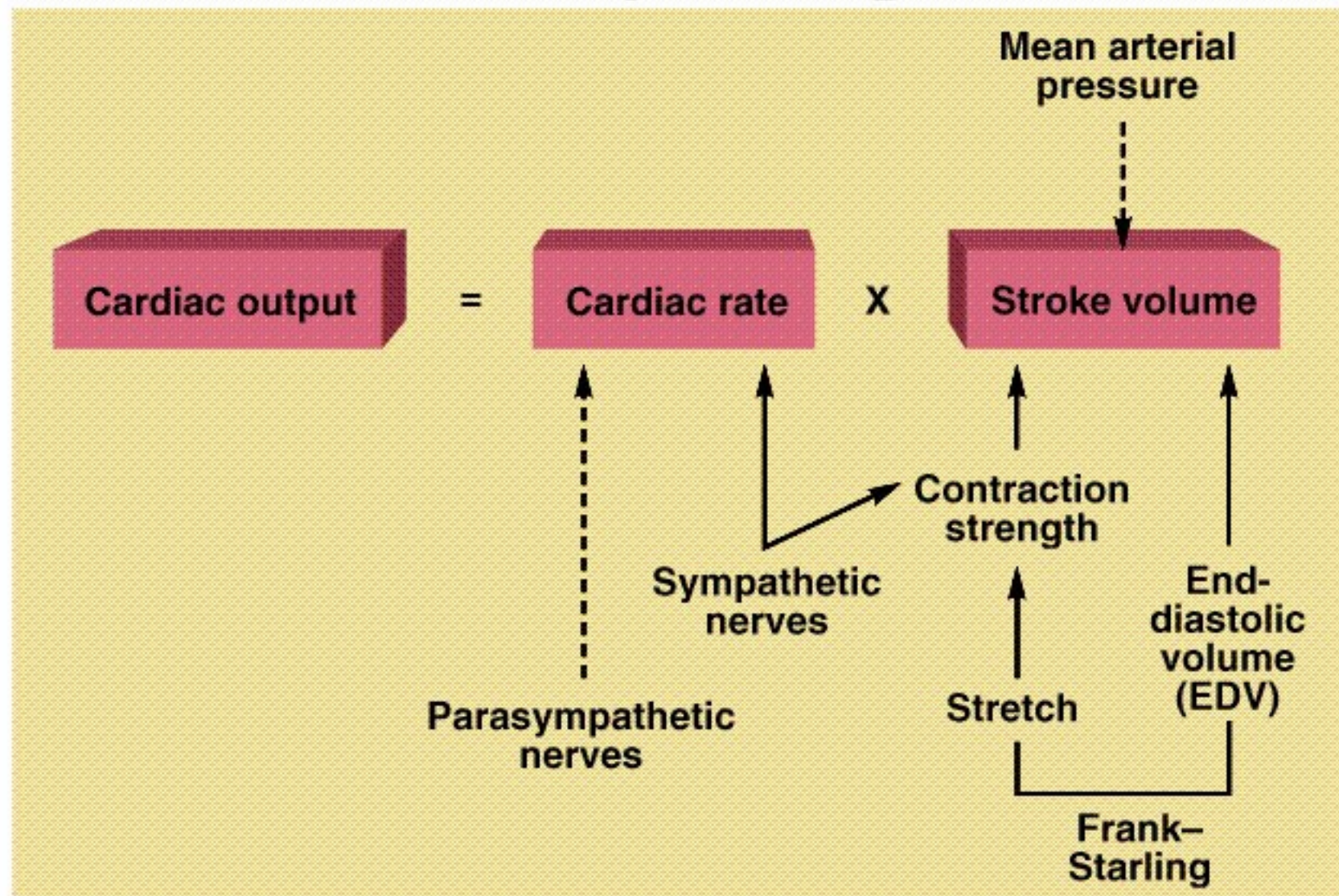
Regulation of Pumping Activity of the Heart

- There are two types of regulatory mechanisms:
 1. **The Intrinsic Auto-regulatory Mechanisms (The Frank Starling Mechanism)**
 - ☐ “Within physiological limits, the contractile force of the heart is proportional to the amount of the blood enters the heart”.
 2. **The Extrinsic Regulatory Mechanisms**
 - ☐ Sympathetic, parasympathetic stimulation
 - ☐ Catecholamine
 - ☐ Electrolytes (Ca^{2+} , K^{+})
 - ☐ Drugs: Cardiac glycosides, xanthenes
 - ☐ Hormones: AD, Glucagon, T3/T4

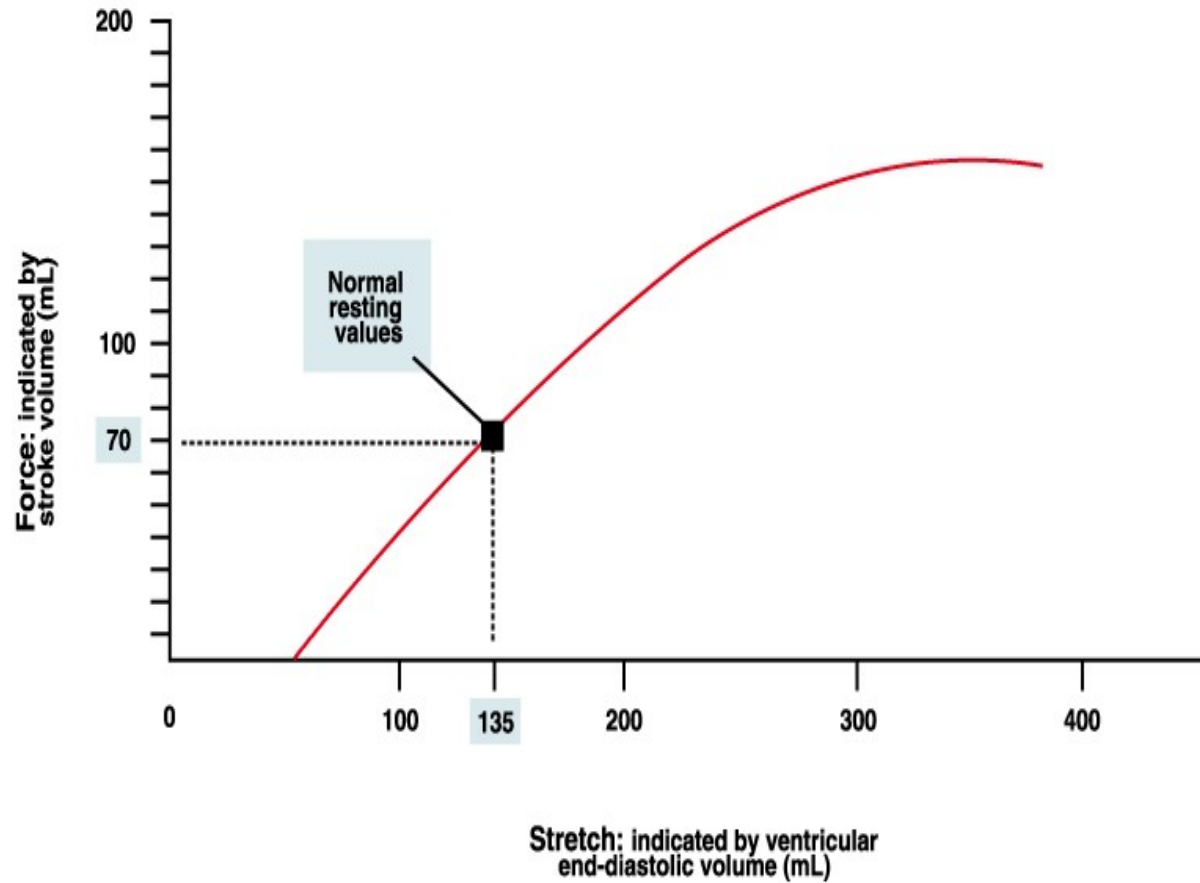
Frank Starling Law of the Heart

- Frank-Starling mechanism means the greater the heart muscle is stretched during filling $\rightarrow$ $\uparrow$ force of contraction $\rightarrow$ $\uparrow$ blood pumped to the aorta.
- Pre-load or degree of stretch of cardiac muscle cells before they contract is the critical factor controlling stroke volume
 - It depends on end diastolic volume
- Slow heart beat and exercise $\rightarrow$ $\uparrow$ VR to the heart $\rightarrow$ increasing EDV that in turn, $\uparrow$ SV
- Blood loss and extremely rapid heartbeat $\rightarrow$ decreases ventricular filling time and decreasing EDV $\rightarrow$ that decreases SV

Cardiac Output Regulation



Starling's Law of the Heart

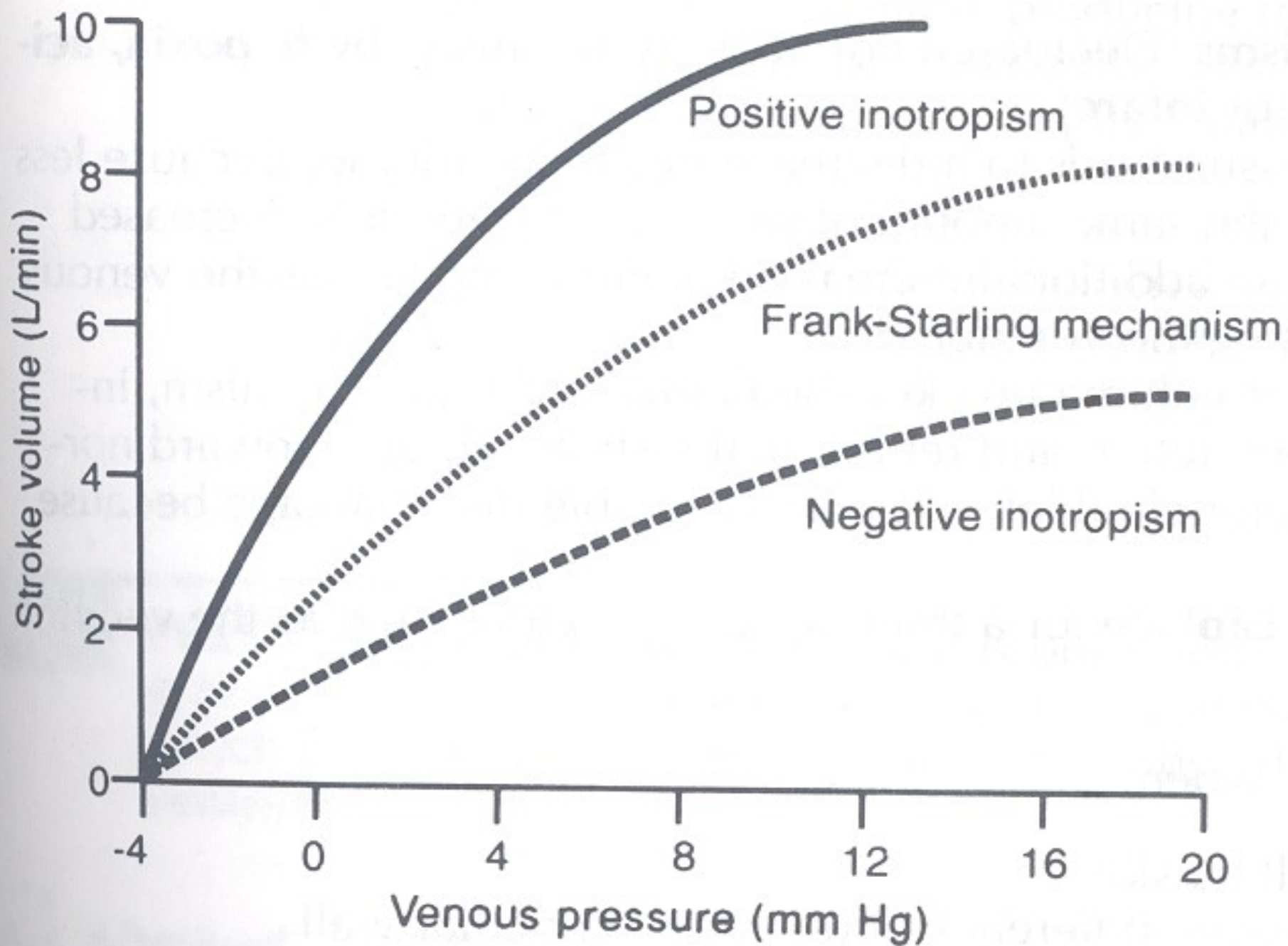


...cont'd

- A few seconds of **heterometric** autoregulation(**pre-load**)→↑ metabolic state of heart→ more forceful contraction and 30-60 sec. after, pressure elevation occurs; the heart expels more blood than is brought into it
- ∇ ↑EDV & ESV persists for 2-3 minutes.
- This increase in contractile force without a change in EDV is referred to as **Homeometric (after-load)** regulation, a condition mediated by a change in metabolic state or contractility of the heart.

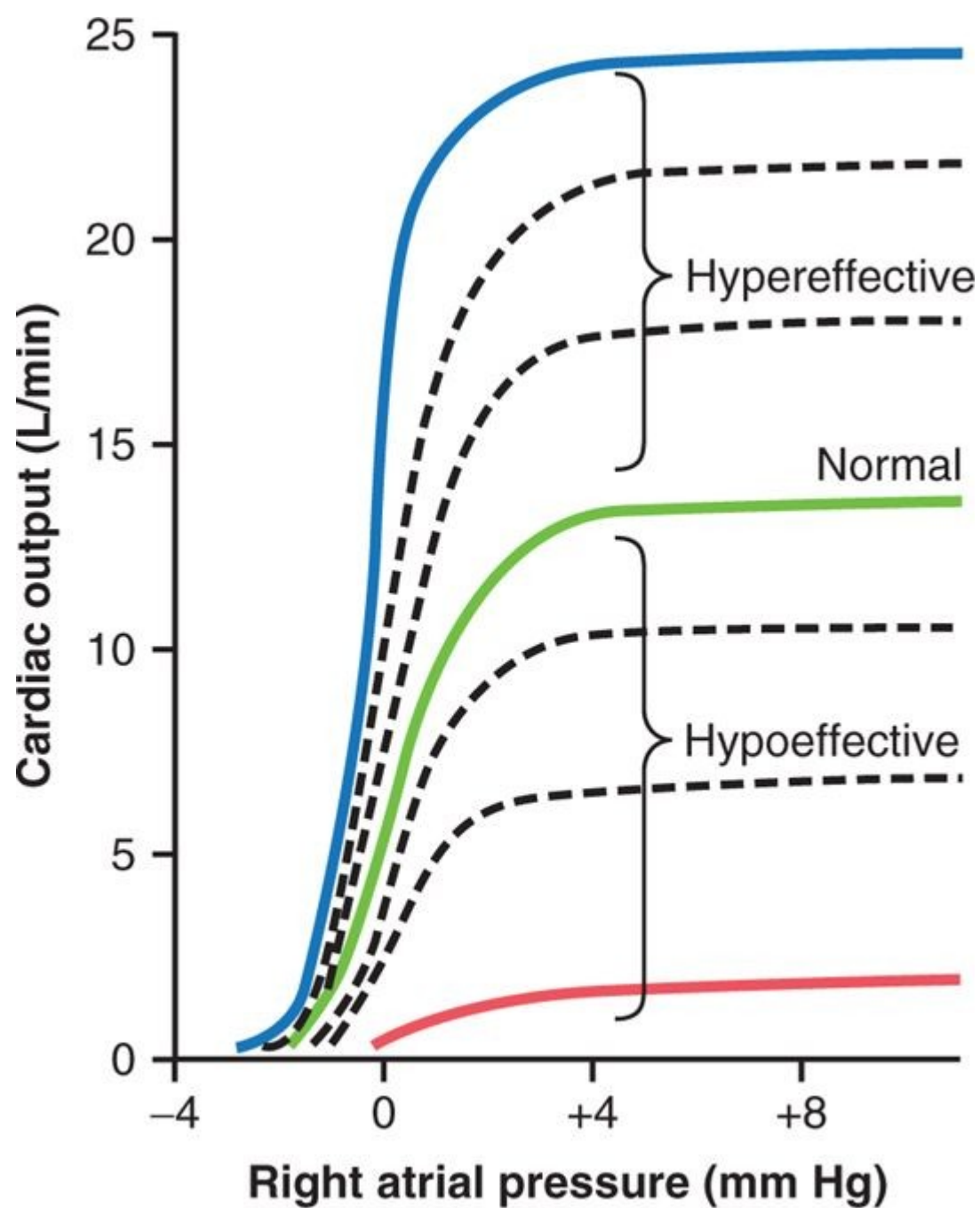
NB:

- **Heterometric autoregulation** is related to filling pressure of the heart
- while **homeometric autoregulation** is related to maximum pressure (contractile force).

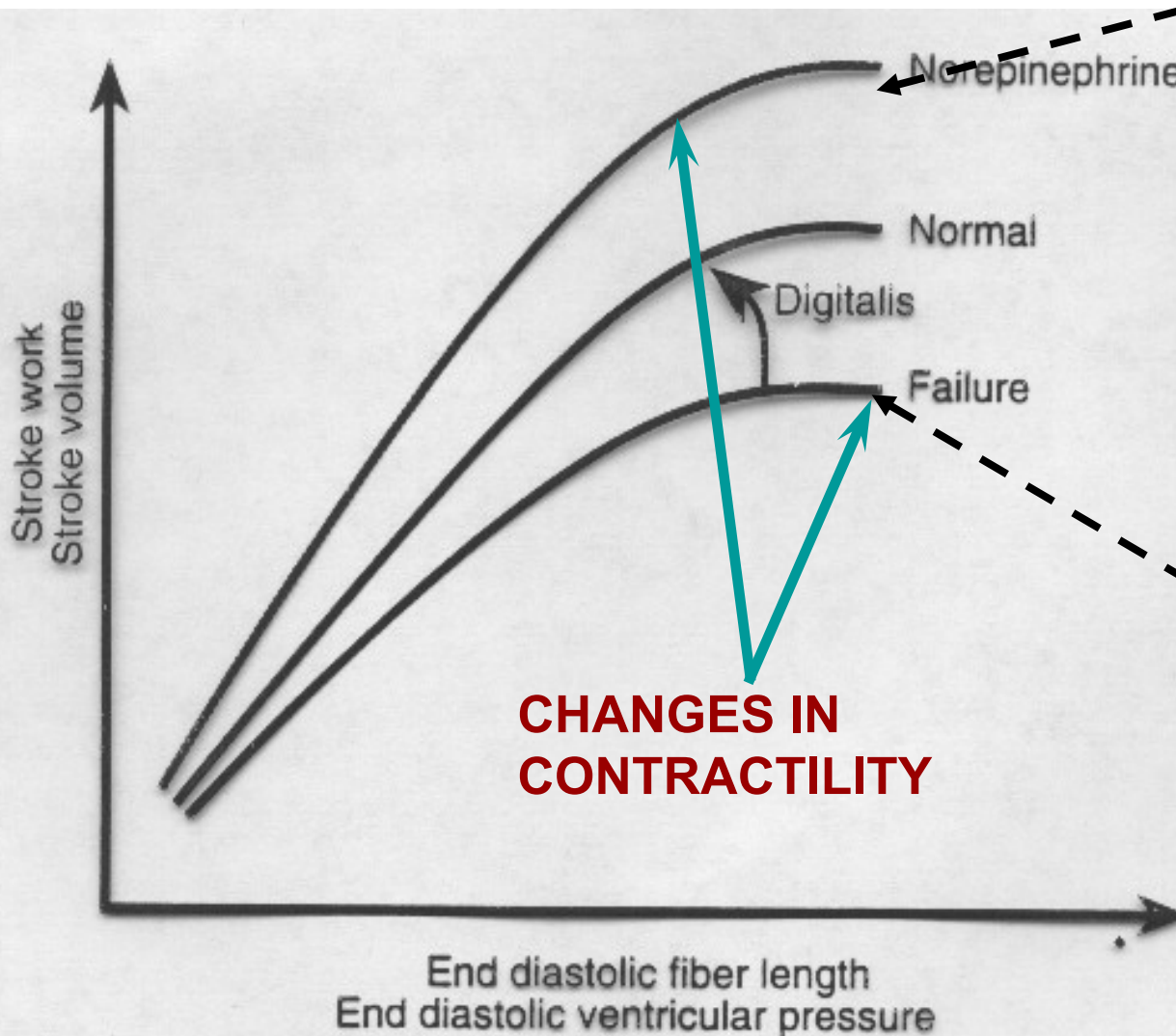


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- A sudden increase in venous return (VR) → ↑ in heart size.
- For example, a change from erect to supine position leads to shift of blood from lower extremities to thorax → ↑ CVP (4-12 cm H₂O) → ↑ in heart size.
- The heart may be:
 - **Hyper-effective** (capacity more than normal) due to sympathetic stimulation or hypertrophy.
 - **Hypo-effective** and pumping capacity is lower than normal due to:
 - Sympathetic denervation
 - Parasympathetic stimulation
 - Atrial flutter or fibrillation
 - Myocarditis
 - Cardiac hypoxia



CONTRACTILITY: THE VENTRICULAR FUNCTION CURVE



+ve inotropic agents

- Sympathetic stimulation
- Hypercalcemia
- Glucagon, T3/T4
- Digitalis, Xanthenes
- Catecholamines

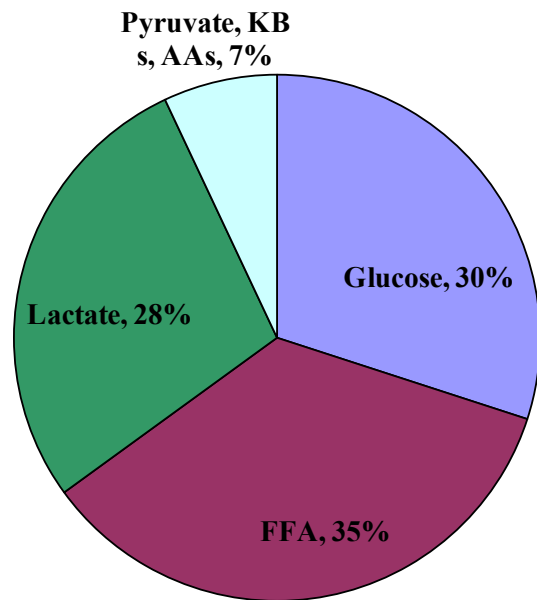
-ve inotropic agents

- Parasympathetic stimulation
- Hyperkalemia
- Hypocalcemia
- Acidosis
- Toxins
- Heart diseases

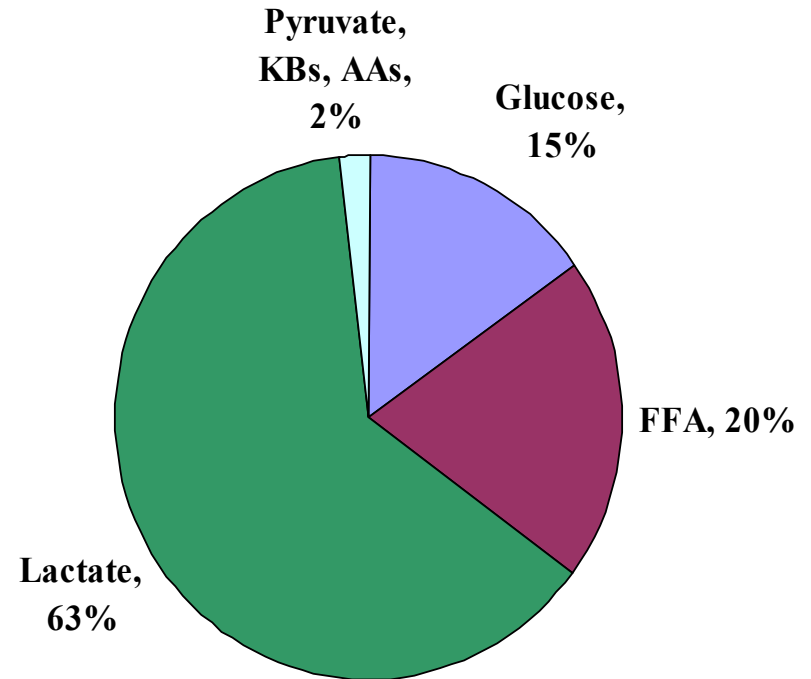
Metabolism and Energetics (work) of the Heart

- Energy provided by conversion of metabolic energy into pressure and kinetic force.
- Sources of energy= glucose or glycogen, fatty acids, lactate, ketone bodies, and amino acids.
- Metabolic processes are as shown in figures below
- **Pumping work**= work done against aortic pressure= $SV \times \text{mean pressure}$ ($\Delta P \times \Delta V$) influenced by metabolic pathways
- **Kinetic work**= energy imparted to the velocity of blood
- Accounts for only some 5% of the external work of left ventricle beating at rest.

Metabolism and energetics of the cardiac muscle



At rest



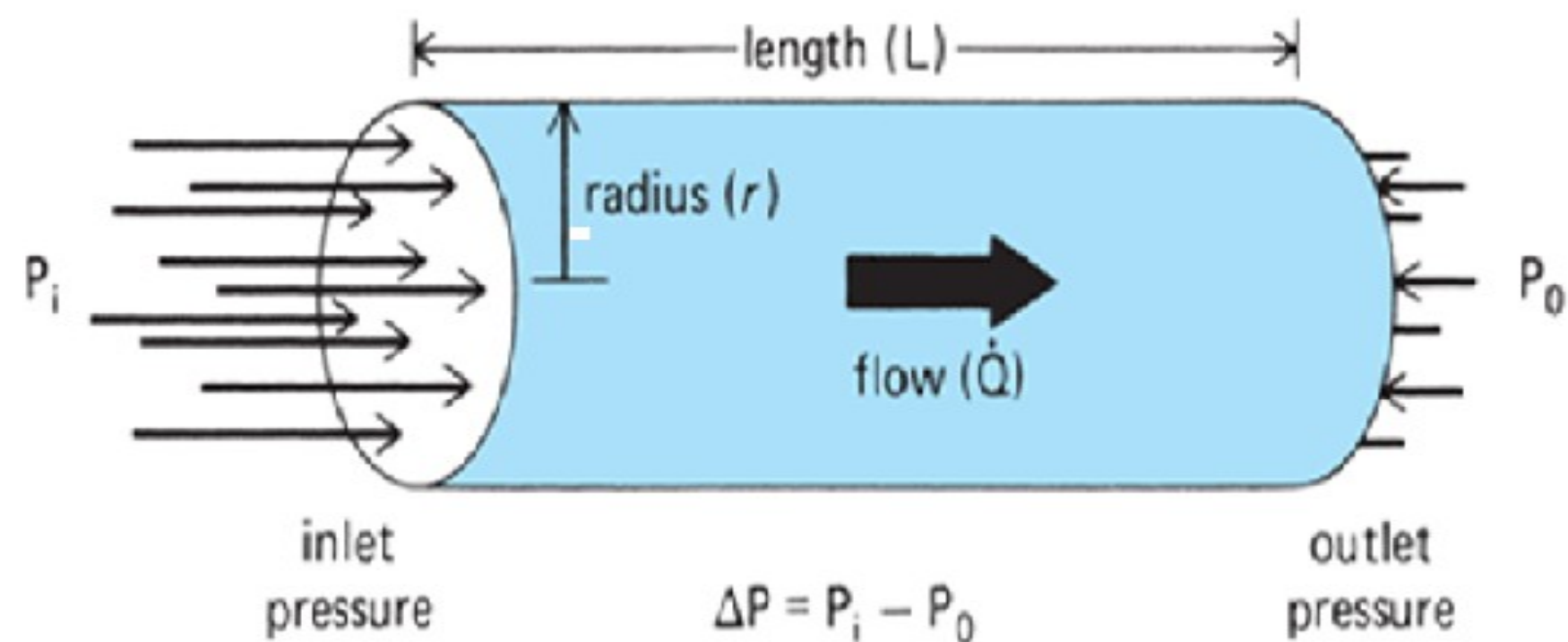
Moderate exercise

HEMODYNAMICS

- It is the study of the relationship between flow, pressure and resistance and other physical principles of blood circulation
- In 1846, Poiseuille, a French physician(1797-1869) and Hagen, a German scientist (1797-1884)
- Described factors governing non-pulsatile flow of a homogeneous (Newtonian) viscous fluid through a rigid cylindrical tube with constant circular cross-section.
- Showed that, with other things being equal, and with the inclusion of the proportionality constants $\pi/8$, the **rate of flow, Q** , through a cylindrical tube of length L , and radius r , would be:

$$Q = \Delta P \pi r^4 / 8 \eta L$$

- This is Hagen-Poiseuille's law.



Resistance to Flow(r)

- This is an impediment to flow provided by a vessel or circulatory bed.
- Vascular resistance to BF is directly proportional to the length of the vessel and viscosity of blood, but inversely proportional to the 4th power of radius of the vessel.

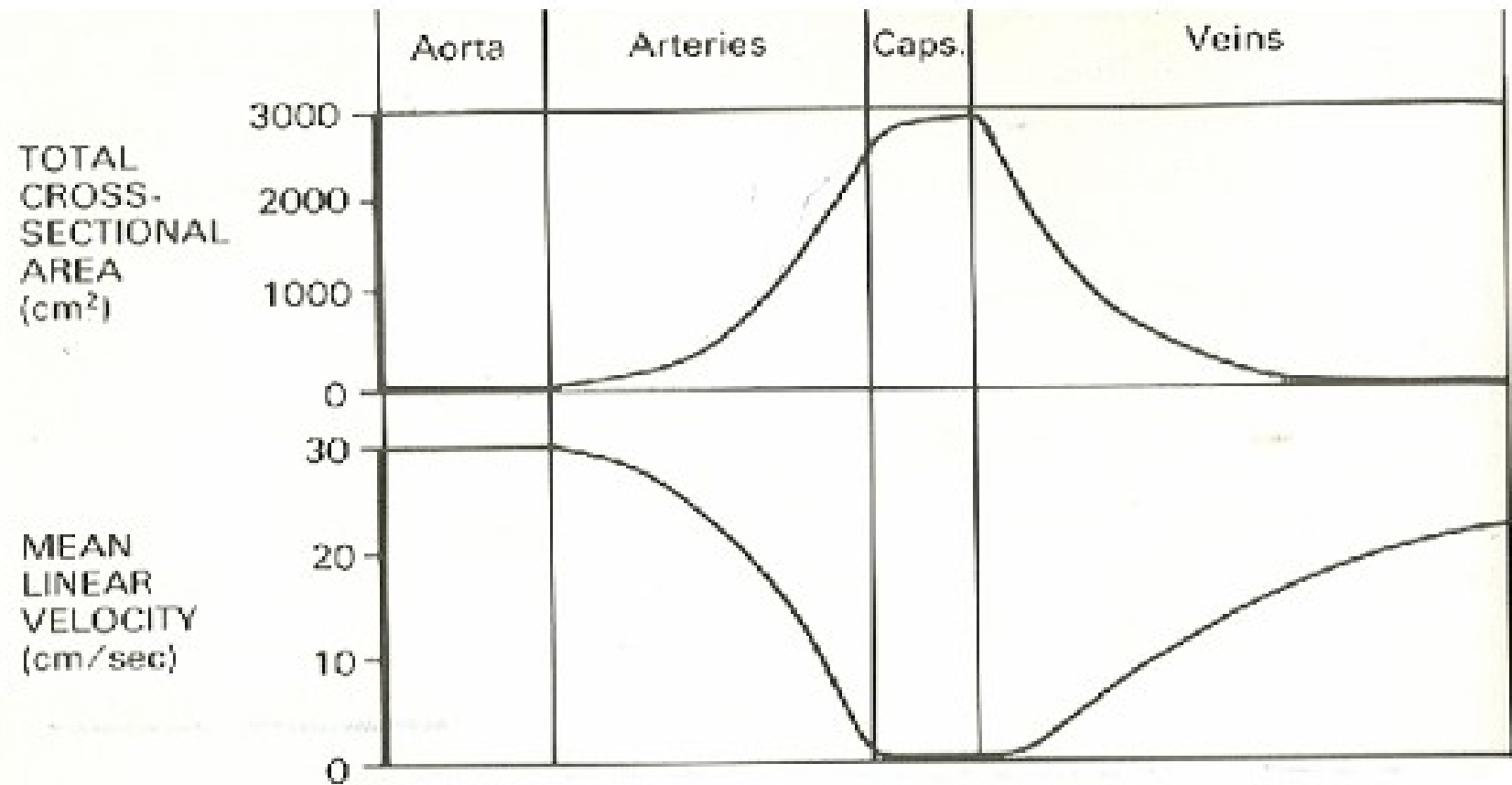
$$\Delta P = Q \times R$$

- **$R = \Delta P / Q$, analogous to Ohm's law ($R = V / \text{current}$)**
 $= \Delta P \div \Delta P \pi r^4 \div 8 \eta L$,
- Using this relationship, resistance in different vascular beds can be assessed.
- E.g. Aortic pressure=100mmHg, Ventricular output =80mL/min
- Total peripheral resistance=100/80 =1.25 PRU
- Analogous to electrical circuits, **vascular resistances** may be in **series or parallel** (fig. below).

Velocity of Blood Flow

- Velocity (v) of blood flow is directly proportional to the pressure difference (ΔP) and diameter of the vessels, but inversely related to the viscosity of blood and length of the blood vessel.
- $V = Q \div \text{Cross-sectional area } (A = \pi r^2)$
 $= (\Delta P \pi r^4 \div 8 \eta L) \div \pi r^2$
- $V = \Delta P r^2 / 8 \eta L$

Velocity profile (Fig...below)



$$\bar{V} = \frac{\text{Flow}}{\text{X-Sect. Area}}$$

Fig.. Relation of cross-sectional area and mean velocity of flow in the systemic circulation.

Viscosity

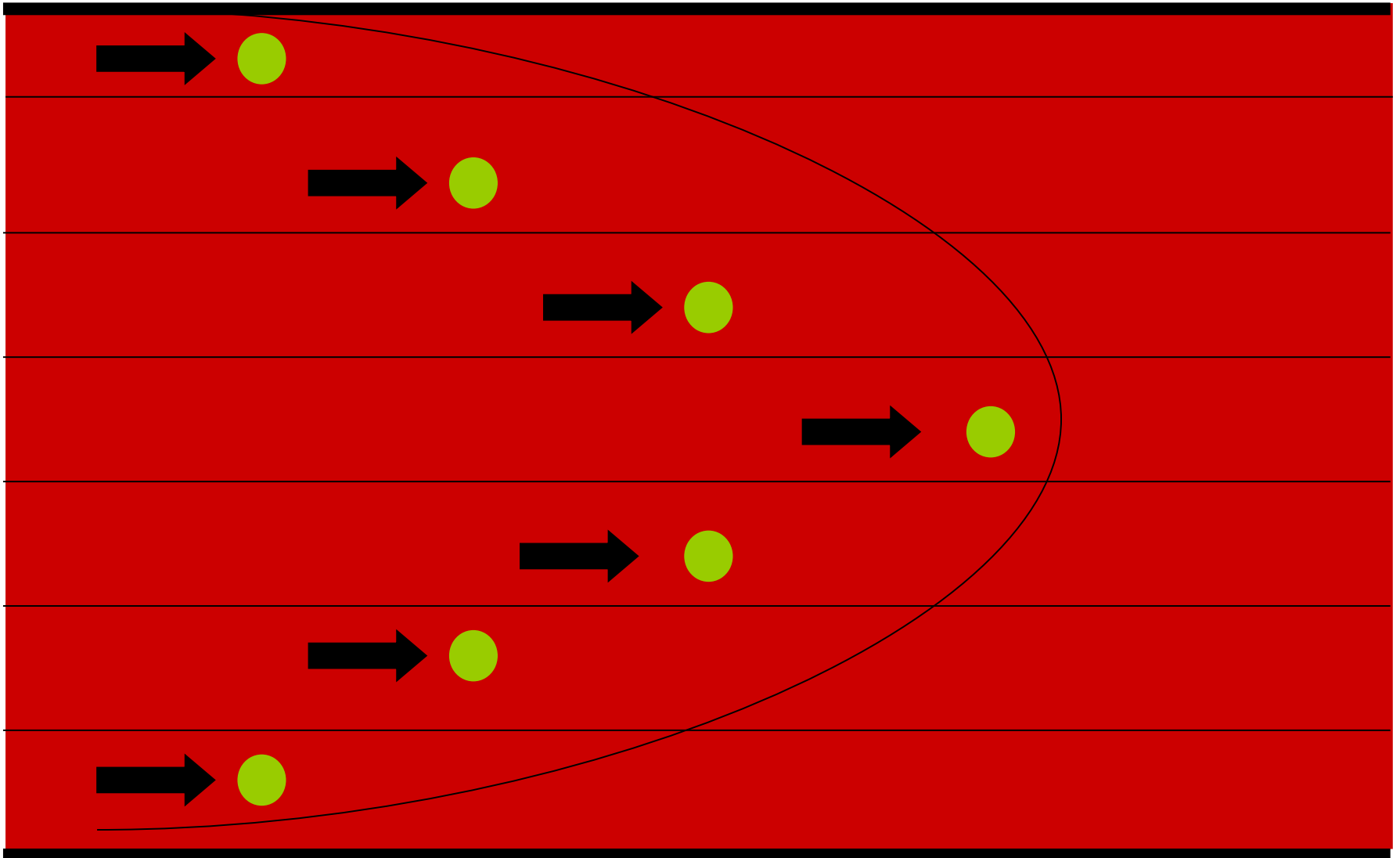
- This is friction of molecules in the moving stream of fluid.
- It depends on:
 1. Concentration of suspended medium (e.g. RBC in blood). Viscosity becomes high in polycythemia, and low in anaemia
 2. Velocity of flow
 3. Radius of vessel
 4. Temperature

RHEOLOGY OF BLOOD (types of blood flow)

1. Laminar or streamline blood flow

- Pulsatile (laminar) flow of fluid along a uniform tube (>2 or $3RBC$) at a fixed rate is laminar, i.e. infinite number of concentric laminae or layers sliding upon one another are seen.
- Innermost or axial layer has the highest velocity and the outermost layer touching the lining of the tube remains stationary.

LAMINAR or STREAMLINE FLOW



2. Steady (non-pulsatile) Blood Flow

- In steady blood flow, there is no variation of flow in time, i.e. flow is non-pulsatile.
- Blood flow is laminar (streamline) in all the smaller vessels of the resistance channels and steady or non-pulsatile in veins.

3. Turbulent Blood Flow

- Flow continues to be laminar (streamline) as long as the product of velocity and diameter does not exceed a critical value.
- It becomes turbulent when the critical value is exceeded.
- This is determined by the **Reynolds number Re (a dimensionless number)**.
- It represents the **ratio of inertia to viscous forces**.

Types of blood flow

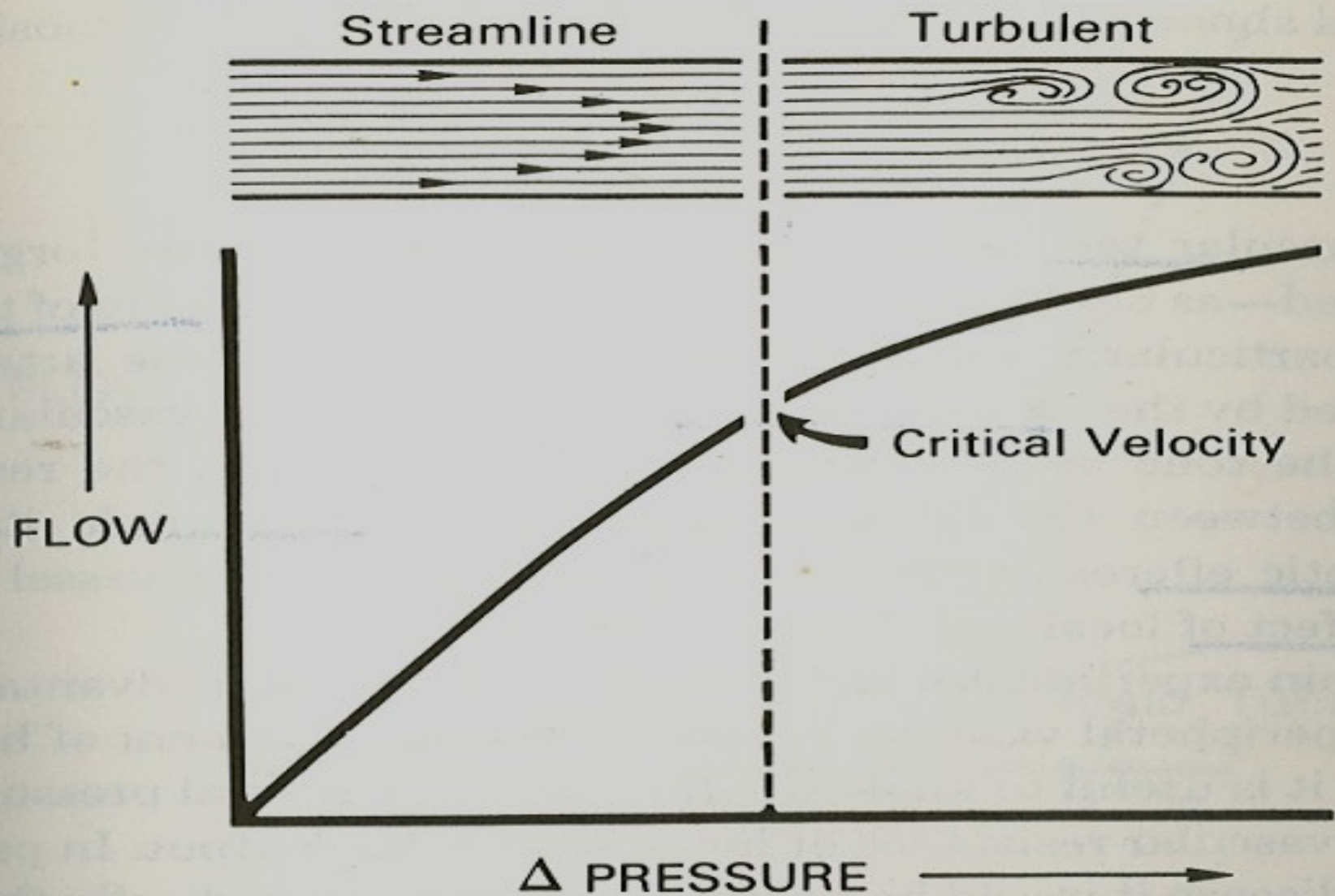


Fig.. Relationship between velocity of flow and turbulence.

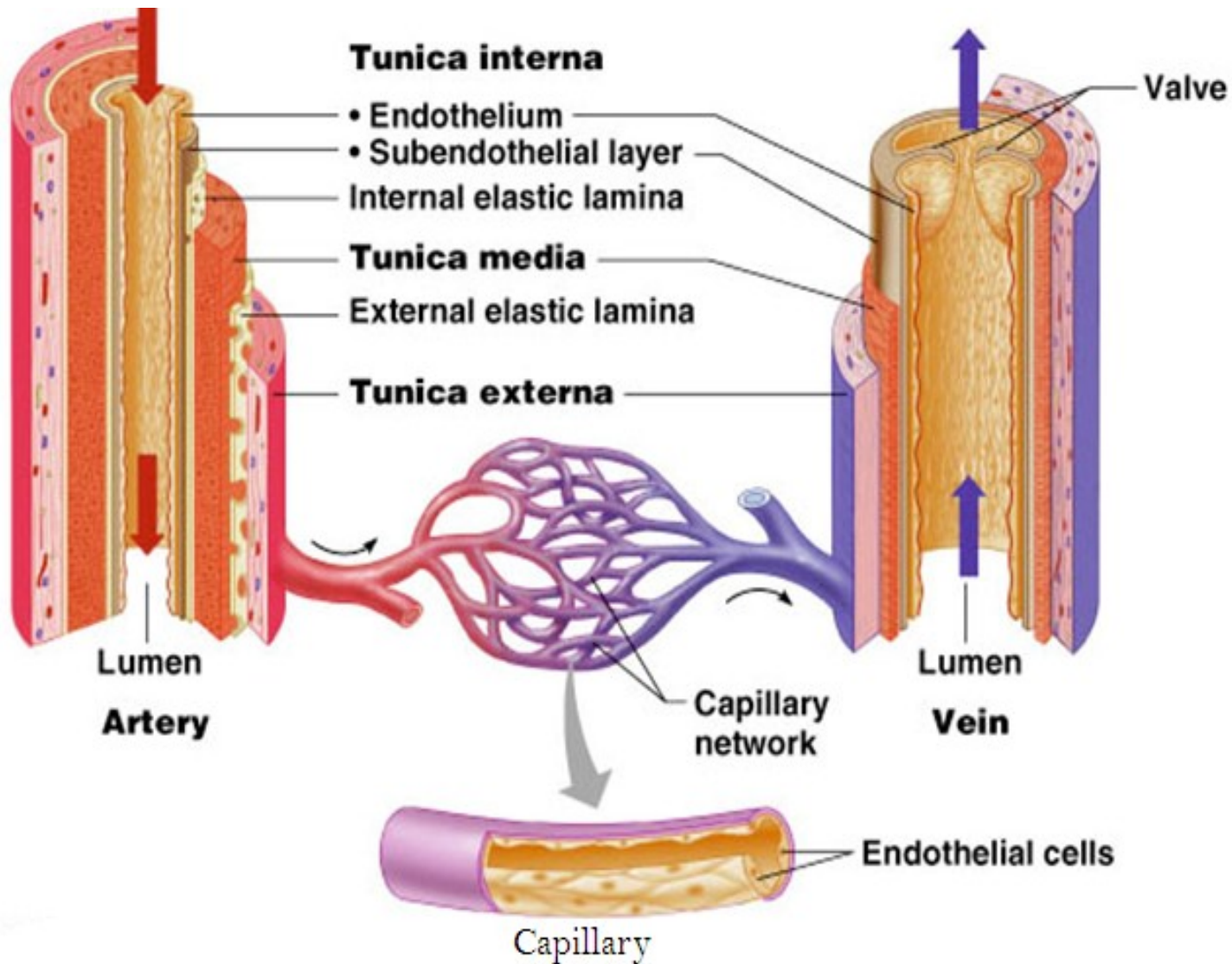
The Structure of Arteries and Veins

- Arteries and veins resemble each other in that their walls contain three coats.
- However, the vessels adapt to their different circulatory tasks by differing in the structure of these coats.
- The **inner coat** (vascular endothelium, tunica intima) consists of a single layer of endothelial cells applied to a thin connective tissue layer, the basement membrane.
- The **middle coat** (tunica media) contains primarily smooth muscle and elastic tissue fibers.
- The **outer coat** (tunica adventitia, adventitious coat) embeds the vessel in its surroundings and consists mainly of connective tissue.
- In addition, the arteries have an elastic, fenestrated membrane.

....cont'd

- Arteries are distinguished by an especially well developed muscle coat, which contains a varying amount of elastic fiber according to its site (predominantly elastic and predominantly muscular arteries).
- This layer is the driving force of the blood vessels by **dilating (vasodilatation)** and **constricting (vasoconstriction)** the diameter of the blood vessels, it regulates **blood flow** and **blood pressure**.
- The arteries near the heart contain a high proportion of elastic fibers and this creates an elastic recoil.
- In the smallest blood vessels, the capillaries, the coats are reduced to one, the tunica intima; this **facilitates the exchange of fluids and gases**.

STRUCTURAL PATTERN OF THE VASCULAR SYSTEM



Lymph Vessels

- The lymphatic system runs parallel to the venous side of the circulation
- It begins near the capillaries as “blind” lymphatic capillaries that reabsorb fluid that has not been taken up from the tissues by the blood vessels (lymphatic fluid).
- Small and large lymph vessels then return the lymph to the venous blood.
- The wall of the lymph vessels consists of an endothelium, and a thin layer of rhythmically contracting smooth muscle cells.
- Similarly to the situation in the veins, numerous valves further the transport of the lymph.
- The course of the lymph vessels is interrupted by **lymph node** stations, which represent a kind of biological filter and fulfill important functions in immune defense.

Functional Classification of Blood Vessels

1. Elastic vessels (Windkassel vessels):

- Example: Aorta, big arteries
- Pressure storing components
- High ability of recoiling

2. Resistance vessels (Stop Cock Vessels)

- Example: small arteries and arterioles
- High muscular component
- Develop high resistance
- Regulate blood flow

Blood vessels: Classification...cont'd

3. **Exchange vessels:**

- Example: capillaries
- Made up of endothelium and basement membrane.
- Thin enough for exchange
- 3 types: continuous, discontinuous and fenestrated capillaries

4. **Capacitance vessels** (big to small veins)

- Very high capacity of distension
- Can accommodate large volume of blood (65% of blood volume)

Arteries

- **AORTA** branches → peripheral vascular bed → 1000 fold increase in cross-sectional area at capillaries
- Large amount of elastin → ↑ expansion and recoil → ↑ pressure storage.
- A velocity of 40 cm/sec at aorta → < 2 mm/sec at capillaries, i.e. $V = Q/A$.
- Beyond capillaries, the situation is reversed and blood flow accelerates as total cross-sectional area of veins decreases.

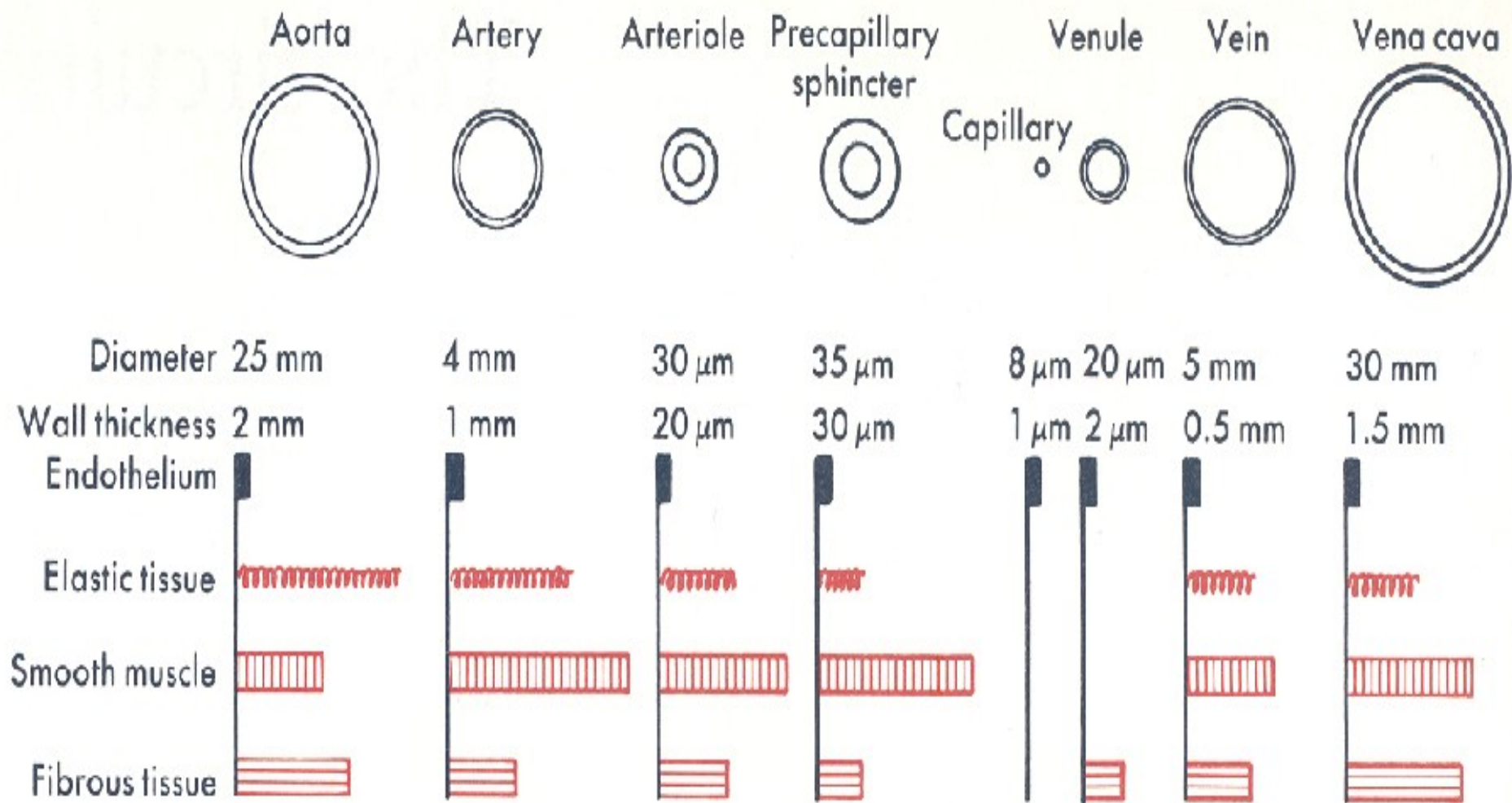
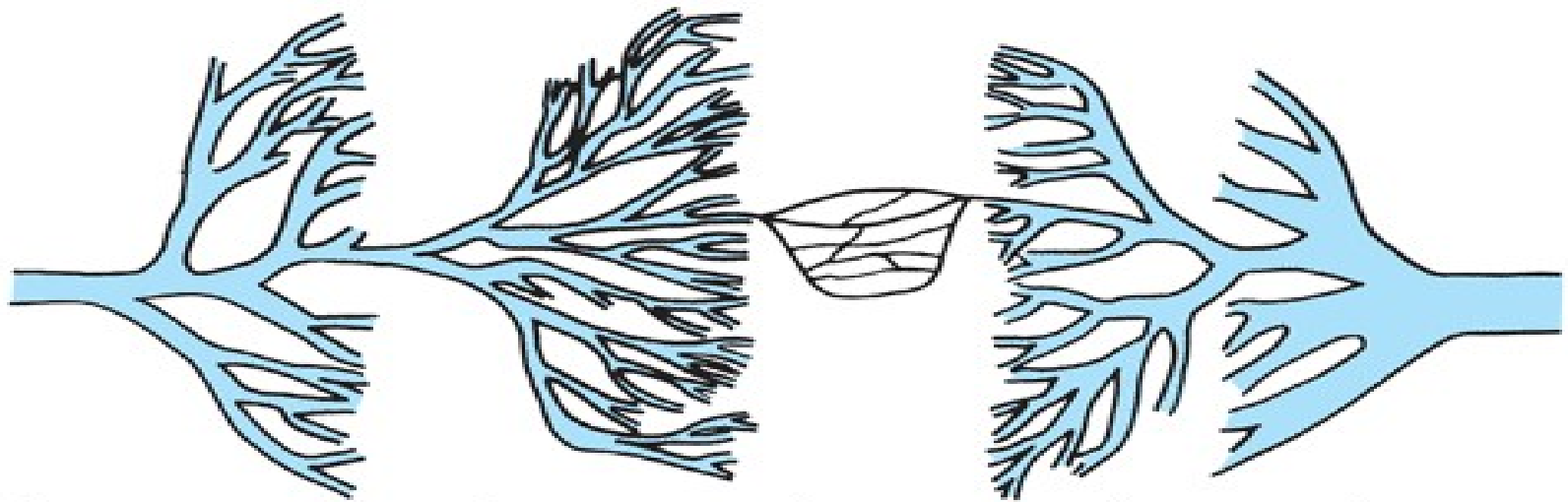


Fig. Internal diameter, wall thickness, and relative amounts of the principal components of the vessel walls of the various blood vessels that compose the circulatory system. Cross sections of the vessels are not drawn to scale because of the huge range from aorta and venae cavae to capillary.



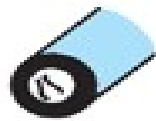
ARTERIES

ARTERIOLES

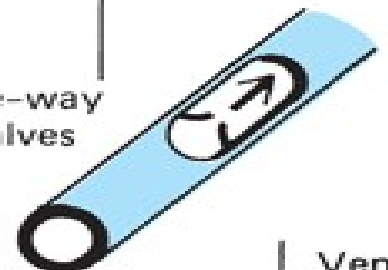
CAPILLARIES

VENULES

VEINS



Aorta



one-way
valves

Venae
cavae

	ARTERIES		ARTERIOLES	CAPILLARIES	VENULES	VEINS	
internal diameter	2.5 cm	0.4 cm	30 μm	5 μm	70 μm	0.5 cm	3 cm
wall thickness	2 mm	1 mm	20 μm	1 μm	7 μm	0.5 mm	1.5 mm
number	1	160	5×10^7	10^{10}	10^8	200	2
total cross-sectional area	4.5 cm^2	20 cm^2	400 cm^2	4500 cm^2	4000 cm^2	40 cm^2	18 cm^2

...cont'd

- Resistance (R), pressure (P) and flow (F) patterns change as we go from arteries to arterioles, small branches of arterial system measuring about 0.10mm in diameter.
- R increases as we go from the aorta to the arterioles and hence arterioles are referred to as resistance vessels and “stop cocks” of the arterial tree.
- Pressure decreases from aorta to capillaries and blood flow becomes steady as it passes the arterioles.
- Decrease in O_{ce} or in transmural pressure → extreme narrowing → active or passive closure in resistance vessels → obliteration of lumen (“critical closure”) → flow ceases.

...cont'd

- “Critical closure” occurs most readily in such vessels as the thick walled **cutaneous A-V shunts** or in the **ductus arteriosus** and in strongly constricted arterioles- **met-arterioles sphincters**.
- One **arteriole** branches to **10 capillaries** leading to the formation of **10 billion capillaries** in the peripheral tissues.
- **Capillaries**, in spite of their thinness and weakness, are capable of withstanding ABP without damage because they have **narrow lumen** and the **tension developed is small**.
- The protective effect of small size is an example of the operation of Laplace's law.
- According to this law, tension in the wall of a cylinder (T) is equal to the product of transmural pressure (p) and radius divided by the wall thickness (w).

$$T = p \times r / w$$

Arterial Blood Pressure (BP)

- The overall strategy of the cardiovascular system is to :
 1. provide all organs with a constant perfusion pressure and
 2. allow each individual organ to regulate its blood flow in accordance with the local needs of the tissue.
- **ABP** includes pressure in the aorta and other large arteries.

NB:

- BP is conventionally written as systolic (peak, ejection pressure)/ diastolic (minimum, filling pressure) blood pressure.

Normal SBP: 90 – 130 mm Hg (120 mm Hg)

DBP: 60 – 90 mm Hg (80 mm Hg)

PP: SBP – DBP

MAP: $P_d + \frac{1}{3}(P_s - P_d)$

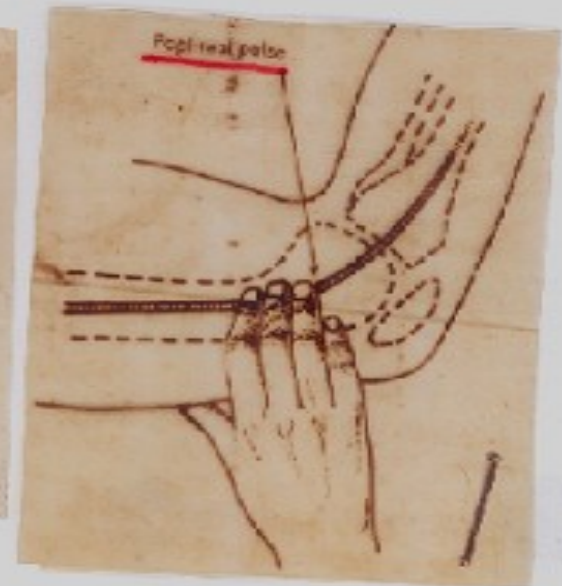
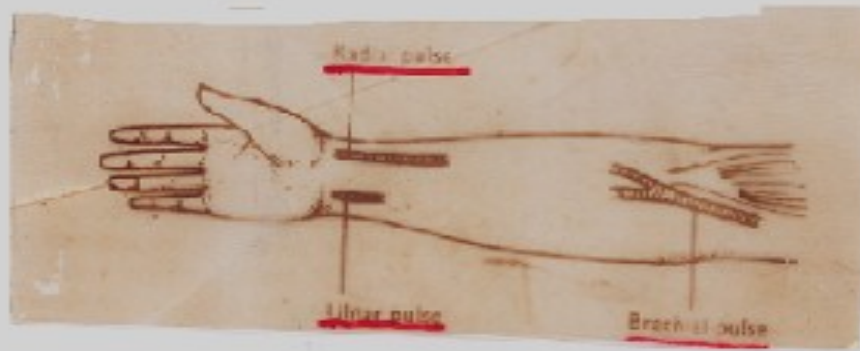


Fig..Pulses

Sites of arterial blood pressure pulsations

Measurement of ABP

1. Invasive (direct) measurement:

- A cannula is inserted into a large artery and pressure measured directly with a manometer or a strain gauge connected to a recorder.
- The record is made as shown below.

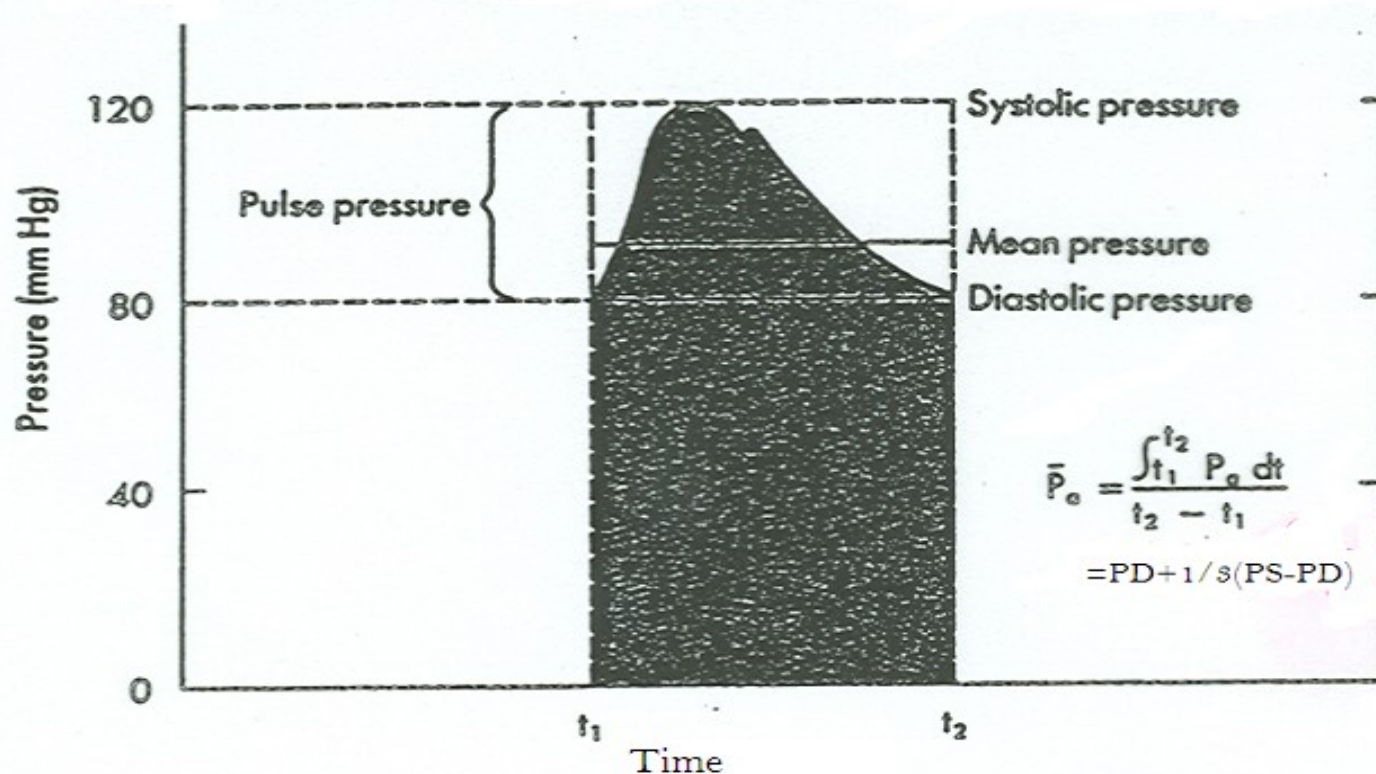


Fig.. ABP: systolic, diastolic, pulse and mean

2. Indirect routine measurement

- This method uses Riva Rocci cuff attached to a sphygmomanometer

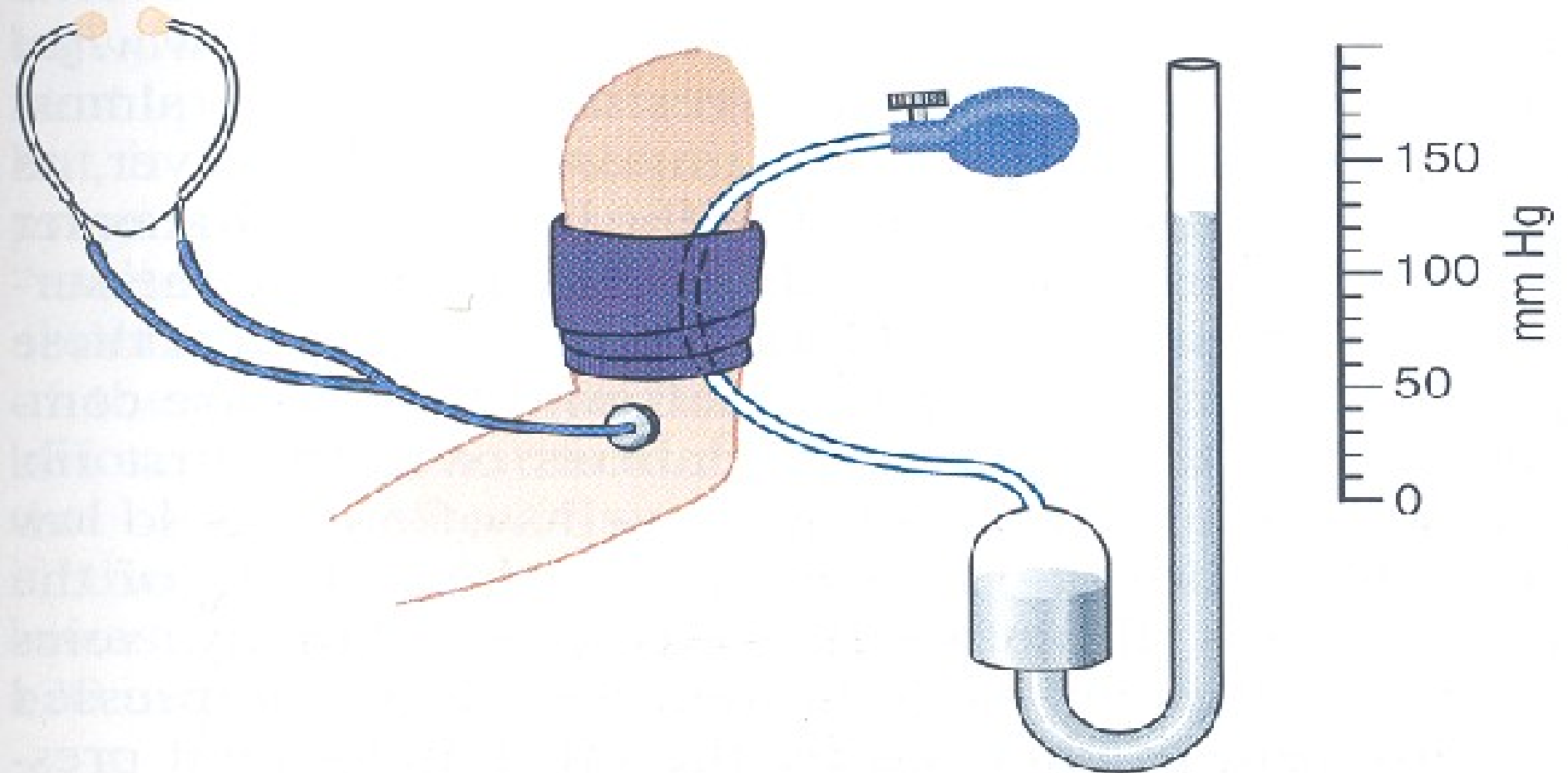
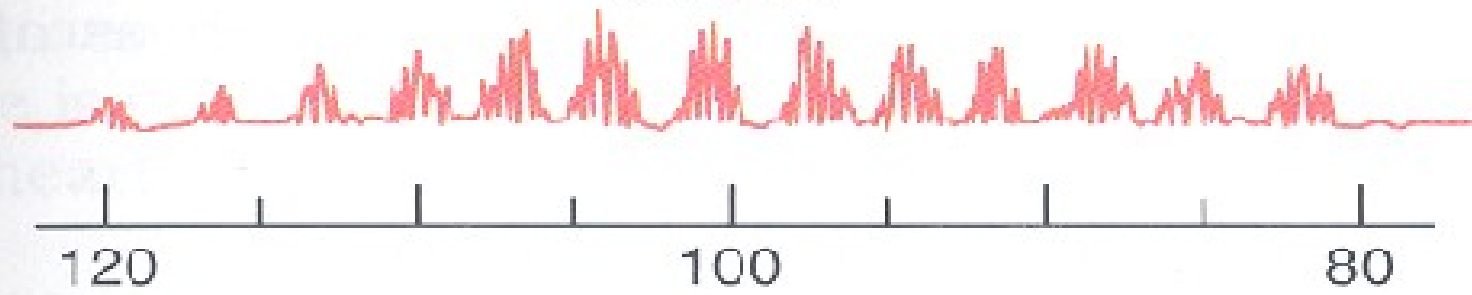
a. Auscultatory method: BP(D/S) is read using Korotkoff's sound.

- BP reading in:
 - **Children:** muffling of Korotkoff's sound for DB
 - **Adults:** complete cessation of Korotkoff's sound for DB

b. Palpation method: used to estimate systolic BP

- Approximation of mean BP

Sounds



Recommended cuff sizes for estimating BP

1. Under one yr of age =2.5cm
 2. 1-3 yrs =5-6cm
 3. 4-8 yrs- 9-10cm
 4. Average adults =12.5cm
 5. Obese adults =14cm
- **Venous pressure measurement:** use a hypodermic needle to be inserted into a vein and connect it to a sensitive aneroid manometer.

General Factors that affect arterial BP

1. **Posture:** In the upright position any vessel below heart level has increased BP,
 - e.g. BP in dorsalis pedis = 200 mmHg
 - Gravitational effect on BP = 0.77 mmHg/cm
2. **Age:** Newborn: 70/50, infants: 90/60, children: 100/70
 - SBP increases by ~ 1 mmHg/yr, secondary to atherosclerosis.
 - DBP increases by ~ 0.4 mmHg/yr secondary to increase in TPR.
3. **Sex:** ♀ sex ≤ 40-50 yrs → low BP, ♀ ≥ 50 yrs → high BP.
 - This BP could be higher in women than in men.
 - probably due to hormonal changes during menopause.
4. **Body weight:** both SBP and DBP are directly related to body wt.
 - ABP is higher by 10-15 mm Hg in obese persons.

5. **Race and socioeconomic status:** Both SBP & DBP are higher in blacks than those in whites at all ages and for both sexes.
 - This difference is attributable to genetic factors and/ or economic status.
6. **Exercise:** SBP and DBP increase with exercise.
 - $\uparrow$ SBP is secondary to $\uparrow$ contractility;
 - DBP may decrease initially secondary to vasodilatation in skeletal muscle vasculature but $\uparrow$ heart rate limits runoff time $\rightarrow$ $\uparrow$ DBP.
 - Exercise increases ABP by 40 – 50 mm Hg
7. **Emotion:** increases ABP by 10– 30 mm Hg
8. **Deep sleep:** decreases ABP by 20 mm Hg
9. **Time of the day:** $\downarrow$ BP in the morning, $\uparrow$ BP in the evening
10. **Thermal stress** decreases ABP

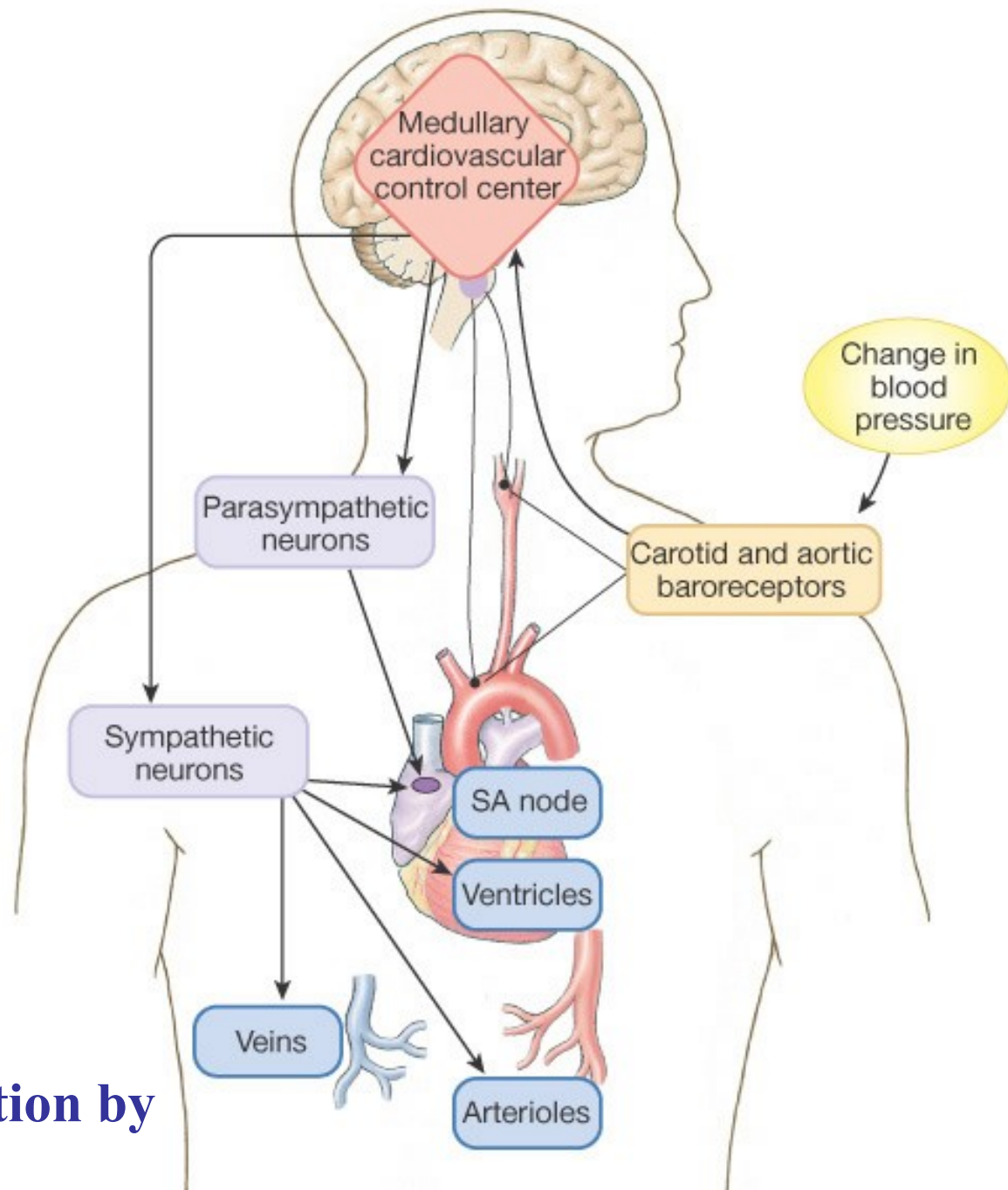
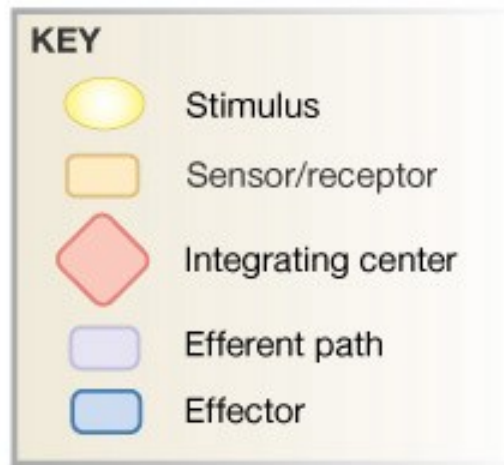
Regulation of ABP

A. Short-term controlling mechanisms

- CNS ischemic response
- Baroreceptor reflex
- Chemoreceptor reflex
- Atrial stretch reflex
- Stress relaxation and reverse stress relaxation
- Capillary fluid shift
- Hormonal mechanisms: AD, NAD, Vasopressin (ADH)

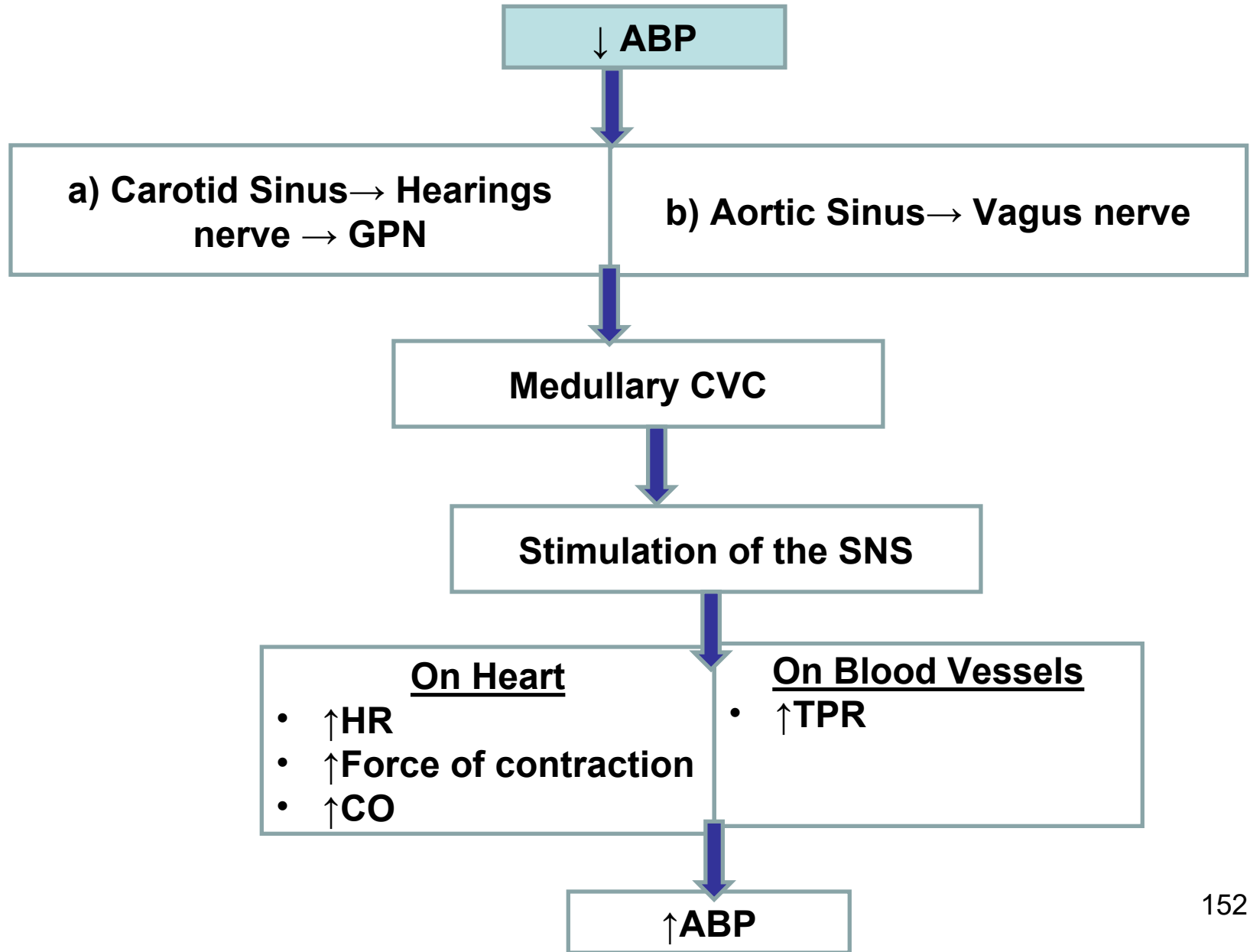
B. Long-term controlling mechanisms

- The renal mechanism
- The Renin-angiotensin-aldosterone system

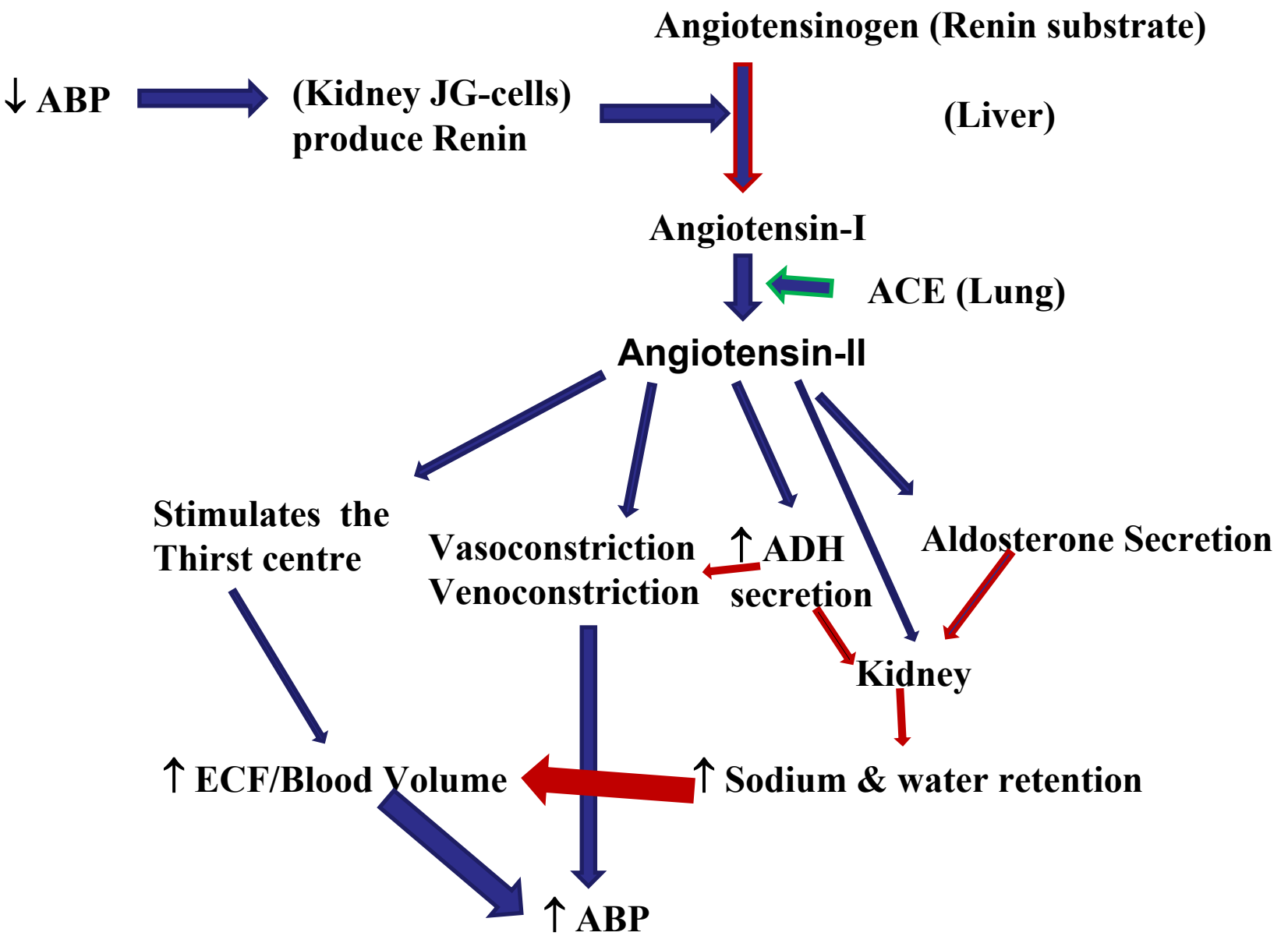


Blood Pressure Regulation by Baroreceptors

Baroreceptor Reflex



RENIN-ANGIOTENSIN-ALDOSTERONE MECHANISM



Blood Pressure Abnormalities

Hypertension:

- Hypertension is persistent elevation of ABP $> 160/95$ mmHg
- Upper limit of ABP in different age groups is different

Age (yrs).....Upper limits of normal

- 10-20..... 140/90 mmHg
 - 20-50..... 150/95 mmHg
 - >50 160/100 mmHg
- Any levels higher than these limits are considered as hypertension.

Types of Hypertension

A. Primary (Essential) Hypertension (90% of the cases)

- ↑↑ ABP with no clear cause, It may be benign or malignant.

B. Secondary (Symptomatic) Hypertension (10%)

Causes:

1. Renal HTN: renal ischemia, excessive renin
2. Endocrine disturbance
 - Pheochromocytoma (↑↑ AD)
 - Cushing's disease (↑↑ Cortisol)
 - Hyperaldosteronism
 - Oral contraceptives
3. Cardiovascular diseases due to changes in CO, PR, arterial compliance
 - Aortic coarctation: congenital constriction of the aorta (↑BP in upper parts of body)
 - Atherosclerosis
4. Drug induced HTN

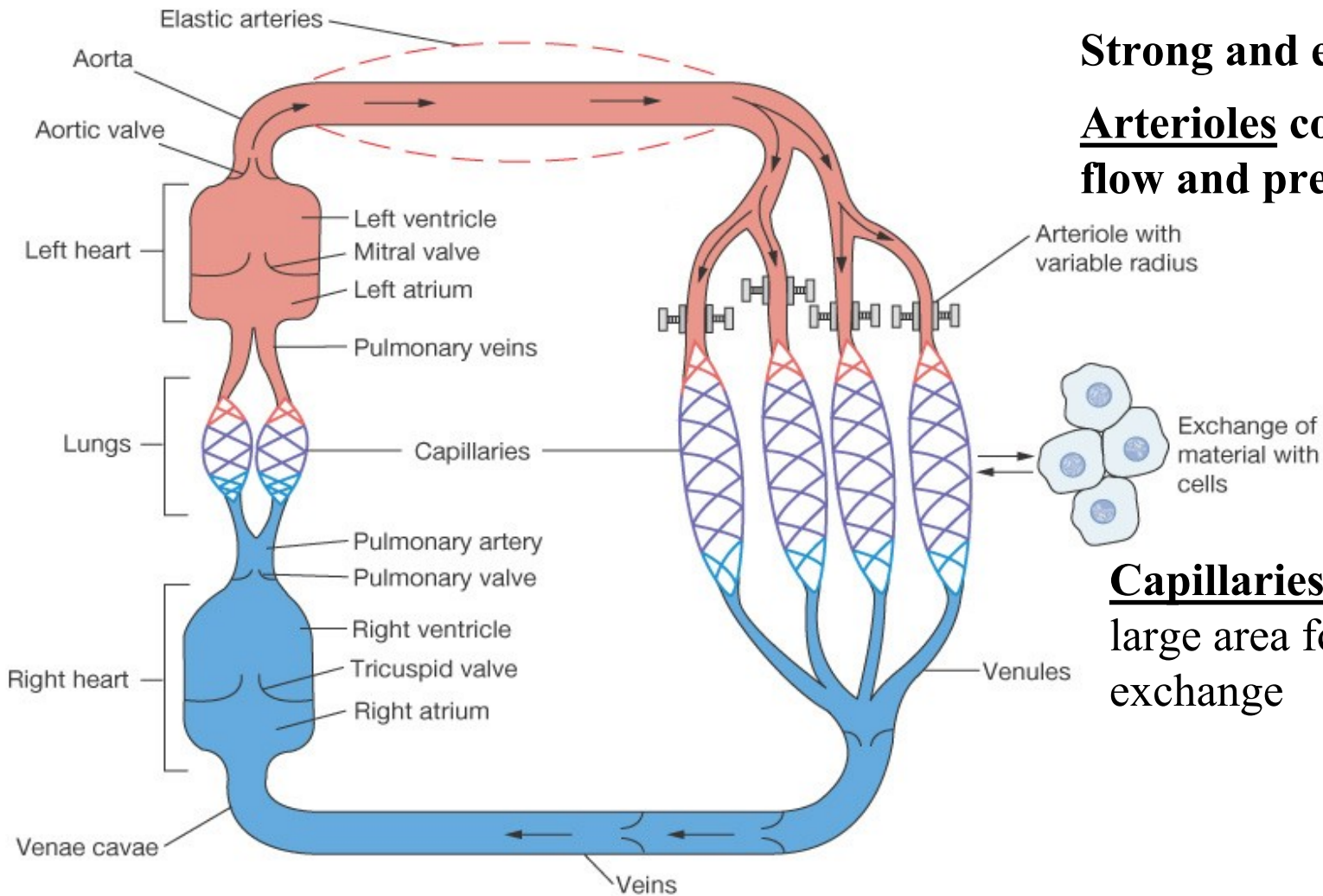
Hypotension:

- SBP= $\leq$ 90 to 110 mmHg
 - a. Essential hypotension
 - b. 2° hypotension
 - Acute fever
 - Myocardial infarction
 - TB
 - Adrenal or pituitary insufficient.
 - c. Orthostatic or postural hypotension
- Reflex mechanisms greatly depressed.

General therapeutic measures of HTN

- Wt. reduction
- Salt-intake restriction
- Mild exercise
- Psychotherapy, relaxation, yoga, transcendental meditation, music therapy, etc.
- Avoid:- Smoking
 - Excessive alcoholic intake
- Patient education
- Drug treatment
 - Diuretics
 - Drugs acting on ANS

Blood vessel functions: overview



Strong and elastic arteries

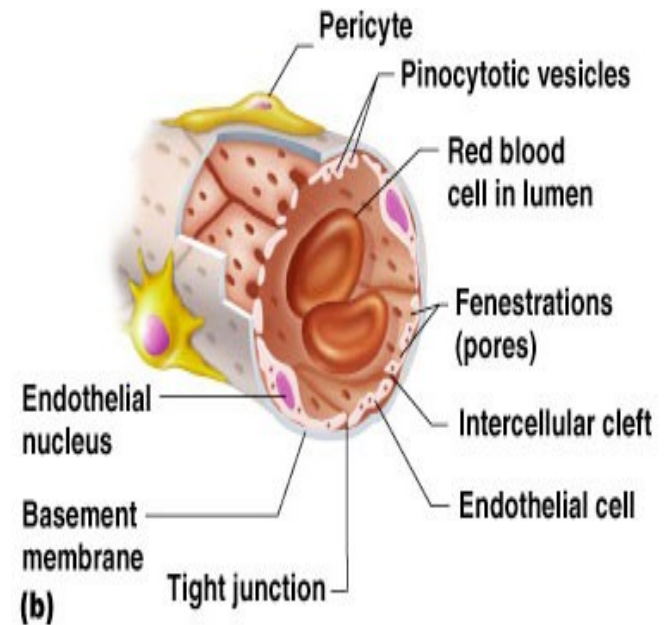
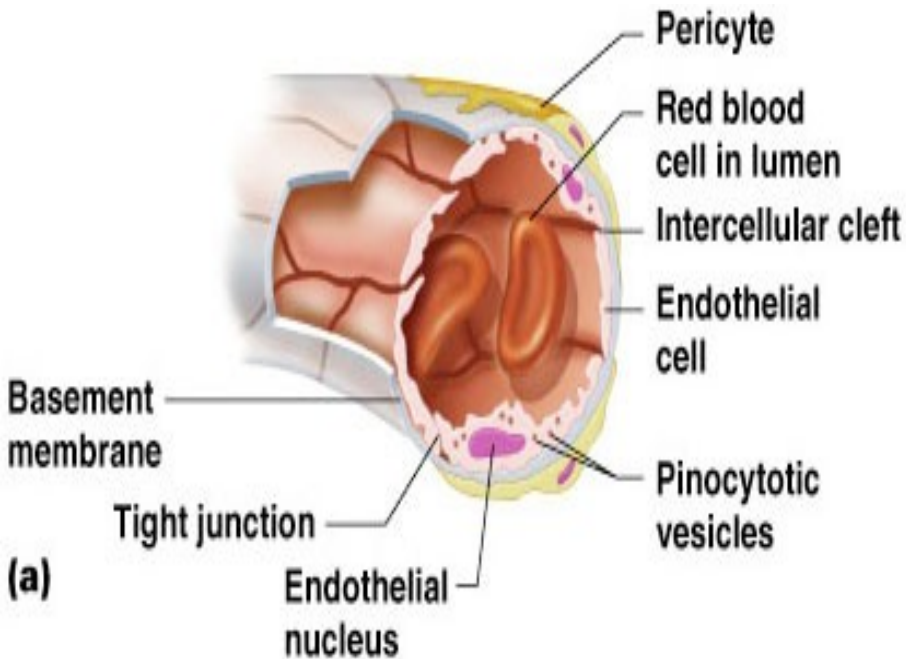
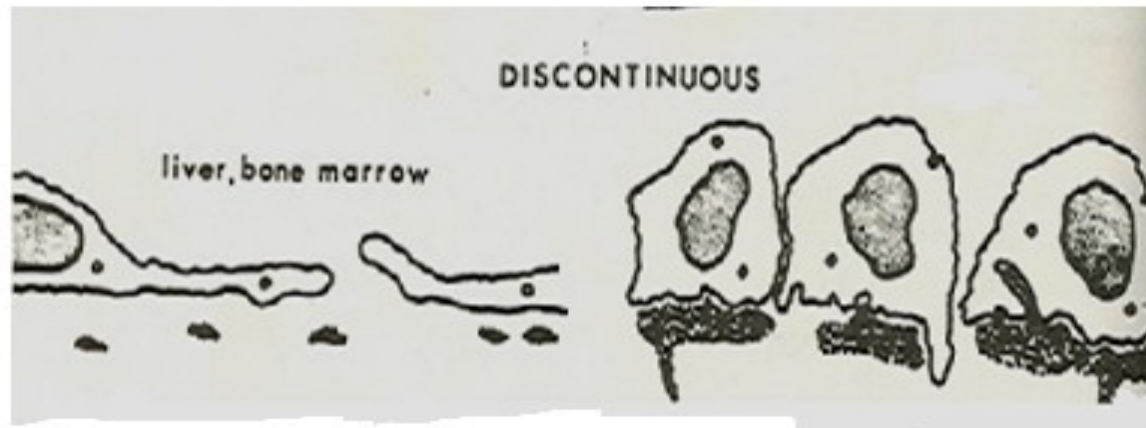
Arterioles control blood flow and pressure

Capillaries: thin and with large area for diffusional exchange

Veins: compliant, large, low resistance
veins have valves & assure blood return to the heart

CAPILLARIES

- Pressure inside is 15 to 35 mmHg
- 5% of the blood is in capillaries
- Necessary for exchange of gases, nutrients, and wastes
- Flow is slow and continuous
- Blood flow controlled by pre-capillary sphincter muscles *via* metabolic regulation
- No smooth muscle, no innervations
- Largest total cross sectional area
- Based on the continuity of their filtration barriers, capillaries have been labelled as **continuous, fenestrated and discontinuous**



Continuous Capillaries

Fenestrated capillaries

Capillaries Exchange

- There are two forces that determine the direction of flow of fluid through the capillaries
 - **The capillary colloid osmotic pressure (25 mm Hg)**
 - **The capillary hydrostatic pressure (32mmHg)**
- Fluid that is leak out of the capillaries is taken up by the lymphatic vessels to prevent edema
- Net outflow into ECF
 - Net filtration – net absorption = net outflow
 - About 2 L/day collected by lymph vessels

Capillary Exchange:

(a) Filtration in systemic capillaries

Net pressure = hydrostatic pressure – colloid osmotic pressure

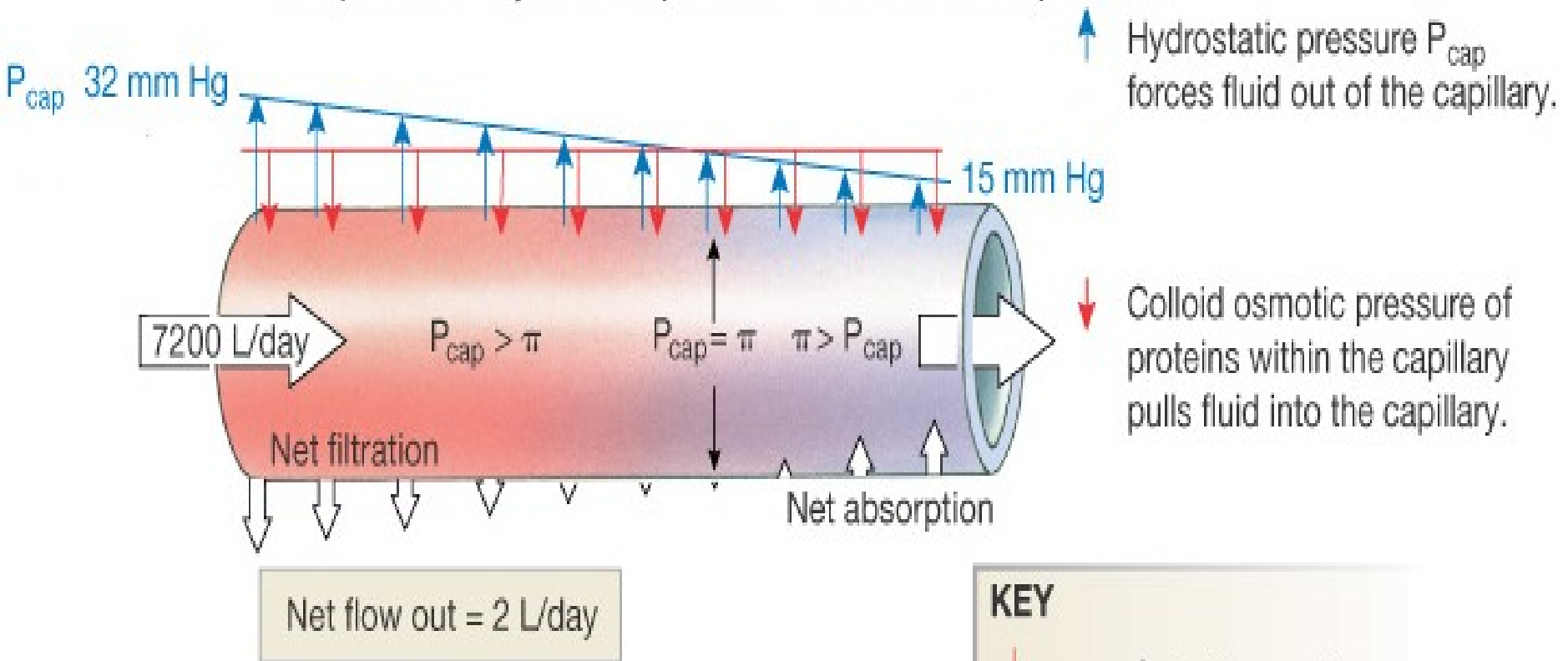
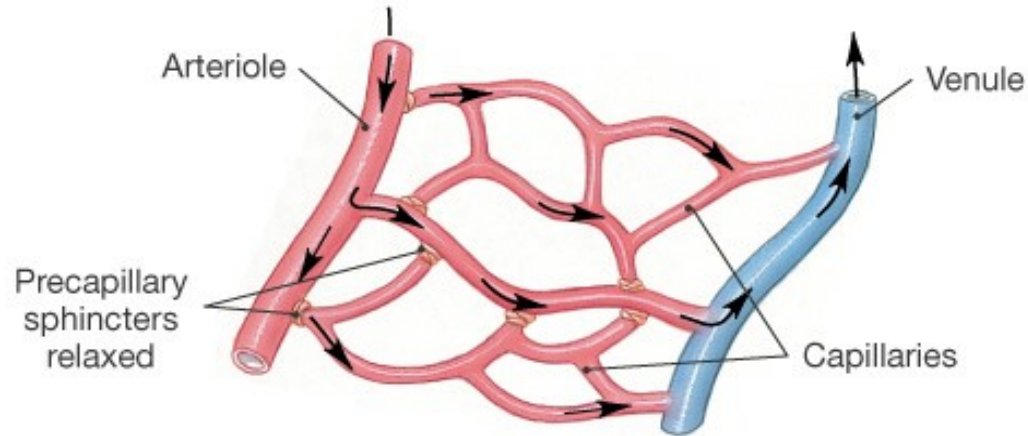
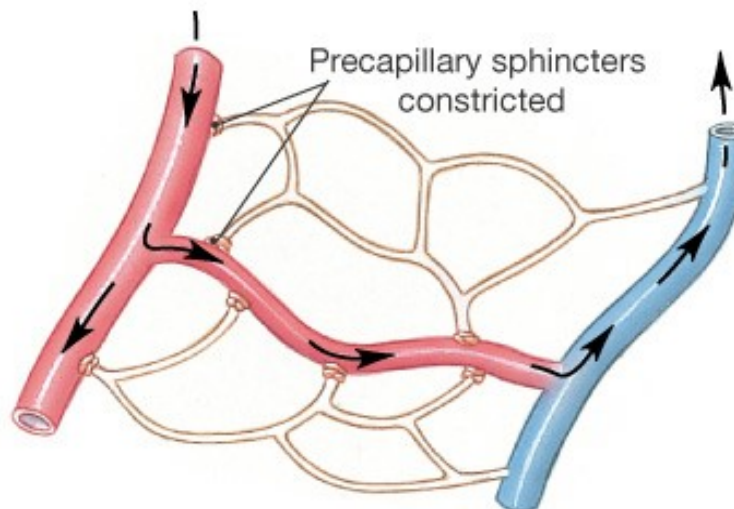


Figure: Capillary fluid exchange

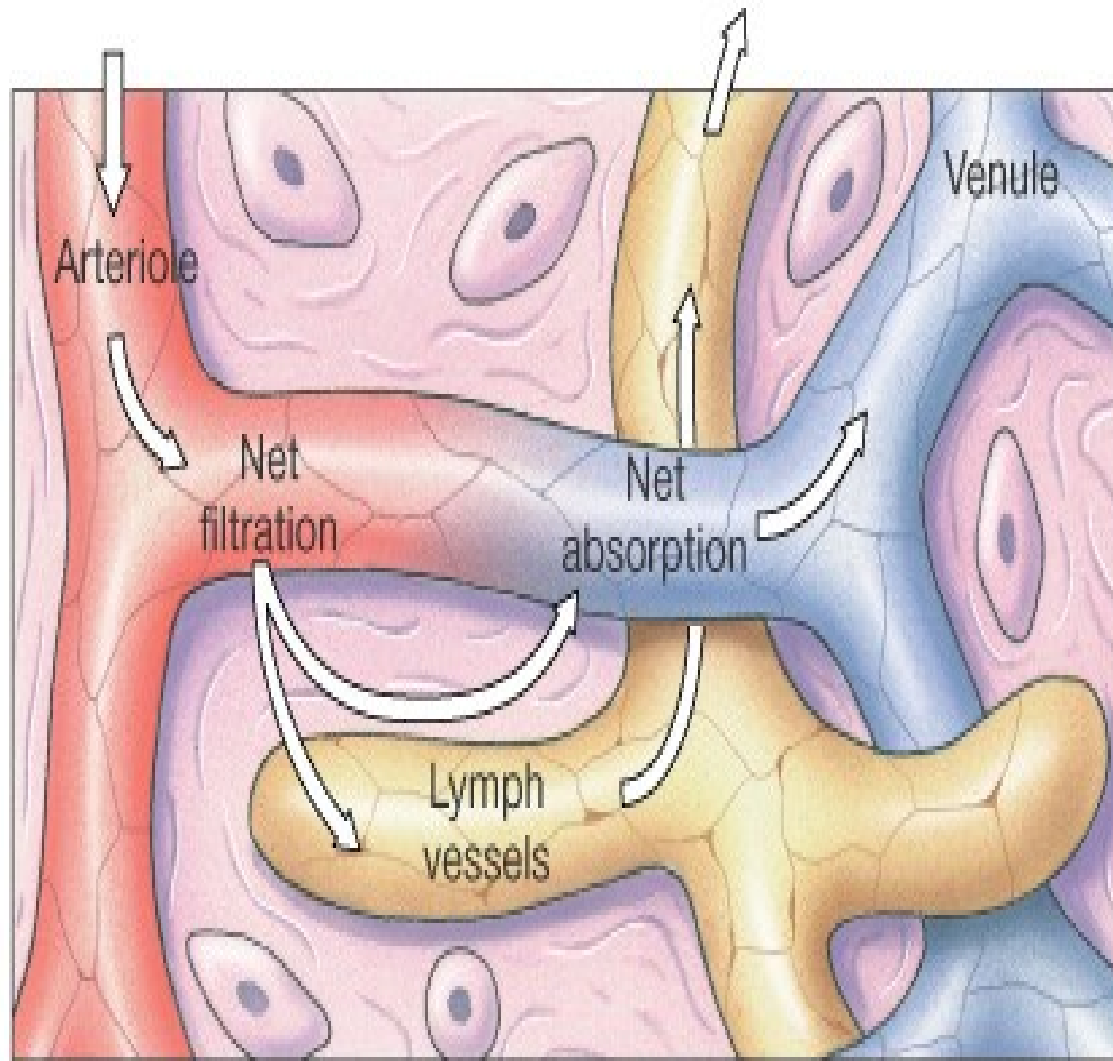
(a) When precapillary sphincters are relaxed, blood flows through all capillaries in the bed.



(b) If precapillary sphincters constrict, blood flow bypasses capillaries completely and flows through metarterioles.



(b) Relationship between capillaries and lymph vessels



The excess water and solutes that filter out of the capillary are picked up by the lymph vessels and returned to the circulation.

Figure: Fluid exchange at the capillary

Capillary Blood Flow-Velocity

- Lowest velocity
- Largest total cross sectional area
- Hydrostatic pressure drops slightly

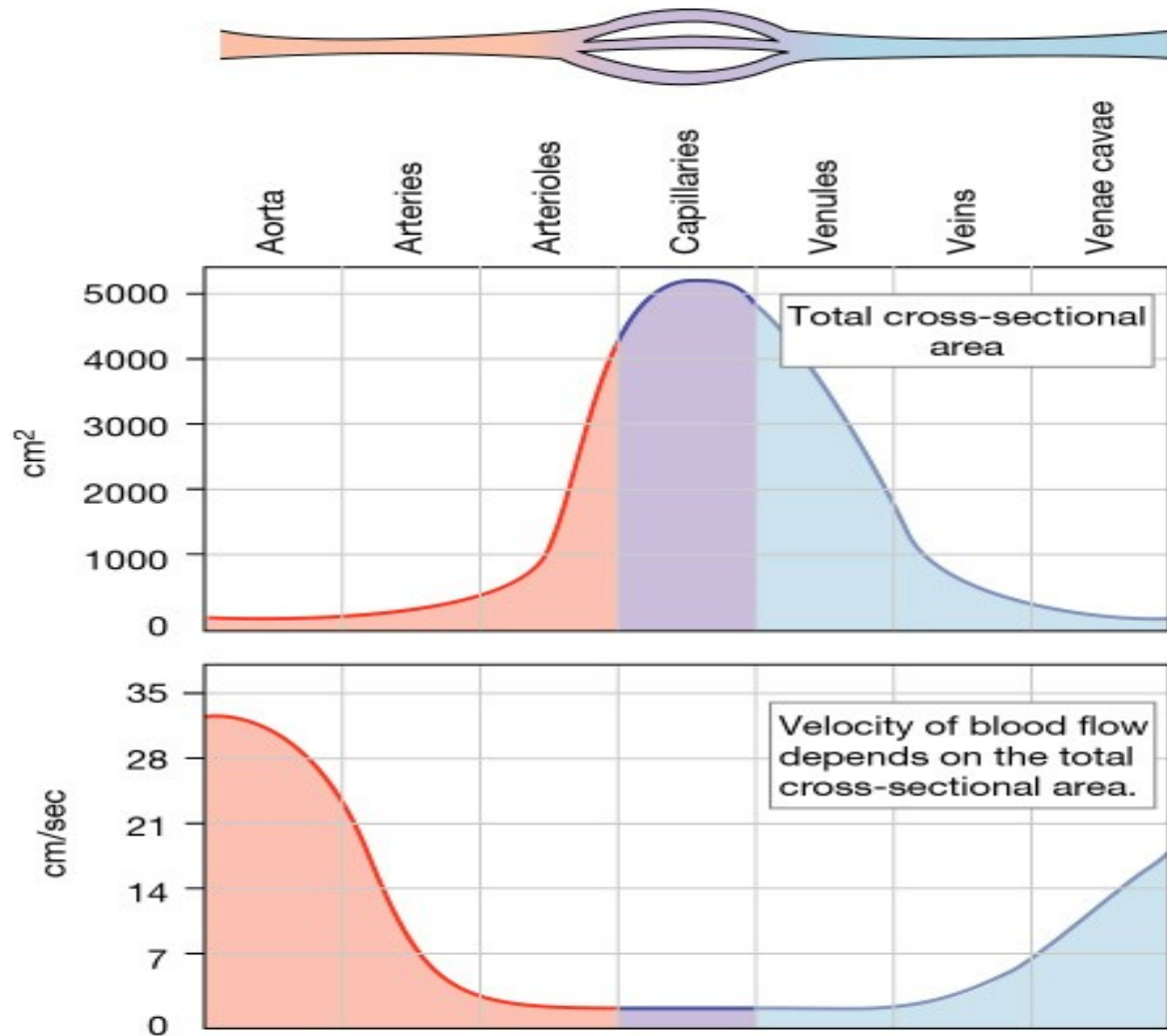


Figure. The velocity of flow depends on the total cross-sectional area

Blood flow

- Blood flow (BF) is the amount of blood that moves to a particular organ in a given time
- Total blood flow is equal to CO; 5-6 L/min
- BF is determined by 2 factors
 1. Pressure difference b/n 2 ends of the vessel
i.e., the force that pushes blood through vessels
 2. Resistance of flow, hindrance to flow through vessels

Where Q = Blood flow

$$Q = \Delta P / R$$

ΔP = Change in pressure

R = Resistance

- **Ohms Law:** states that BF is directly proportional to the ΔP but inversely proportional to resistance (R).

Vascular resistance

- **Poiseuille's Law**: Vascular resistance to BF is directly proportional to the length of the vessel and viscosity of blood, but inversely proportional to the 4th power of radius of the vessel.

Where, R = Resistance

$$Q = \frac{\Delta P \pi r^4}{8 \eta l} \quad R = 8 \eta l / \pi r^4$$

Q = Blood flow

(3.14)

Radius

Factors affecting viscosity of blood

- Velocity of BF: $\uparrow$ Velocity = $\downarrow$ Resistance
- Hematocrit: polycythemia = $\uparrow$ Viscosity = $\uparrow$ PR
Anemia = $\downarrow$ Viscosity = $\downarrow$ PR
- Diameter of blood vessels
- Plasma protein concentration

PR = Peripheral resistance

l =

η =

π =

r =

Resistance to blood flow

- It is the hindrance to blood flow
- Has no direct means of measurement, but can be calculated from values of BF and ΔP in vessels
- If the ΔP is 1 mm Hg and BF 1 ml/sec, then vascular resistance is said to be 1 peripheral resistance unit (PRU)

Total peripheral resistance and total pulmonary resistance

- The rate of BF through the circulatory system at rest is 100 ml/sec or 5 L/min.
- The ΔP between systemic arteries and veins is 100 mm Hg
- Resistance of the entire systemic circulation, called TPR is $100/100 = 1$ PRU.
- TPR ranges between 0.2 and 4 PRU during vascular dilation and constriction

TOTAL PERIPHERAL RESISTANCE

SYSTEMIC CIRCULATION:

RAP=Right atrial

$$\text{TPR} = \frac{\text{Aortic Pressure} - \text{RAP}}{\text{FLOW}}$$

pressure

$$\text{TPR} = \frac{100 - 0 \text{ mmHg}}{83.3 \text{ ml/sec (5 L/min)}} = 1.2 \text{ PRUs}$$

PULMONARY CIRCULATION:

$$\text{Pul. R.} = \frac{\text{Pul. Art. P.} - \text{LAP}}{\text{FLOW}}$$

$$\text{Pul. R.} = \frac{15 - 5 \text{ mmHg}}{83.3 \text{ ml/sec}} = 0.12 \text{ PRUs}$$

Oedema

- Water and solutes accumulate in IF compartment→ oedema,
- It is pitting if firm pressure on swollen area leaves an impression- occurring in dependent parts- extracellular oedema.
- If the oedema is non-pitting, it is cellular.
- Oedema in general occurs when the increase in tissue fluid volume is 30% or more of the basal volume or when filtration is >reabsorption + lymphatic drainage

Oedema may be evident from:

- Puffy face (eyes) in the morning
- Puffy feet
- Sacral oedema when patient is lying in bed.

Generalized Oedema (Anasarca):

- Expanded IF volume
- ∇ ↑Total body sodium

Possible causes of pitting oedema

1. Increased **hydrostatic pressure** in capillaries secondary to heart failure
2. Decreased **osmotic pressure** due to low plasma proteins secondary to conditions like nephrotic syndrome
3. Increased **capillary permeability**
4. **Lymphatic obstruction** (non-pitting edema). . Localized

Cellular or non-pitting oedema:

- may be caused by:
 1. Depression of cellular metabolism $\rightarrow$ $\downarrow$ Na^+ - K^+ pump $\rightarrow$ $\downarrow$ Na^+ influx into cell $\rightarrow$ Water following Na^+ $\rightarrow$ Cellular oedema
 2. Inflammation $\rightarrow$ $\uparrow$ membrane permeability to sodium and other ions that diffuse into cell with water $\rightarrow$ oedema.
Accumulation of water inside cells $\rightarrow$ **Water intoxication.**

Water intoxication: occurs

- when excessive volumes of water are absorbed too quickly leading to nausea, vomiting and shock--conditions caused by undue drop in plasma osmolality before adequate inhibition of ADH secretion occurs.

Other related problems

- **Depletion of ECF volume:** low plasma vol. → low BP, rapid pulse; low IF vol. → poor skin turgor, dry tongue, sunken eyes.
- **Depletion of ICF Volume:** Brain goes mad with thirst due to intense stimulation of hypothalamic osmoreceptors

Characteristics of the Venous System

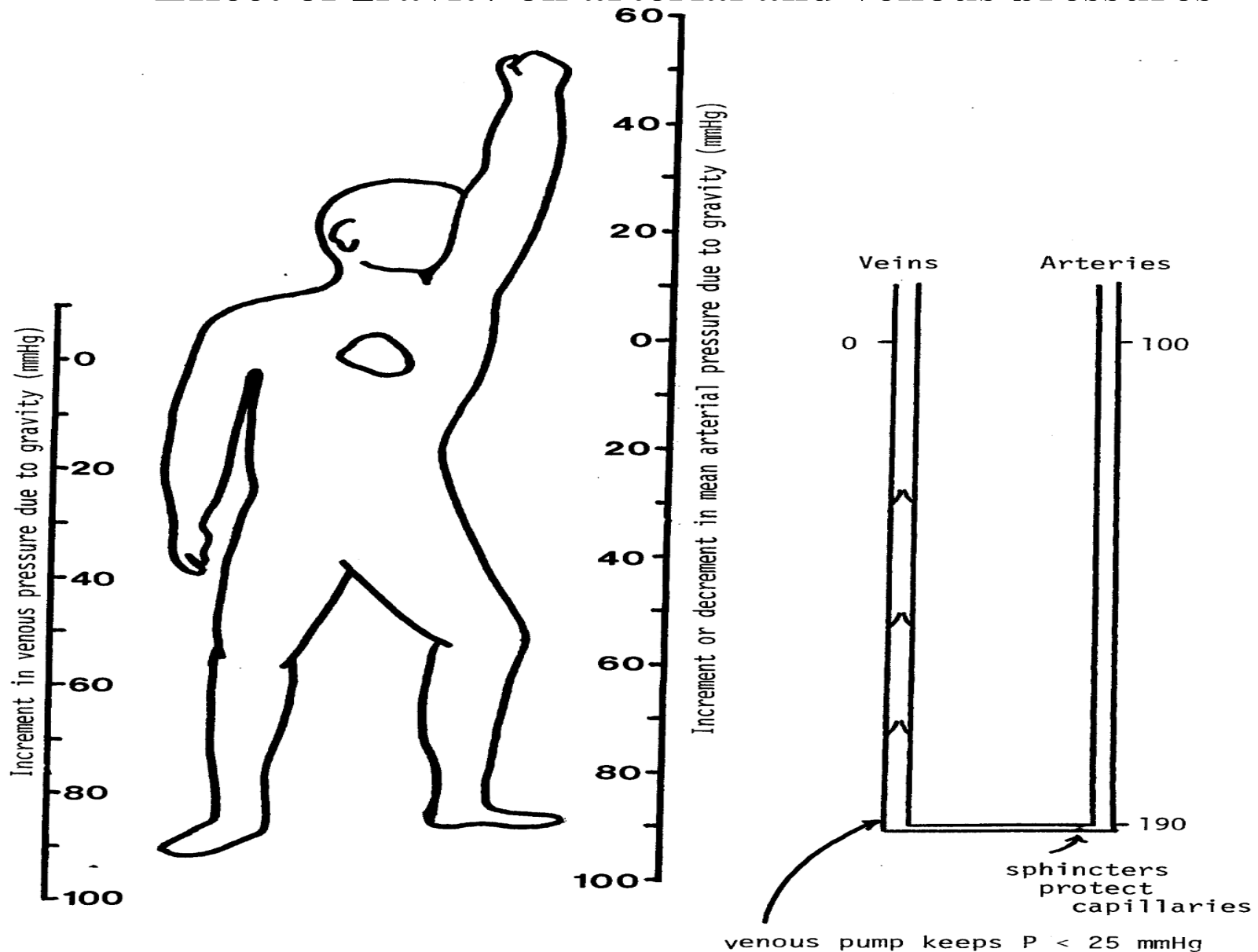
Veins:

- Constitute the low pressure system and extend b/n smallest venules ($>10\text{mm Hg}$) and largest terminal veins (venae cavae = 0 mmHg).
- Are **wide-bored** and **thin-walled**
- Are **volume storer** (high capacitance) of the circulation
- Are low pressure vessels
- Have **great distensibility**
- Have 5-6 times greater distensibility and 2-3 times greater capacity than the arteries.
- Have **one-way valve** to break up columns of blood and to prevent excessive venous distension.
- Faulty valves may lead to development of varicose veins.
- Use **sympathetic supply to facilitate circulatory homeostasis**.

Venous Pressure

- 12-18 mmHg in venules
- 5.5 mmHg in large veins
- Mean pressure in ante-cubital vein is 7.1mmHg
- At entrance of large veins into RA (CVP), the average pressure is 4.6 mmHg
- Like ABP, peripheral venous BP is affected by gravity; it is increased by 0.77mmHg for each cm below RA and decreased by a similar amount for each cm above RA (Fig..).

Effect of gravity on arterial and venous pressures



Measurement of venous pressure

- Pressures, either central (CVP, low) or peripheral (relatively high), can be estimated by inserting a catheter into the thoracic great veins.
- The pressure in peripheral veins is greater than central venous pressure by an amount proportional to the distance away from the right side of the heart.
- CVP can be estimated simply by noting the height to which the external jugular veins are distended when the subject lies with the head slightly above the heart.
- The vertical distance b/n RA and the place the vein collapses (pressure=0) is the venous pressure in mm of blood.
- CVP is decreased during –ve pressure breathing and shock. (inspiration)
- It is increased by +ve pressure breathing(expiratioon), straining, expansion of blood volume and heart failure.

CENTRAL VENOUS PRESSURE (CVP)

- Normal venous pulsations are weak and usually do not appear at levels higher than the sternal angle.
- To make them visible in the neck veins, the subject has to recline on his back with the head supported on pillows (Fig..below). In this position, pulsations in internal jugular vein can be seen and examined.
- Maximum jugular pulsations are observed when the trunk is inclined by less than 30°.
- The vertical distance b/n the right atrium (sternal angle) and the place the vein collapses (the place where the pressure in it is zero) is the venous pressure in mm of blood.

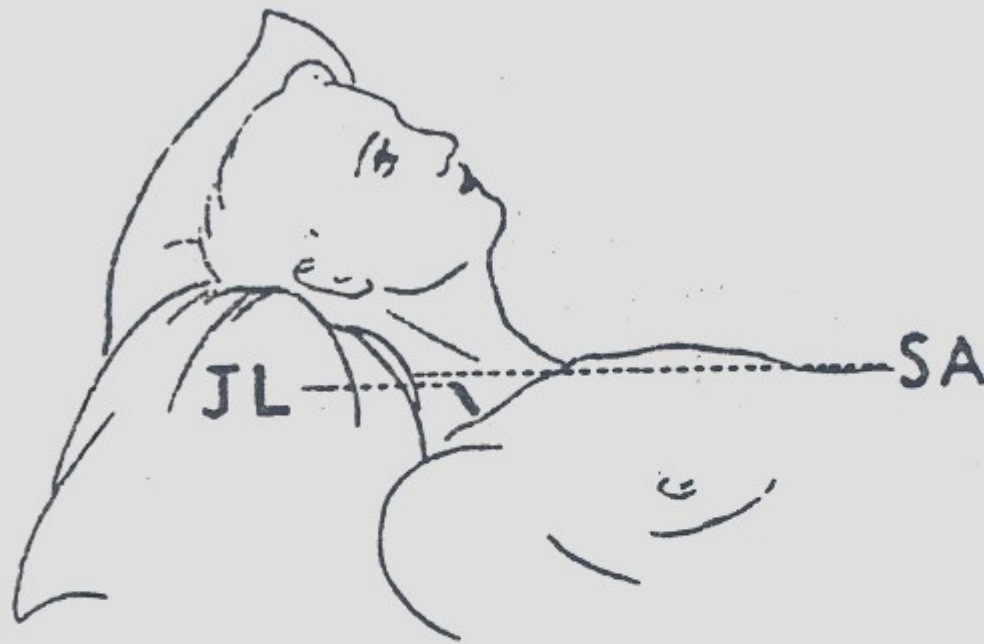


Fig.. The position to examine normal jugular venous pulsations. SA = sternal angle level; JL = upper level where jugular pulsations appear (Jugular level).